

Master Thesis

Enhancing User Confidence in Report Timeliness and Data Accuracy through Data Quality and Availability Checks

Course of study	Master of Science MSc in Engineering: Profile Data Science
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- School of Engineering and Computer Science
- Master of Science in Engineering

Abstract, TODO

Todo, Work for Future Roland

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1. Introduction

Since mankind first began making plans, they were based on the available description of the environment. This approach continued into modern times, although the nature of the predictions has changed. People no longer plan which animal to hunt or when to produce stone tools, but instead how many people the postal service should hire for the Christmas season. These decisions require data, and in order to make the right decisions, the data must be available in the right time, in the right context, for the right people and naturally be correct. This makes data one of the most important assets in a company [The Economist 2017]. Data allow companies to gain an edge over competitors, analyse clients' wishes, or improve internal processes. This is where the relevance of this project emerges.

Swiss Post requires data availability and data quality in its reporting for decision-making. This project analyses the current state of a specific data area of Swiss Post and provides steps towards improvements.

This chapter familiarises the reader with the topic. First, this document's position in the setting of a longer research initiative is described. Second, it presents the background of the project at Swiss Post and delves into the motivation behind it. Third, the business relevance is shown, demonstrating why it is both important and beneficial for Swiss Post to invest in the topic. Fourth, the problem statement is addressed, explaining what this project aims to solve. Fifth, the scope and limitations of the project are outlined more clearly. Sixth, a brief overview of the expected outcomes of the project is provided. Finally, this chapter concludes with a preview of the document's structure.

1.1. Study structure and previous work

The master's thesis is part of a research initiative of Swiss Post to improve data quality and data availability in analytical reporting. It is concluding the initiative, which started one year prior to the start of this research.

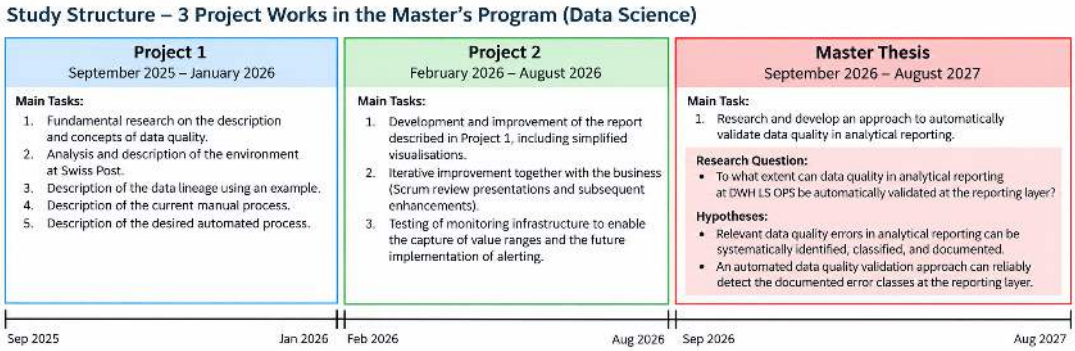


Figure 1.1.: Visualisation of the research initiative at Swiss Post

- **Project 1, Enhancing User Confidence in Report Timeliness and Data Accuracy through Data Quality and Availability Checks** September 2025 to January 2026, [Roccaro 2026b]
  - Fundamental research on the description and concepts of data quality.

  - Analysis and description of the relevant environment at Swiss Post.
  - Description of the data lineage using an example.
  - Description of the current manual process.
  - Description of the desired automated process.

- **Project 2, Automating Data Quality Validations in Amazon Athena: An MVP Based Approach** February 2026 to August 2026, [Roccaro 2026a]
  - Development and improvement of the report described in Project 1, including the development of several simplified visualisations.
  - Iterative improvement in collaboration with business stakeholders, including Scrum Review presentations and subsequent improvements.
  - Testing of monitoring infrastructure to enable the collection of value ranges and support future alerting.

The research results of the previous projects continue to be an essential foundation for this thesis and are thus directly kept in this concluding document. This will also help the reader get a full understanding of the topic without needing to read into details within the other two works.

1.2. Background

Swiss Post is Switzerland's national service provider, active not only in mail and parcel logistics but also in passenger transport (PostAuto), financial services (PostFinance), and various digital and e-government solutions. Its workforce of approximately 45'000 employees and nationwide infrastructure move millions of items and people every day, supported by highly automated sorting centres and extensive data flows [Post 2024]. In such a diversified organisation, analytical reporting is essential for strategic steering, operational planning, and maintaining service quality across business units. The organisational structure of Swiss Post is shown in the following diagram.

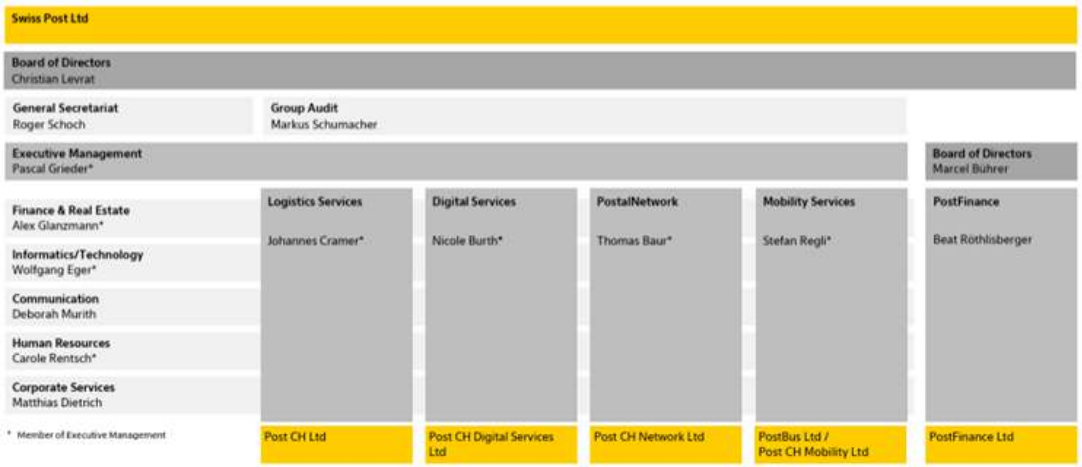


Figure 1.2.: Organisational structure of Swiss Post (online version)

The full mail piece lifecycle

Within this large organisation, the project takes place within a specialised team supporting the entire lifecycle of logistics for mail and parcel delivery. In German, this lifecycle covers every step from A to Z, ASTVZ, which stands for:

- **A**nnahme: Receiving, where parcels and mail are accepted and registered by Swiss Post to be sent through its network
- **S**ortierung: Sorting, the process during which received mail pieces are separated by destination and routed to the next process step, either further sorting or delivery
- **T**ransport: Transport includes the movement between sorting plants and delivery sectors
- **V**erzollung: Customs clearance handles international parcels and letters, including the required customs checks and documentation
- **Z**ustellung: Delivery is the final process step for every item; it is delivered by the staff of Swiss Post

However, this process is more easily visualised using the future vision of Swiss Post. In particular, the version tailored for the IT domain is well suited to describing the project environment. Starting from the top left with the A or V step, followed by S, T, and Z, the process progresses towards the bottom right.



Figure 1.3.: Future way of work illustrating the full postal delivery workflow from receiving an item to delivery.

DWH LS OPS

The project team supports the **D**ata **W**are**h**ouse of **L**ogistic **S**ervices **Op**eration**s** (DWH LS OPS). This **Busi**ness **Dev**elopment **Op**eration**s** (BizDevOps) team is located within the organisational structure of the Logistics Services vertical and the Informatics/Technology horizontal. It represents the single source of truth for analytical reporting at Swiss Post. In contrast to operational reporting, which focuses on real-time guidance for operational processes, analytical reporting focuses on aggregated insights for evaluation and planning. Examples of analytical reporting include working time, process time, and aggregations of sorted items. The stakeholders of the team distribute these reports to various task groups within postal services, such as sorting and delivery.

There are two reasons why a deeper understanding of analytical reporting is currently very important. First, the infrastructure for analytical reporting is transitioning from on-premise systems to the cloud. Second, in addition to the migration of reporting, the infrastructure at all Swiss Post sorting center locations for parcels is undergoing fundamental changes. These changes may include complete modifications of data pipelines, which can make comparisons with previous figures impossible. Therefore, at this critical point in time, it is of utmost importance to maintain users' trust in the data during this change process. The following figure illustrates the shift from the on-premise setup on the right-hand side to the target architecture using cloud hardware on the left-hand side.

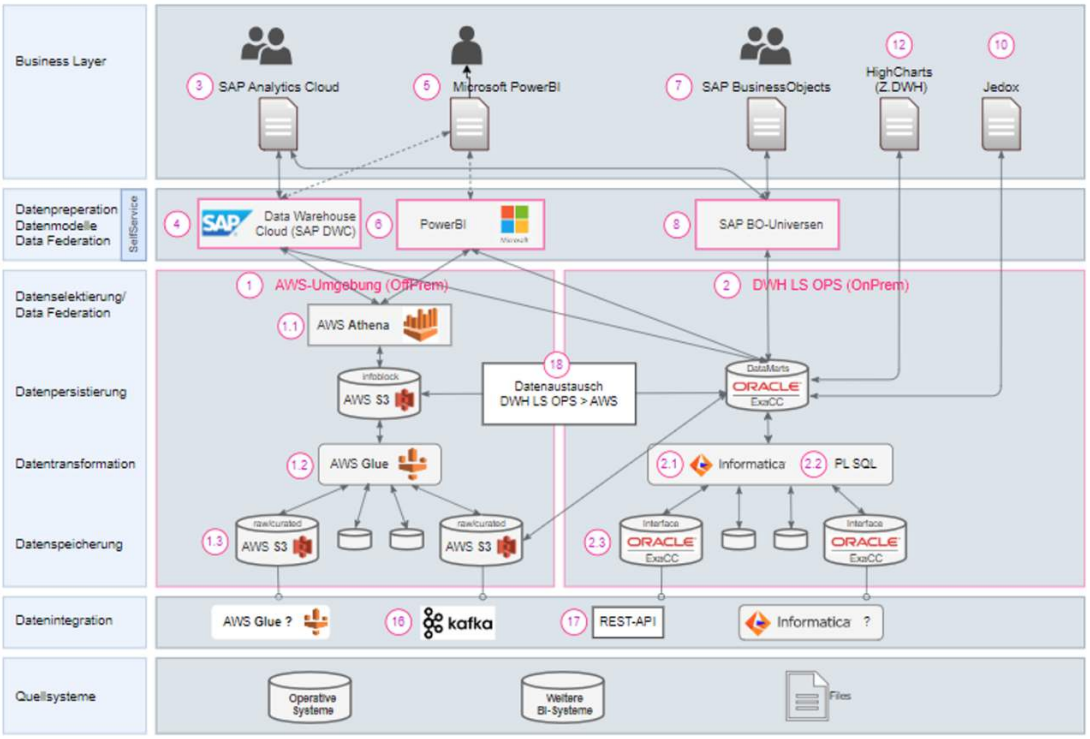


Figure 1.4.: The transition from an on-premise setup to a cloud-based target architecture.

Swiss Post sorting centers are cyber machines

Swiss Post has built and operates a number of massive sorting centers located across Switzerland. These are far more than just mechanical warehouses or factories. Rather, these sorting centers are massive "cyber facilities" that include high-tech hardware but also high-tech IT systems to keep them running.

While the team handles use cases across the entire postal delivery workflow, this project initially focuses on supporting sorting data flows. Sorting at Swiss Post is the process of consolidating received items in centralised plants, to which letters and parcels are sent from regions across Switzerland. One such plant is located in Härkingen:



Figure 1.5.: Aerial view of the Swiss Post sorting center in Härkingen.

Mail pieces are handled by workers and placed onto the automated sorting hardware of Swiss Post. This hardware enables the handling of up to 10'000 parcels per hour [Post 2023]. The system scans the received items and transports them through the plant via conveyor belts to their correct destinations, where they are discharged through chutes. These chutes are shown in the second image, including staff placing the parcels into roll boxes for further processing.



Figure 1.6.: Interior view of the automated sorting system in a Swiss Post sorting center.



Figure 1.7.: Discharge chutes inside a Swiss Post sorting center, where parcels are released to their designated zones. Then put into rollboxes to get them to next processing steps.

1.3. Motivation

Data quality and **data availability** are central to the successful operations of any company. Even a company working in low-tech or non-IT industries [Pas 2025] has basic but essential data needs, for example, information on who is working where and the addresses of clients and staff. Not having this data in the right quality or available to the right people at the right time will cause problems whenever the data needs to be used.

In the context of ongoing changes to the Swiss Post reporting infrastructure, keeping track of data availability and data quality is challenging. These infrastructure-related changes comprise the construction of new reports, one-to-one migrations, and the re-engineering of existing reports. Since the topic of data quality in the Swiss Post operations area could be built from scratch, as if starting on a greenfield, it became an ideal subject for a master's thesis.

Furthermore, the topic was not of the highest priority for Swiss Post at the start of this initiative, but it was considered a valuable idea to pursue. The medium-priority nature of this project also allowed the topic to be divided into multiple steps, such as working from a PoC to an MVP setup. Additionally, the medium priority regarding feature development in this area allowed for well-grounded research on the topic. All these aspects combined made it a suitable project leading to a master's thesis.

Moreover, due to the lack of automated data accuracy and data quality checks in the DWH LS OPS environment at the time of defining this project, this topic is highly relevant to the team responsible for both the operational and analytical reports that help guide the company. If the data is incorrect, wrong decisions are made; and if the data is unavailable, management cannot properly lead the staff. Thus, the topic is of high importance and business critical: data quality and availability help Swiss Post continue to deliver reliable services on a daily basis and enable IT to provide better assistance to staff throughout the postal process.



Figure 1.8.: Main customer touch point, delivery of a parcel

1.4. Problem Statement

The two key problems addressed in this research initiative are as follows. First, data quality in this area of Swiss Post has so far been handled in an ad hoc manner, without formal definitions, frameworks, or an agreed set of KPIs. Second, data quality is handled manually by the business at a high level rather than being automated by the IT department. Both of these problems lead to issues not being detected early and to longer reaction times. As a further consequence, delays in data processing are not identified and targeted for systematic resolution, which in turn reduces management confidence in the reports. All of this prevents continuous improvement, as the checks were neither documented nor stored in a way that allowed follow-up analysis or structured refinement.

Thesis update

Up until the start of this project, however, much of the initial situation had changed. Data quality was handled both within Swiss Post, which was searching for company-wide solutions, and within the team which is closely followed in this project. In addition, the two preceding projects added to the initiative in the following way:

Project 1: Researched the basis of data quality and data availability management. Furthermore, it described the current steps taken by Swiss Post to handle data quality within the environment described in the above background section. Finally, it laid out a path for implementing a data quality check in a rule-based check tool [Roccaro 2026b].

Or, as a quote from the document's problem statement:

> The data quality and data availability reporting at DWH LS OPS is in its infancy, both in regards to data quality as well as automation. This shows itself as: When reports are not executed or contain missing values, IT and the business are not automatically alerted. Due to the lack of automation, no data is gathered on the number of errors and outages.

Project 2: This research helped the stakeholders by providing them with an updated view of the data described in Project 1. This was required due to the ongoing changes in the data structure. The solution provided was introduced during a Scrum review meeting and put into production. In addition, the project also tested the feasibility of data quality monitoring using repeated counts [Roccaro 2026a].

Or, as a quote from the document's problem statement:

> Based on this groundwork, the next logical step was to adapt the existing reports to the new reporting platforms. However, in the meantime, the reporting data pipelines were updated. The old data pipelines remained available, but all further updates were performed only on the new pipelines. As a result of this development, the focus of this project shifted towards supporting these changes and helping the team to improve the monitoring and assessment of the existing data. Furthermore, an important objective was to establish the groundwork for the upcoming master's thesis by evaluating whether the available tools could serve as the basis for a use case to be explored in greater depth during the thesis.

Current problem statement

Finally, the current problem statement addresses the next but not final step in the research initiative. The problem statement will be described in more detail within the research chapter. At the start of the project, it was defined as:

To what extent can data quality in analytical reporting at DWH LS OPS be automatically validated at the reporting layer?

1.5. Scope and boundaries

This research focuses on the data availability and data quality challenges of the DWH LS OPS BizDevOps ASTV teams at Swiss Post described in Section 1.2. To simplify the scope of this work, the first step of data retrieval, such as how data is stored from Kafka topics, is excluded. The scope begins with the curated and reporting layers in the ETL process of the retrieved data. The aim of this project is to specifically provide solutions for the mentioned teams and not to offer a general solution or framework for the entirety of Swiss Post. Also, the research question confirms this boundary by focusing on automated validation at the reporting layer.

1.6. Expected outcome

The high-level goal of this research initiative is to validate the problem statement. Additionally, in the best case, it provides a solution or steps towards a solution capable of proving the problem statement's use case. In the worst case of not being able to fulfil the problem statement, this research provides the foundation for further improvements in data quality within the defined boundaries or researches alternative ways to improve the situation within these boundaries. In any case, this work will show a structured approach to the topic, enabling the reader to understand the challenges during the project and the decisions taken over the course of its duration.

1.7. Structure of the document

As this document is part of the larger research initiative which started before this specific project, some of the research and solution design from the first research projects which are relevant to understanding this one have been kept in this document.

This document is intended to guide the reader through the project in meaningful steps, explaining how the project is set up and executed, and then leading into the research and conclusions. It is structured in the following way. First, the document begins with the project management chapter, which explains the methods used during the project. It provides reasoning for tool choices where applicable and lists the tools and teams with which collaboration took place.

Next, the research chapter describes the problem in more detail. It clearly defines the specific and measurable goals of the project and presents the research sources used. This allows the reader to understand the decisions made and establishes the basis for the solution design.

The subsequent solution design chapter establishes the relevant terms and definitions valid throughout the project. It documents how the problem statement is addressed and provides the architectural groundwork for the implementation. Building on the solution design is the implementation chapter, which presents the implementation of this design. It contains the work carried out by the project team and outlines any issues encountered. Next, the results chapter lists the outcomes of the project without yet evaluating them. The following discussion chapter compares the expected outcome with the achieved outcome, assessing how many goals were reached and listing their fulfilment status.

The document concludes with the reflection and outlook, summarising what went well and what did not,

and offering specific suggestions on which methods to retain or replace. This will be valuable to any follow-up to the research initiative. Finally, additional materials are provided in the appendix.

2. Project management

The project management chapter summarises the environment of this project and how it is conducted. The following sections cover the relevant topics.

- The initial planning at the start of this project is laid out.
- The teams in which the project staff are engaged are listed, allowing the reader to see the relevant surroundings.
- The different methods for effort and task tracking are introduced, so that the project management methods applied are understood, together with their reasoning.
- The toolsets used by the different teams are described, which allows the variety of tools and therefore the complexity of the environment to be grasped.
- The phases and milestones which were set at the beginning of the project are described. Whether they were reached is evaluated within the results and discussion chapters.
- Time tracking is introduced and explained for transparency, while the tracked time is provided in the appendix.
- The risk management section describes the risks handled and the measures taken to counteract them.
- A review of how all these methods worked out is provided, showing which worked and which adaptations were made throughout this project.

2.1. Initial Planning

Before jumping into the master thesis, an initial planning was layed out to help split different stages over its full timeline. This initial planning served as a means of agreement and showing the right amount of preparation and mindset was there for a one year research work with many changes to be expected. Furthermore, this document described the setting of this research in relation to the bigger initiative of Swiss Post. Additionally it defined different phases and steps without fixing on one specific solution way to a solution but staying open to change. These specific contributions such as the phases and milestone planning are described in more detail in the upcomming sections of this chapter. This full document is part of the appendix for reference B.1.

2.2. Team constellation

This section explains the team environments in which the project takes place. This serves to provide an understanding of the direct project team working on this project and other teams at Swiss Post that contribute to the goals of the project.

Relevant Swiss Post teams

At Swiss Post, three teams are active and related to this project. In what follows we elucidate their tasks at Swiss Post, how they work together, and their contributions to this project. These teams are:

- BizDevOps team DWH LS OPS
- AWS Modernisation team
- Data Quality Initiative (DQM team)

DWH LS OPS team

The team **D**ata **W**are**h**ouse of **L**ogistic **S**ervices **Op**eration**s** (DWH LS OPS) has a long written-out name, and its role in Swiss Post is very important. The key mission of this team is to deliver accurate reports guiding decision-making in the postal services of Swiss Post. The DWH LS OPS team is responsible for developing and operating the Swiss Post data warehouse (DWH) for logistics services, which is the business unit that handles all postal services. The DWH is used for analytical reporting, which includes reports not used in direct daily operations but used for various forms of reporting, such as productivity analyses based on volumes and working time, informed long-term decision-making, as well as daily handovers and near-term planning. Also, this team includes the main project resource and the author of this document, namely Roland Roccaro, a Business Analyst in the MPP sub-team. MPP is the abbreviation of **M**engen**p**lanung **P**akete, which translates to volume planning for parcels. This is a project team title aimed at introducing new planning methods for parcel volumes, which in turn will help Swiss Post to plan the resources required to handle these volumes more efficiently, ultimately resulting in lower costs for handling them.



Figure 2.1.: Picture of DWH LS OPS team

Within the Swiss Post this team is also referred as a **Busi**ness **Dev**elopment **Op**eration**s** (BizDevOps) team, because it contains all required personnel, to create, deploy and maintain reporting solutions. This starts with business analysts to evaluate existing reports and their usage, and to document and specify them for further updates or re-engineering. This continues through the implementation, which is carried out first by data engineers and then by business intelligence (BI) engineers. Finally, the business side is involved in testing these reports and also serves as the main source when it comes to questions of how or what to implement. Due to the size of this team and the different responsibilities, the DWH LS OPS team consists of three separate sub-teams. These teams are ordered along the process of postal services as introduced in the full mail piece lifecycle being: Receiving, Sorting, Transport, Customs clearance (German acronym

is ASTV), **D**elivery **& F**inance (D & F), and finally the planning focused Parcel Volume Planning and Forecasting (german acronym is MPP).

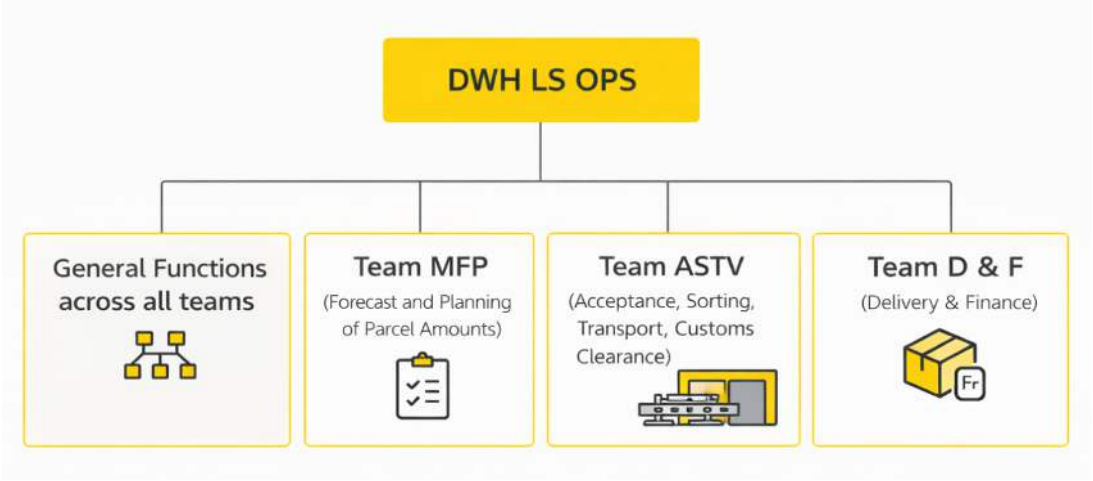


Figure 2.2.: Team and sub-team organisation

AWS modernisation team

Swiss Post is currently in a process the modernisation of the AWS cloud environment, which the Swiss Post itself describes as follows:

Swiss Post is currently pursuing a strategic modernisation of its AWS-based data lake with the objective of improving developer efficiency, increasing operational reliability, and establishing a modern, data-driven architecture. According to internal documentation, the initiative is aligned with defined key performance indicators and aims for completion by the end of the first quarter of 2026 [Swiss Post 2025].

This modernisation is performed with the assistance of consultants as part of improving AWS usage in DWH LS OPS. The improvements are detailed in 15 KPIs across topics such as AWS Glue job optimisation and deployment pipeline improvements. With the assistance of the consultants, best practices are applied to the development of ETL, as well as general improvements to deployment pipelines. Most importantly for this project, the initiative also includes DQM improvements. As a reference, a screenshot of the Confluence landing page describing the initiative is shown below:

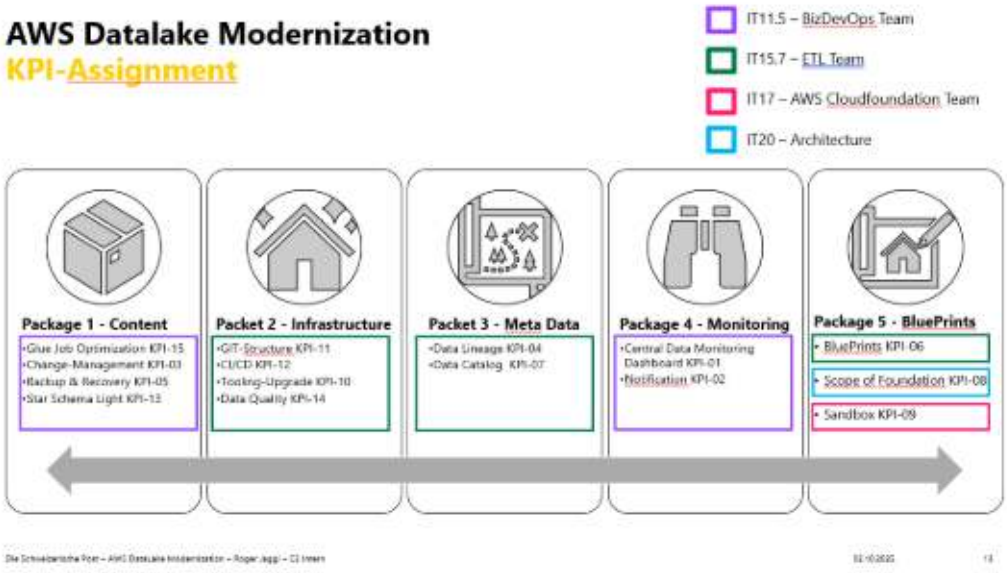


Figure 2.3.: The start page of the AWS modernisation, listing KPI and latest news

This team consists mainly of consultants with long-term experience in assisting companies in becoming proficient with AWS. They support data engineers at Swiss Post by improving existing code and reviewing it for potential enhancements. This is relevant for Swiss Post, as the development of the cloud infrastructure started without full architectural support and therefore has room for improvement.

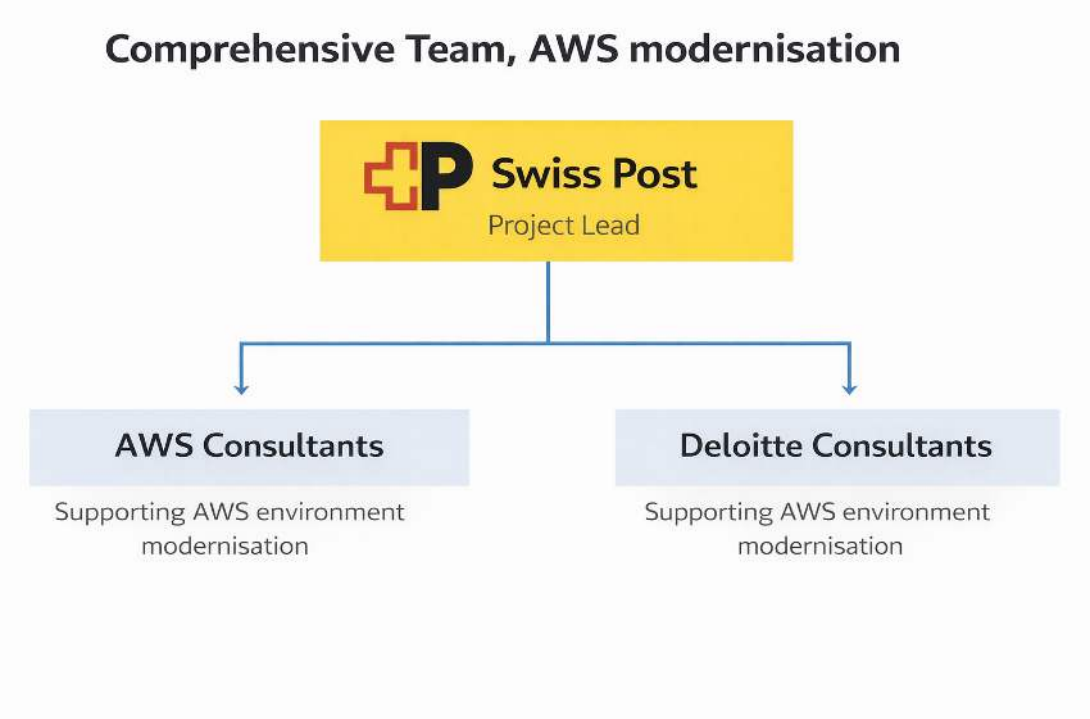
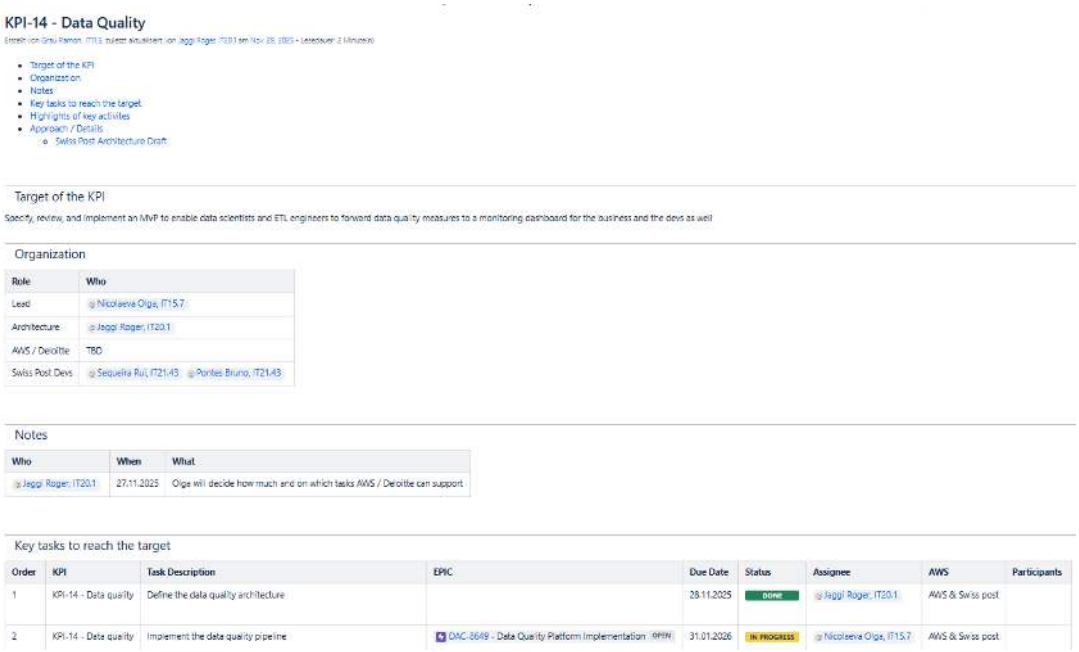


Figure 2.4.: The team setup of the AWS modernisation

DQM team

The third team relevant to the project is the DQM team. The team serves KPI 14 Data Quality within the AWS modernisation. Their goal is described on Confluence as follows: specify, review, and implement an MVP to enable data scientists and ETL engineers to forward data quality measures to a monitoring dashboard for both business users and developers.



KPI-14 - Data Quality

- Target of the KPI
- Organization
- Notes
- Key tasks to reach the target
- Highlights of key activities
- Approach / Details
  - Swiss Post Architecture Draft

Target of the KPI

Specify, review, and implement an MVP to enable data scientists and ETL engineers to forward data quality measures to a monitoring dashboard for the business and the devs as well

Organization

Role	Who
Lead	Nicolaeva Olga, IT15.7
Architecture	Jeggi Roger, IT20.1
AWS / Deloitte	TBD
Swiss Post Devs	Sequeira Rui, IT21.43 Pontes Bruno, IT21.43

Notes

Who	When	What
Jeggi Roger, IT20.1	27.11.2025	Olga will decide how much and on which tasks AWS / Deloitte can support

Key tasks to reach the target

Order	KPI	Task Description	EPIC	Due Date	Status	Assignee	AWS	Participants
1	KPI-14 - Data quality	Define the data quality architecture		28.11.2025	DONE	Jeggi Roger, IT20.1	AWS & Swiss post	
2	KPI-14 - Data quality	Implement the data quality pipeline	DAC-3649 - Data Quality Platform Implementation OPEN	31.01.2026	IN PROGRESS	Nicolaeva Olga, IT15.7	AWS & Swiss post	

Figure 2.5.: The KPI 14 Data Quality start page on DWH LS OPS Confluence

The team consists of three developers with the assistance of a lead data engineer. This setup ensures that the developers can deliver results efficiently, while the lead data engineer provides design guidance and solution structure.

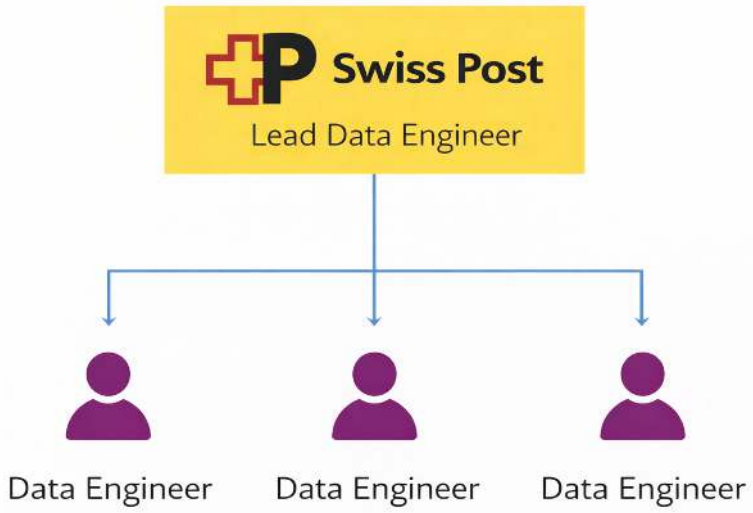


Figure 2.6.: The DQM team setup

Academic project team

The academic team consists of the author and the academic advisor. The author conducts the practical work, which includes analysis, implementation, and evaluation. The advisor provides expertise in the academic process and acts as a sparring partner regarding the writing process and project management, offering methodological and academic supervision.



Figure 2.7.: The academic team setup

2.3. Project management methodologies

This section details how the different teams manage themselves. This is done for reasons of transparency, so that the reader understands which project management methods the individual teams use. This highlights an additional level of complexity in the working environment. Not all teams are addressed in every subsection, as the project team was not involved in all teams' methods to the same depth. First, project management methods are highlighted, followed by the different timeline management approaches. In the following figure, the different boards and management methods are aligned around the central academic team to illustrate the broad set of methods.

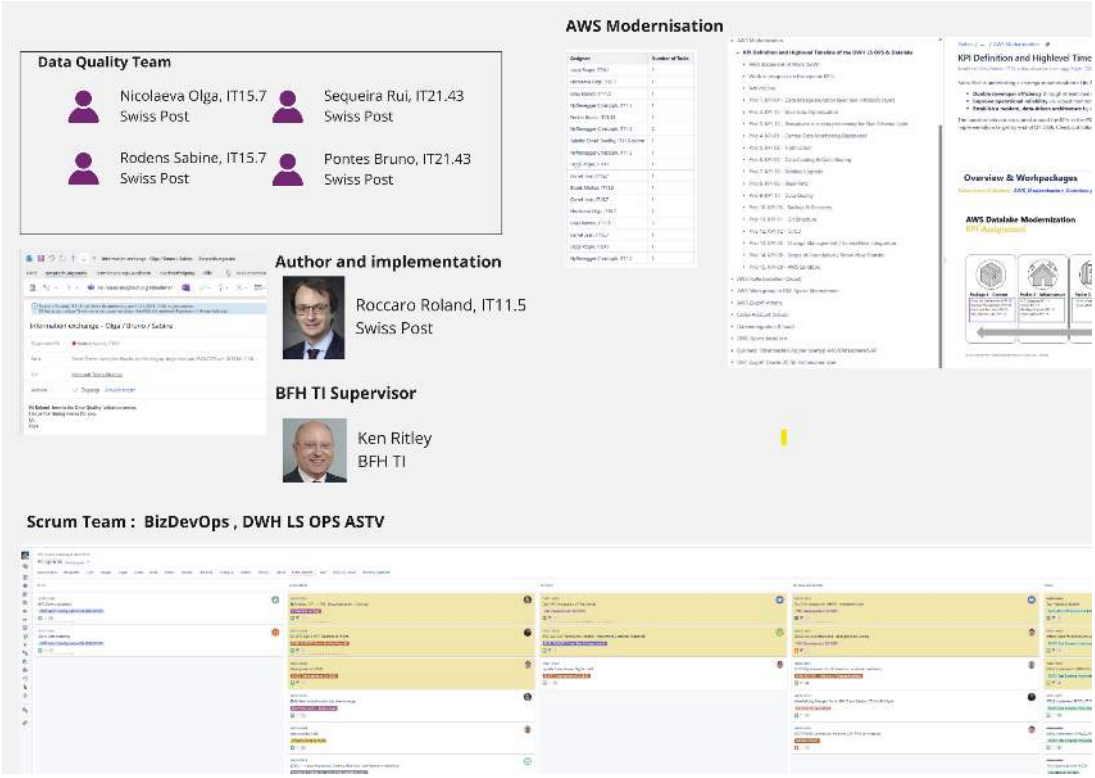


Figure 2.8.: All methods shown in one figure. Top left: DQM team with weekly meetings. Top right: AWS modernisation using a Confluence page. Bottom: DWH LS OPS and academic team using Jira.

Project management

Project management for this project consists of two main elements. These are Scrum and weekly meetings.

Scrum

The DWH LS OPS team is organised as a BizDevOps Scrum team (see Section 2.2). The AWS modernisation teams join these Scrum teams (see section 2.2). The full set of rituals defined for Scrum is used, including daily meetings, sprint retrospectives, sprint planning held every two weeks, and reviews held every four weeks. In addition, a weekly Scrum meeting is held to maintain communication between the ASTV, D&F, and MPP teams. The meetings are shown in the first screenshot, while the second illustrates a noteworthy team definition: the rules of engagement established for collaboration, which are a result of the retrospectives.



Figure 2.9.: Scrum ceremonies of DWH LS OPS BizDevOps team



Figure 2.10.: Rules of engagement of DWH LS OPS BizDevOps team

Weekly Meetings

Contact with the DQM team is handled through weekly meetings (see section 2.2). These serve as minimal update sessions during which weekly progress is communicated to all team members. While the developers in the DQM team are themselves also part of a Scrum team, their sprint management is out of scope.

Adapted Scrum

The project team handles task planning and progress management using Scrum, however, in an adapted fashion. The DWH LS OPS board is used for a dual purpose. As a result, the sprint lengths are also two weeks. The board is linked to the shared Miro board for easier management within the academic team. The Scrum events are mainly held within the DWH LS OPS team. The academic team, however, performs a separate planning confirmation every two weeks and retrospectives, which happen less frequently. The dates for retrospectives are noted in the Miro calendar.

Jour Fixe

The academic team holds a Jour Fixe every week. This serves to maintain close communication and functions as a check-up meeting. Each meeting follows a defined structure; as an example, the meeting held on 12.10.2026 is described below.

- Project check-in and clarification of objectives

- Overview of the project and completed deliverables
- Current deliverable in progress
- Update of initial project description
- Status of milestones
- Other topics

First, a general introduction to the project is provided to serve as a check-in and to allow participants to focus on the project. Second, the deliverables currently being worked on are listed based on the deliverables overview. Third, the status of the milestones is reviewed. Fourth, this is followed by individual agenda items, which can differ from meeting to meeting. In this meeting, these included updates to the initial project description, as well as a check-in regarding open administrative tasks such as handling contracts and payments. Finally, the meeting is closed with an open section in which any other relevant topics may be discussed. These meetings are documented in Miro (see below), structured using a three-block template, and most often recorded using handwritten notes that are later uploaded. Depending on the week's topics, some information, such as the sample below, is added. It shows the question of how to cite sources properly and the options provided.

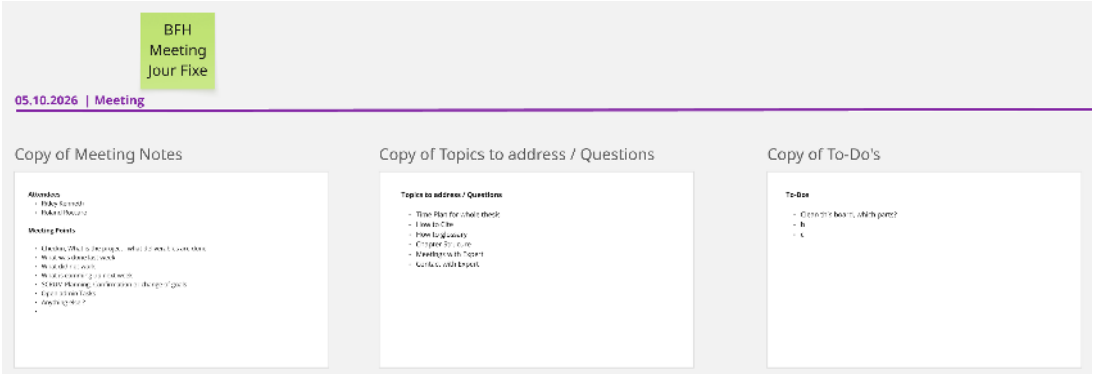


Figure 2.11.: Jour Fixe of Academic team, meeting structure and rework

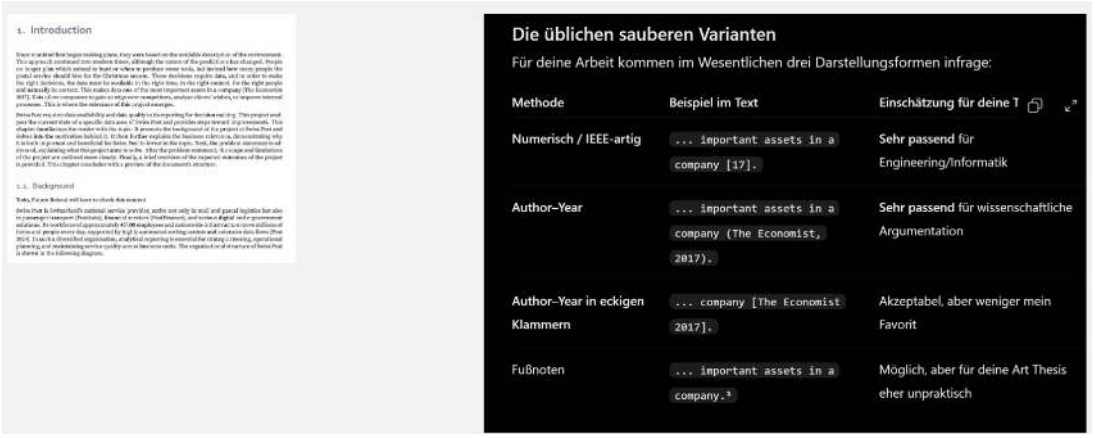


Figure 2.12.: Jour Fixe documentation of the academic team. Example of a question to be discussed: How should citations be formatted, and is the current citation style appropriate?

Timeline management

Timeline management is performed in different ways by the teams. In this subsection, the planning of actions is described, as well as how staff availability is tracked.

Scrum Events

All teams use Scrum events to manage their weekly progress. Progress on stories is monitored in daily meetings, during which each team member updates the team on progress and issues and indicates whether assistance is needed or deviations from the previously planned approach are required. Sprints encompass the stories the team works on during a two-week cycle. During sprint planning, the stories for the next sprint are defined. The stories added to a sprint represent the commitment the team makes for the sprint, together with the sprint goal. Planning over longer time horizons is conducted through quarterly planning sessions. Every three months, all teams in DWH LS OPS plan which epics should be completed and how many story points should be delivered. These quarterly planning sessions also serve to prioritise epics and allow higher management to introduce priorities.

Holidays

Holidays are noted in the calendars used; this applies to the DWH LS OPS team as well as the academic team. In both cases, every holiday must be recorded with an absence notification. This allows other team members to quickly see whether a team member will be available in the future and to adjust their planning accordingly, or to confirm in daily operations that the person is not available.

Phases and milestones

The academic team also uses phases and milestones to split the work into logical blocks and track its progress over the full duration. The details of these phases and milestones are described in their respective sections.

2.4. Tools to support the project management

This section lists the tools used during the project for reasons of transparency. It also allows the reader to evaluate different ways in which these tools can be used.

Miro

Miro is part of the toolset of both the academic team and the Swiss Post teams. However, in this project, it was used only by the academic team. The reason for using Miro is that it is a lightweight tool that allows combining different data structures on a single board. Miro provides an infinite canvas on which a wide variety of information can be placed. The information can be structured and easily rearranged to regroup related elements. This allows the tool to be used both for early documentation of information and for later restructuring when additional ideas emerge, particularly in the context of long-term documentation. The ability to restructure information visually supports the decomposition and understanding of complex environments.

A full set of screenshots illustrating changes on the Miro board has been moved to the appendix, while the final one is included here (B). The following screenshot represents the Miro board during the finalisation

of the document, zoomed out so that all elements are visible. While this view prevents individual details from being seen, some relevant parts are shown in detail in the following subsubsections.

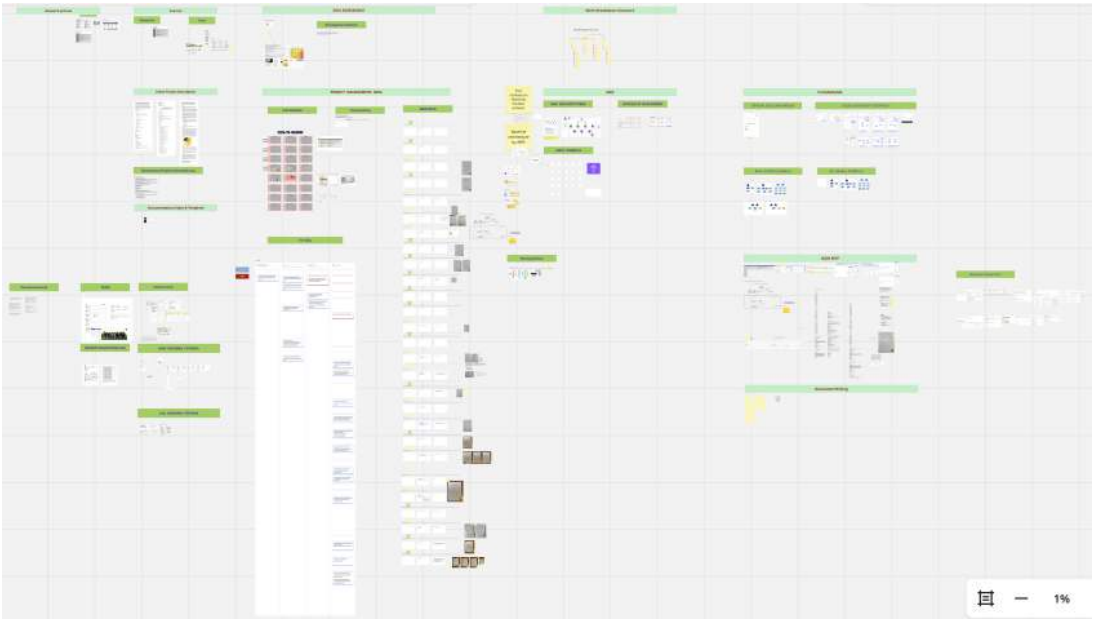


Figure 2.13.: Miro board documentation, view of 23.12.2025

Miro calendar

The calendar in Miro was primarily used to schedule the milestones. These are shown with green tiles on the corresponding dates, including their number and milestone title. Jour Fixe meetings are marked in yellow whenever they occurred, including adjustments in cases of unavailability. Actual unavailabilities at scheduled Jour Fixe meetings and other times are documented in black. Finally, holidays are marked in teal. In the case of this project, this applied only to the Christmas and New Year holiday period.



Figure 2.14.: Miro board calendar used for coordination and tracking of milestones and related events

Meetings documentation

The meetings are documented using a layout with three sections, including a coloured tag at the top in yellow for Swiss Post meetings and green for academic ones. From left to right, these sections are structured as follows. First, the meeting points to be discussed are listed; these are copied from the Outlook invitations. Second, the topics addressed are summarised, mostly after the meeting. Third, the to-do block describes any actions that arose during the meeting. Not every meeting includes all sections. Furthermore, meetings are often documented using handwritten minutes. These are uploaded to the right of the meeting blocks without further refinement and serve as references.

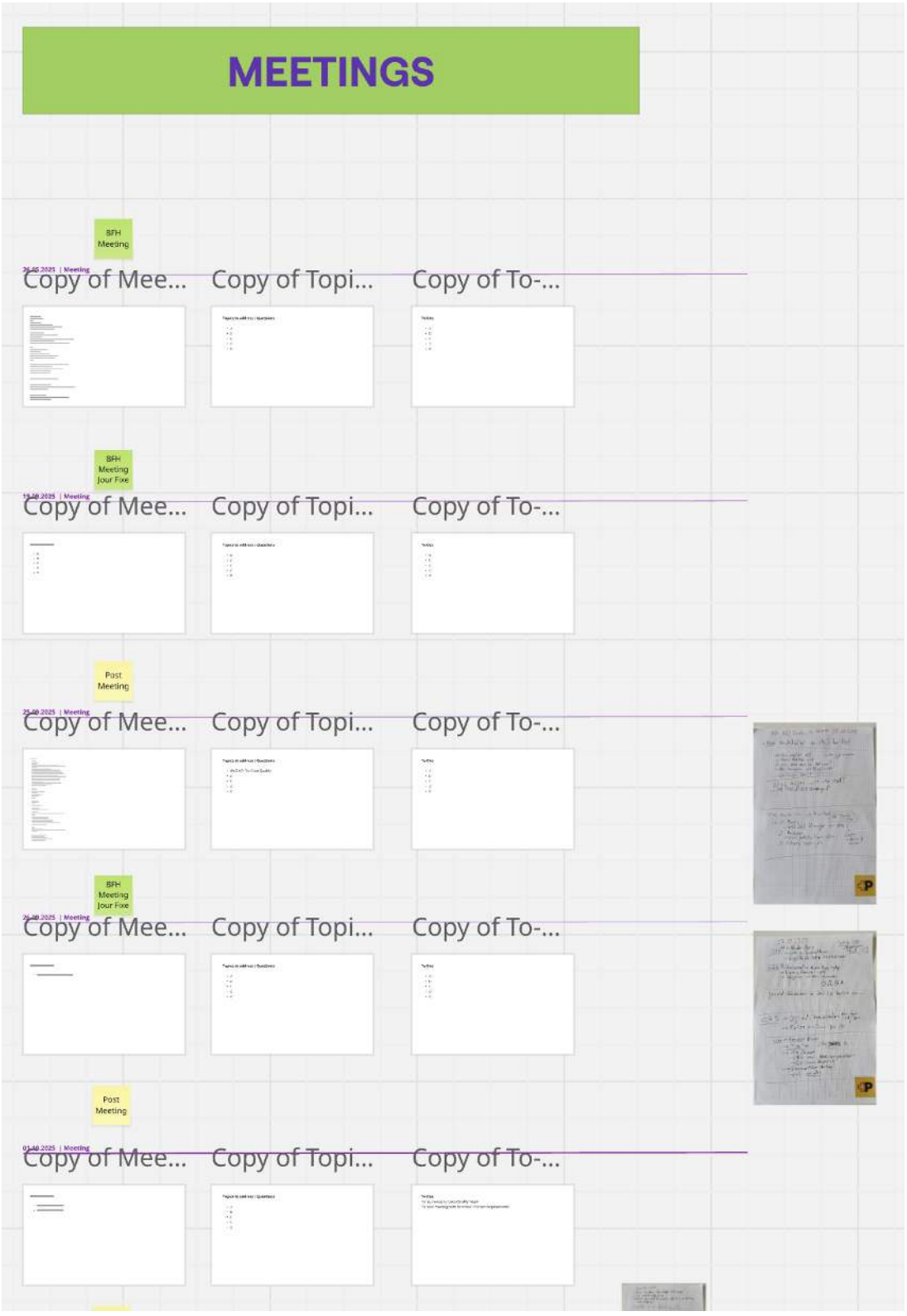


Figure 2.15.: Miro board, tracking of meetings: when, who, what

Documentation of the Implementation

The implementation was first documented in Miro, as the rapid creation of screenshots did not disturb the flow of information gathering. In the following figure, the information-gathering process can be seen, starting from the top left, moving to the right, and then restarting from the left. First, screenshots of the SAC report are shown, followed by navigation towards DSP. Further details on the tools are described in the Implementation chapter.

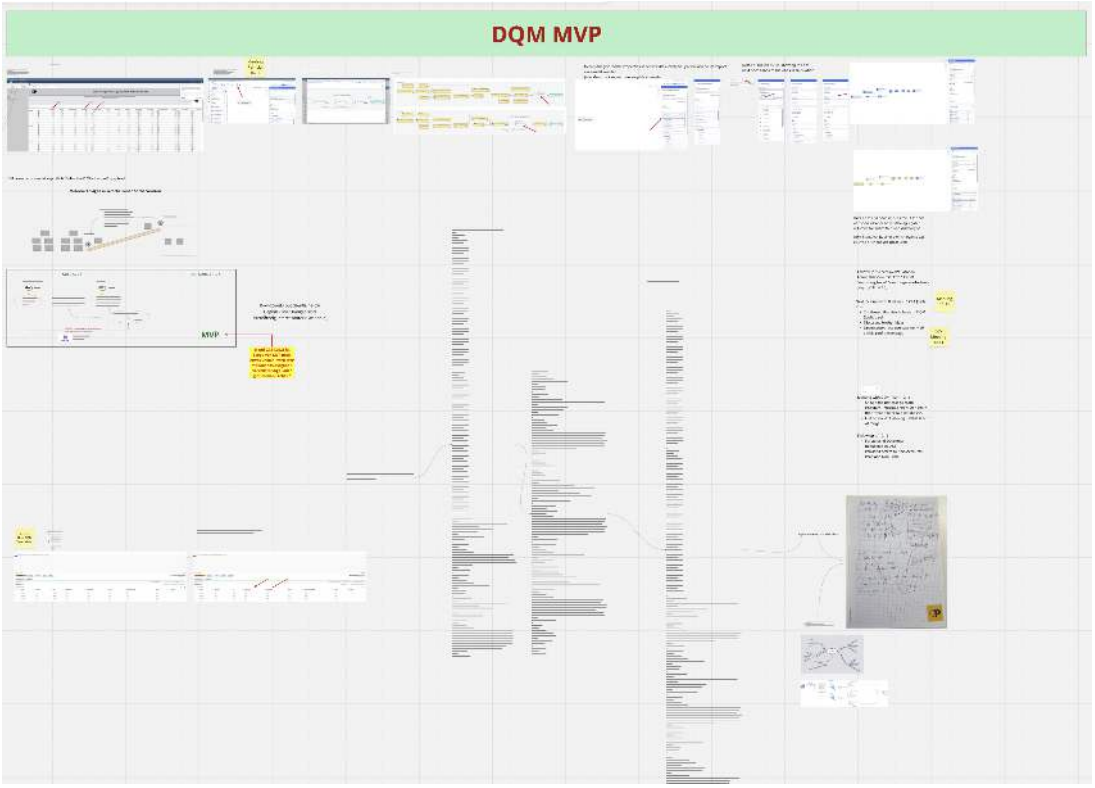


Figure 2.16.: Miro board usage for quick documentation during implementation

Jira and Confluence

Jira (mainly for issue tracking) and Confluence (mainly for documentation) are both tools provided by Atlassian and are therefore well integrated with each other. The tools are installed on-premises, which limits sharing possibilities outside the Swiss Post network. They are used by the three Swiss Post teams presented in the Teams section. The project team also used Confluence when research results had to be presented as a first use case. While Jira is used for the short-term management of Scrum artefacts, Confluence is used for long-term documentation. An example of such long-term documentation is shown in the following screenshot, illustrating the documentation of a data lineage.



Figure 2.17.: Example of project documentation in Confluence used to record architectural decisions and their rationale for future reference

Microsoft Teams and Microsoft Outlook

MS Teams and MS Outlook are used by both the academic and Swiss Post teams. They are organisational standards at Swiss Post and are not optional. The same applies to the academic context. Both tools serve to support focused long-term (MS Outlook) and short-term (MS Teams) communication.

Shared chat

For the academic team, the chat was used for short-term communication. However, a challenge arose due to the involvement of two distinct organisations, BFH and the Swiss Post. In order to keep communication between Swiss Post accounts and BFH accounts in one place, a chat with three users was required. In this context, the cross-organisational capabilities of MS Teams proved useful. After the Swiss Post account was invited to participate in the project team, it was able to communicate directly with BFH accounts. A sample of the communication involving three accounts is shown below.

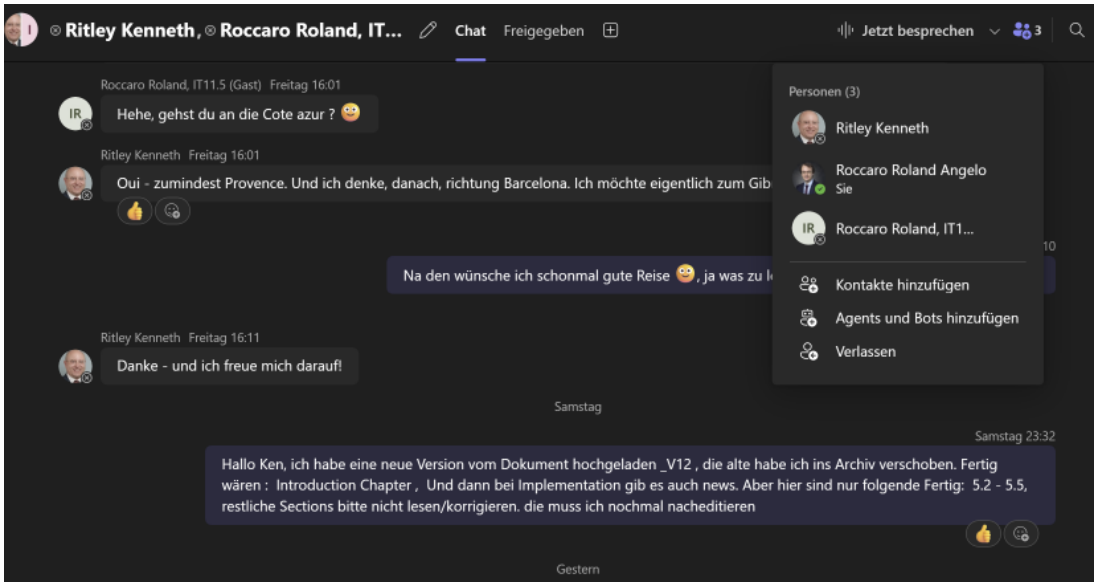


Figure 2.18.: Shared chat using Teams

A second important application of MS Teams is a document management system. This includes source documents, document versions, and related materials. All relevant documents are stored within this structure, which is designed to grow as needed.

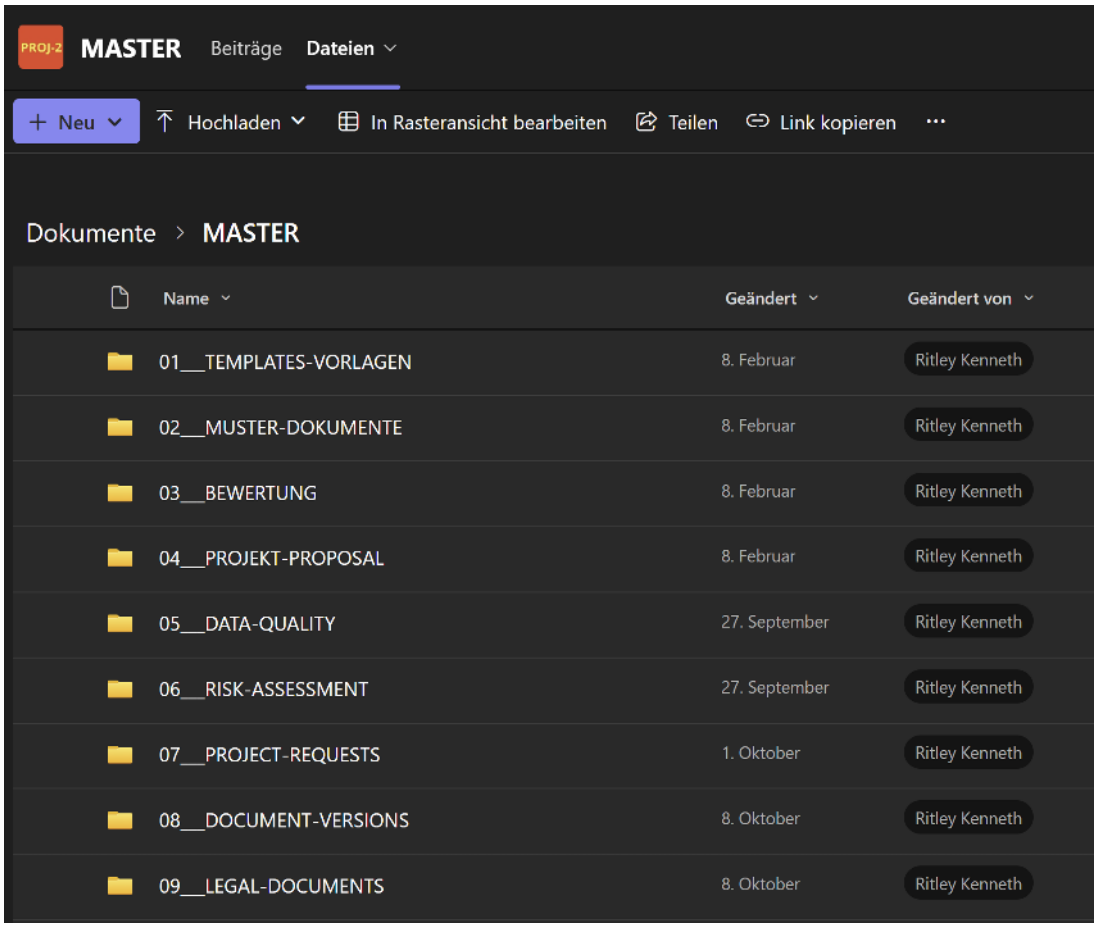


Figure 2.19.: Project document store used with Teams

LaTeX

LaTeX was the tool chosen for the main writing process by the academic team. The primary reason for selecting this tool was the consistency of the document structure. It allows changes to be made in a large document without the document becoming unstable. The tool was specifically selected for this higher consistency compared to Word. The fact that the creation of some stylistic elements takes more time than with a WYSIWYG editor such as Word was accepted, particularly as design was valued lower than document content. Finally, the consistency of being able to reference text sections within the document and the ability to automatically generate lists of figures and tables were seen as additional benefits. Furthermore, the automatic glossary and citation features using Biber were considered advantages for which the slower writing process without a WYSIWYG editor was accepted. Due to the learning effort involved in using LaTeX, the following badge was earned during the course of the research.



Figure 2.20.: Humorous illustration symbolising the extensive use of LaTeX throughout the preparation of this report and the associated learning experience during the project.

GitHub

For maintaining version control of any created artifacts, the public GitHub was used with the private account of the academic team. It was mainly used as a repository. This Git distribution was chosen because it is free to use and already well known to the team, with all required accounts in place, so no additional adoption or learning effort was required before starting. GitHub was not only used to store the document and track its progress, but also to manage all required sources such as tables and images used in the document.

For the writing process, the use of GitHub as a backup mechanism was particularly important. As all sources are stored in GitHub, the project was safeguarded against hardware failures of the working device, which is one risk mitigation measure. Continuously committing and pushing changes made the progress of the writing process traceable and supported the iterative writing approach which was chosen. This proved useful when the document PDF could no longer be generated, as the last version that compiled without errors could be identified. Finally, storing the sources on GitHub allowed access from different devices, which was useful when transferring materials such as screenshots from the work device to the study device.

ChatGPT

LLMs are now an indispensable part of work. To note how ChatGPT was used in this project, it was given the following prompt:

> Hi ChatGPT, please look at the other chats I have with you and tell me in which ways you supported me during my academic project over the past months since September 2025. Please create a subsection for my scientific work which I can reference.

The following subsection was completely AI generated content by ChatGPT, without further editing.

ChatGPT

To document the support provided by ChatGPT, a summary of the assistance received during the academic project was generated directly by the tool itself: The request asked ChatGPT to review prior interactions and describe the ways in which it supported the academic project over the past months since September 2025. The resulting subsection is intended for inclusion in the scientific work and can be referenced accordingly [OpenAI 2025].

The following subsection is entirely AI-generated content produced by ChatGPT, without further editing [OpenAI 2025].

Use of AI Assistance in the Research Process

During the preparation and initial phases of this Master Thesis, starting in August 2026, OpenAI's ChatGPT was used as a supporting tool for academic writing, project organisation, methodological discussions, and document preparation.

The use of ChatGPT can be divided into several areas:

- ***Academic writing and proofreading:*** *ChatGPT was used to review English texts for spelling, grammar, readability, and academic style. It also suggested alternative formulations, improved transitions between paragraphs, and helped identify unclear or insufficiently supported statements. The author reviewed and selected the proposed changes.*
- ***Thesis structure and research planning:*** *ChatGPT supported discussions concerning the organisation of the thesis, the formulation of research objectives and questions, and the distinction between previous aca-*

demic projects and the new contributions of the Master Thesis. It was also used to critically discuss the scope of the work and possible approaches to scientific evaluation.

- ***Research methodology and scientific reasoning:*** *ChatGPT was used as a discussion partner to explore possible research approaches, evaluation criteria, and methodological considerations. Its suggestions served as input for further examination rather than as independently verified scientific findings.*
- ***Project management and organisation:*** *ChatGPT assisted in preparing project plans, structuring activities, defining milestones, and developing an Excel-based semester planning tool. It also supported the organisation of separate discussion topics for different aspects of the thesis.*
- ***LaTeX and document preparation:*** *ChatGPT provided assistance with LaTeX syntax, document structure, figure captions, references, formatting, and troubleshooting. It was also used to convert and adapt existing text into LaTeX-compatible formats.*
- ***Review of existing academic work:*** *ChatGPT supported the examination and comparison of material from the two preceding academic projects. Particular attention was given to identifying which results constituted previous work and which activities would need to provide a new scientific or technical contribution.*

ChatGPT was used as an assistive tool rather than as an independent source of scientific evidence. The responsibility for research decisions, technical implementation, verification of information, interpretation of results, and the final content of the thesis remained with the author. AI-generated suggestions were subject to review and were not considered substitutes for academic literature, empirical evidence, or expert feedback.

2.5. Phases and Milestones

This master's thesis research stretches over one year; therefore, structured planning is important. The research was split into individual phases at the beginning in order to define time blocks for different activities. This is important so that the focus is kept on moving forward rather than trying to complete a chapter perfectly, which may lead to insufficient effort being allocated towards the end of the thesis.

As an additional check-in during the phases, milestones have been set up. The following figure provides an overview of the setup explained in the upcoming section.

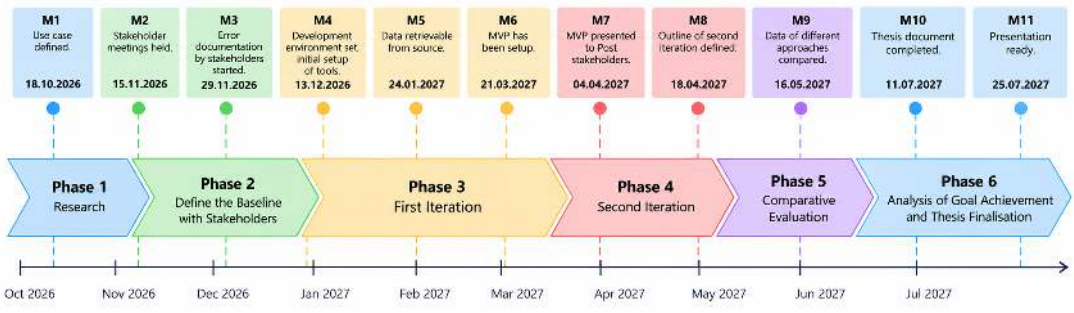


Figure 2.21.: Milestones and phases defined at the start of the project

Phases

The following phases define blocks that can be closed after their completion. The conclusion and start of each phase are used to review and adapt the current state. Furthermore, the phases create distinct blocks that build on top of each other, and they allow for later analysis of how much time was actually spent during

each phase and comparison with the expected effort according to the number of weeks. The following list shows the phases initially set up, with closure conditions laid out for each of them.

The milestones were set to structure the work blocks, but primarily to define how much effort should be invested in each individual block. The time allocated per block is not estimated in terms of how long a task is expected to take. Instead, it is defined by how much time the team is willing to invest in a topic in order to ensure sufficient capacity for subsequent tasks. For example, documenting the Swiss Post environment could have taken two to three weeks; however, only three days were intentionally allocated to this task.

- **Phase 1, Research**
  - Refine the problem statement.
  - Define and narrow the scope.
  - Define the use case.
  - Conduct the literature review.
  - Define the basic outlines for P2 to P3 and specify possible alternatives.
  - Closure conditions:
    * Research based on problem statement initially closed.
    * Basis for P2 and P3 defined.
    * Methodology for thesis defined.
    * Review of baseline document parts from project 1 and 2 performed.
- **Phase 2, Define the Baseline with Stakeholders**
  - Adapt P2 based on the findings from P1.
  - Document the current stakeholder knowledge regarding error detection.
  - Attempt to systematically identify and document relevant errors.
  - Define how the results will be evaluated against future iterations.
  - Closure Conditions
    * Stakeholder knowledge documented
    * Stakeholders have means to document errors.
    * Defined test window during which manual and automatic errors are compared.
    * Error documentation setup so that it is comparable to future Phase 3 implementation.
- **Phase 3, First Iteration**
  - Review the results of P2 and adapt P3 accordingly.
  - Define a minimal outline of the P3 objective while keeping the specific approach open to findings from the preceding research.
  - Develop the data retrieval from AWS source.
  - Develop the first/basic implementation.
  - Summarise the results.

  - Closure conditions
    * The development environment is setup.
    * Permissions required to work have been setup
    * The MVP is setup and running.
    * Data for test window retrieved.
    * The Post stakeholders received presentation of current state.
    * Defined test slot and registered test results

- **Phase 4, Second Iteration**
  - Review the achievement of the objectives from P3 and, if necessary, pursue an alternative approach.
  - Define a minimal outline of the phase objective while keeping the specific approach open to findings from the preceding research.
  - Develop an adapted implementation.
  - Summarise the results.
  - Closure conditions
    * Second iteration outline defined.
    * Data for test window retrieved.
- **Phase 5, Comparative Evaluation**
  - Compare the results of all investigated approaches.
  - Identify factors that may limit a direct or fair comparison, including specific differences between the approaches.
- **Phase 6, Analysis of Goal Achievement and Thesis Finalisation**
  - Describe which objectives were achieved and which were not.
  - Finalise the thesis document.
  - Finalise the required deliverables:
    * Video
    * Poster
    * Presentation and Defence
  - Thesis document delivered
  - Presentation held

Milestones

The milestones are events during the course of the phases. Each milestone event is set as a check-in where the project team evaluates whether the event has been completed or not, which is then used in an upcoming analysis block. The list below shows the milestone planning at the start of the thesis.

- **Milestone 1:** Use case defined. – **18.10.2026**
- **Milestone 2:** Stakeholder meetings held. – **15.11.2026**
- **Milestone 3:** Error documentation by stakeholders started. – **29.11.2026**
- **Milestone 4:** Development environment set, initial setup of tools – **13.12.2026**
- **Milestone 5:** Data retrievable from source – **24.01.2027**
- **Milestone 6:** MVP has been set up. – **21.03.2027**
- **Milestone 7:** MVP presented to Post stakeholders. – **04.04.2027**
- **Milestone 8:** Outline of second iteration defined. – **18.04.2027**
- **Milestone 9:** Data of different approaches compared. – **16.05.2027**
- **Milestone 10:** Thesis document completed. – **11.07.2027**
- **Milestone 11:** Presentation ready – **25.07.2027**

Deviations and Unavailabilities

The academic team tracked deviations from milestones or Jour Fixe meetings. This served to ensure transparency and traceability. For this use case, the dates of all Jour Fixe meetings were planned in advance. If a meeting had to be rescheduled, this was noted in the calendar. In the case of shifted milestones, these changes also had to be documented in the calendar, including the reasons why a milestone was not achieved or had to be moved. For all phases and milestones, an evaluation of whether they were completed and what measures were taken if they were not is provided at the end of the project management chapter.

2.6. Effort planning and review

Effort planning and review are carried out to plan and confirm the time allocated to each phase. Additionally, this provides confirmation of the effort structure. Furthermore, it establishes the framework for a later review of the planned hours.

Effort planning

The effort was planned according to the duration of the phases. These were laid out in the previous section and provide a start and end time for each phase. Thus, each phase contains a defined number of weeks. The planned effort is therefore first measured as an absolute number of weeks. Using this count, a relative effort is obtained by comparing the number of weeks per phase with the total number of weeks planned. Which results in the following formula to get the relative time allocation per phase.

$$\text{Relative time allocation}_i = \frac{\text{Weeks allocated to phase } i}{\text{Total planned weeks}} \times 100\%$$

In the upcoming table, the relevant data for the calculation of the phase efforts is detailed. Phases always end according to the two-week sprint rhythm but do not necessarily end with the phase's last milestone. The table shows the following information:

- **Phase:** The number of the phase for each line

- **Milestone:** This is either the milestone number or "end" in case it is the end of a phase.
- **MS date:** The date of the milestone or phase end date
- **Name of milestone:** The abbreviated name of the milestone or the end of a phase
- **Phase weeks total:** The total number of weeks spent in that phase
- **Relative percent:** According to the above formula, what percentage of the total time is spent in this phase.

Phase	Milestone	MS date	Name of milestone	Phase weeks total	Relative percent
0	0	14.09.2026	Start of thesis		
0	End	**03.10.2026**	**End of startup phase**	2	**4.26**
1	1	18.10.2026	Use case defined		
1	End	**01.11.2026**	**End of phase 1**	4	**8.51**
2	2	15.11.2026	Stakeholder meetings held		
2	3	29.11.2026	Error documentation started		
2	End	**29.11.2026**	**End of phase 2**	4	**8.51**
3	4	13.12.2026	Development env. set		
3	5	24.01.2027	Data retrievable		
3	6	21.03.2027	MVP setup		
3	End	**21.03.2027**	**End of phase 3**	15	**31.91**
4	7	04.04.2027	MVP presented		
4	8	18.04.2027	Outline of second iteration		
4	End	**02.05.2027**	**End of phase 4**	6	**12.77**
5	9	16.05.2027	Data of approaches compared		
5	End	**30.05.2027**	**End of phase 5**	4	**8.51**
6	10	11.07.2027	Thesis document complete		
6	11	25.07.2027	Presentation ready		
6	End	**22.08.2027**	**End of phase 6**	12	**25.53**
			Total weeks	**47**	**100.00**

Table 2.1.: Distribution of planned project duration across project phases

Two ways of recorded time

First to some rules as to how time can be registered. Regarding work efforts its important to note that the academic team has an agreement with Swiss Post defining which efforts are considered work time and which not. As per agreement, the research part and the writing part of this document are not to be noted as work time, however the remainder of activities such as implementation, documenting work state in Jira or confluence etc. are considered work efforts and are noted into the time tracking at swiss post as such.

This agreement leads to two different ways of tracking time. At Swiss Post, time is tracked directly with their SAP time-tracking MIAR system, where the efforts are recorded against a specific project account. This makes it easy to export and track them for monitoring purposes in the thesis.

Second, academic efforts committed to the project outside the agreement were recorded separately. The work was then registered in blocks of continuous work on project-related tasks, including short breaks. However, longer breaks, such as lunch, resulted in a new block being created.

Planning review

At the end of the thesis, the number of hours tracked is summed up per phase, based on the start and end dates that were actually reached, taking into account any deviations. These totals per phase and the overall total lead to a second relative measure. The two relative measures per phase are then compared and evaluated. A summary of this is provided along with the analysis of the planning process at the end of this project management chapter.

2.7. Risk management, TODO

Todo, Future Roland will have to check and eventually re-write here...

etc...

In this section, the approach taken to risk management is described. Risk management is essential to the success of any project, as it enables planning for and potentially preventing dangers that may occur during its course. When such risks are not handled, they can endanger the progress of the project. The risk management process consists of several steps [Wikipedia 2025]. First, the risks are identified and classified as to their nature. Next, they are weighted according to how they might impact this project; this includes a consideration of the probability that the risks may arise, and the level of damage that may occur. After this, the specific handling of each risk is defined; these are the concrete measures that are taken to lower how the risk may impact the project. Finally, ongoing monitoring and review of the risks is carried out on a regular basis throughout the project.

Risk identification and classification

The identification of risks has been carried out for this project, following an extensive brainstorming approach. Each risk has been grouped into a category (e.g. goal, time and organization, etc.), and each risk has been classified according to the likelihood of is occurance and severity it would cause to the project. Both criteria are evaluated based on the project team's risk appetite and ranked according to the following definitions.

- **Likelihood:** What is the expected likelihood of a risk coming into effect
  - **1** Rare
  - **2** Unlikely
  - **3** Possible
  - **4** Likely
  - **5** Almost Certain
- **Severity:** How severe are the consequences of the risk coming into effect
  - **1** Insignificant
  - **2** Minor

- **3** Moderate
- **4** Major
- **5** Severe

The identified risks, along with their likelihood and severity, are listed in Table 2.2.

Table 2.2.: Project Risk Overview

No.	Group	Risk	Likelihood	Severity
1	Goal	Unclear question	2	2
2	Goal	Unclear goals	2	3
3	Time and Organisation	Insufficient expertise	4	2
4	Time and Organisation	Underestimated time/effort	4	4
5	Time and Organisation	Other study projects	3	3
6	Time and Organisation	Work projects	4	3
7	Time and Organisation	Procrastination	3	2
8	Human Resources	Unavailability of main project staff, e.g. due to illness	5	4
9	Human Resources	Unavailability of advisor, e.g. due to illness or accident	3	3
10	Tech and Infrastructure	Software version/library/dependency	4	3
11	Tech and Infrastructure	Hardware is damaged	2	4
12	Data and Research	Data related	4	4
13	Data and Research	Non-reproducible work	1	5
14	Ethical and Legal	Plagiarism or incorrect citation	2	5
15	Ethical and Legal	Copyright	1	5
16	Communication and Cooperation	Bad communication with stakeholders or advisor	1	4
17	Communication and Cooperation	Insufficient exchange with external partners	3	5
18	Force Majeure	Unforeseen family crisis	2	5
19	Force Majeure	General force majeure	1	5
20	Unknown Process	The required management process is unknown	4	2
21	Communication and Cooperation	Required resources are not available when needed, or responsibilities are unclear	3	4
22	Organisational	Changes pushed by the organisation	3	4
23	Data and Research	Insufficient scientific contribution	1	5

Risk heat map

After carrying out the initial identification and classification, it is not always easy to determine which risks require the most attention. To support this task, a Risk Heat Map is used, in which each risk is plotted in a

two-dimensional matrix according to its likelihood and severity. This visualisation highlights the risks that pose the greatest threat to the project and should be considered for priority treatment. The colours in the Risk Heat Map 2.7 correspond to these descriptions.

- **Green** Low
- **Yellow** Medium
- **Orange** High
- **Red** Very High

Update at start of master thesis

Risk Matrix		Severity				
		Insignificant	Minor	Moderate	Major	Severe
Likelyhood	Almost Certain				8	
	Likely		3, 22	4, 6, 10, 12		
	Possible		7	5	9, 21	
	Unlikely		1	2	17, 23	11, 13, 15, 19
	Rare					14, 16, 18, 20

Figure 2.22.: Initial risk assessment at the beginning of the Master Thesis.

Risk Matrix		Severity				
		1: Insignificant	2: Minor	3: Moderate	4: Major	5: Severe
Severity	5: Almost Certain	15 17 19	11 14	20		
	4: Likely		16	9	8	
	3: Possible			5	4 6 10 12	
	2: Unlikely		1	2 7	3	
	1: Rare					

Table 2.3.: Risk Matrix (Likelihood × Severity)

As can be seen in the Risk Heat Map, one risk is classified as red and therefore requires special attention, namely, the unavailability of the main project staff. How this risk, as well as the others, is addressed will be described in the following section.

Risk strategies and mitigation strategies

Risk strategies are the high level approaches to dealing with the risks, and the risk mitigations are the concrete plans developed to manage identified risks. In the following table, an overview of the individual risks is provided, including their selected treatment and a brief justification. BNaturally, it is neither practical nor feasible to completely eliminate every risk; therefore, based on likelihood and severity, as well as the team's risk appetite, each risk was assigned one of the following strategies:

- **Accept** – the risk and its potential impact are acknowledged, but no measures are taken
- **Reduce** – the risk is mitigated by applying additional measures to lower its impact or probability
- **Transfer** – the responsibility for handling the risk is assigned to another party
- **Avoid** – the project is adapted in a way that prevents the risk from materialising

Table 2.4.: Risk Strategy and Treatment Plan

No.	Risk	Risk Description	Treatment	Treatment Description
1	Unclear question	The topic description is too vague or contains too many goals	Reduce	Define SMART goals
2	Unclear goals	The goals are not clearly defined or cannot be measured	Reduce	Describe which goals are part of scope
3	Insufficient expertise	Not knowing required tools or programming languages	Accept	
4	Underestimated time/effort	Some tasks take more time than initially expected	Accept	Document end of tasks in Miro calendar; re-evaluate after project
5	Other study projects	Other classes demand more attention, less time for project	Accept	Use minimum time for projects/classes
6	Work projects	Other projects at work demand urgent attention, less time for project	Reduce	Reduction of risk due to starting early; defining blockers for study work
7	Procrastination	Tasks are pushed back	Reduce	Start early
8	Unavailability of main project staff	One or multiple project staff are not available	Accept	Document in case it happens
9	Unavailability of advisor	The advisor is not available	Transfer	Advisor to define emergency replacement
10	Software Version/Library/Dependency	Mismatch in versions; new library versions break the code	Accept	Run the code multiple times; have test cases (e.g., unit tests)
11	Hardware is damaged	The laptops or other material are damaged and need repairs or replacement	Reduce	Second laptop; code on GitHub/Miro/Teams/Outlook
12	Data related	Bias in data; not enough data	Accept	
13	Not reproducible work	The work provided cannot be reproduced	Reduce	Document steps; upload versions on GitHub
14	Plagiarism or wrong citation	Unallowed or undeclared usage of sources	Reduce	Use citation; bibliography

Continued on next page

No.	Risk	Risk Description	Treatment	Treatment Description
15	Copyright	Used literature or materials which are restricted	Accept	
16	Bad communication with stakeholder or advisor	Advisor or stakeholders are not heard; their inputs are ignored	Reduce	Weekly advisor meetings
17	Insufficient exchange with external partners	External partners do not react, are delayed, or communication towards them is late or poor	Reduce	Stakeholders in Scrum team
18	Unforeseen family crisis	Unexpected crisis in family	Accept	To be documented in Miro calendar in case it happens
19	General force majeure	Events which cannot be influenced are very rare and very disruptive	Accept	To be documented in Miro calendar in case it happens
20	Unknown process	Events occurring during the project can block its execution or, in the worst case, lead to its cancellation	Reduce	Always start administrative processes between Swiss Post and BFH without delay. Always send reminders after three weeks at the latest.

Implementation of risk mitigation strategies

Identifying the key risks is the first step in the overall risk management approach; this is followed by defining measures for each of the risks. In the following table, the implementation of these measures is documented. The measures have been implemented according to the strategies and descriptions outlined above. Table 2.5 provides an overview of the actions taken and when they were carried out.

Table 2.5.: Risk Strategy and Action Overview

No.	Risk	Strategy Decision / What	When
1	Unclear question	In the research section SMART goals have been setup	01.03.2026
2	Unclear goals	The introduction describes which parts of the initial project description are part of this document	
3	Insufficient expertise	Accept: Nothing done	01.03.2026
4	Underestimated time/effort	Accept: Nothing done	01.03.2026
5	Other study projects	Accept: Nothing done	01.03.2026
6	Work projects	Added blockers in Work calendar, Work account invited to meeting series (Jour Fixe); study account: set blockers on weekend; researched with ChatGPT for optimal/minimal study plan	01.03.2026

Continued on next page

Table 2.5 – continued from previous page

No.	Risk	Strategy Decision / What	When
7	Procrastination	Work on the project started 25.09.2025 when attending data quality meeting at AWS, being part of AWS Modernisation initiative	01.03.2026
8	Unavailability of main project staff (e.g., due to illness)	Contact of partner provided in Teams and added to Miro board	01.03.2026
9	Unavailability of advisor (e.g., due to illness, accident, etc.)	Asked for contact, confirmed to be Jürgen Vogel	01.03.2026
10	Software Version/Library/Dependency	Accept: Nothing done	01.03.2026
11	Hardware is damaged	Second laptop is available at home; the LaTeX document is a GitHub project; documentation is on Miro. Work laptop can also be used for most tasks, but has no LaTeX installed	01.03.2026
12	Data related	Accept: Nothing done	01.03.2026
13	Not reproducible work	GitHub project for LaTeX document	01.03.2026
14	Plagiarism or wrong citation	JabRef used as GUI for bibliography; Biber used for build	01.03.2026
15	Copyright	Accept: Nothing done	01.03.2026
16	Bad communication with stakeholder or advisor	Jour Fixe setup	01.03.2026
17	Insufficient exchange with external partners	Keep exchange channels active when changing team	01.03.2026
18	Unforeseen family crisis	Accept: Nothing done	01.03.2026
19	General force majeure	Accept: Nothing done	01.03.2026

Ongoing risk monitoring

A critical step in the risk management process is ongoing monitoring. Monitoring the risks ensures that the defined measures remain valid and that any risks that may materialise are identified. If a new risk is identified or causes a disruption to the project, it will be documented. Additionally, an appropriate measure to handle its impact will be defined. Monitoring of the risks will be performed every second Friday of each month. Detailed information about the risk monitoring process is provided in the appendix C. These risk check-ins will contain the following information:

- Number of risk: same as in the tables above
- Name of risk: same as in the tables above
- Occurred: did the risk occur (yes or no)?
- Details: what happened?

- Cause: brief analysis of why it happened
- Measures taken: what measures have been taken?
- Measure date: on what date were measures taken?

Risk review at project completion

Finally, a review of the risks will be held at the end of the project. This will address which risks came into effect and which did not. Additionally, changes to the perception of the risks will be documented at the start of the master thesis.

2.8. Execution and review, TODO

Todo, Work for Future Roland He will have to describe the How well project management did work out... e.g. Phases Milestones Tools... Risks...

2.8.1. Project plan and work packages, TODO

Todo, Work for Future Roland He will have to describe the project plan and the planned workpackages See at following content to take the needed out of this

3. Topic of this research

Todo, Future Roland will have to write here... The Research Chapter will need rework... Read what is to be kept... What should be removed... etc...

This chapter establishes the research foundation for the project. It first presents the problem addressed and derives the research question. Based on this, the project's SMART goals are defined to provide clear guidance for the subsequent work. The chapter then identifies the research gap that motivates the project. Finally, core definitions related to data quality and data availability are introduced to provide the theoretical basis for the analysis and implementation presented in the following chapters.

Todo Future Roland will have to rework this Or at least review and restructure

3.1. Problem statement

This work builds upon the previous research [Roccaro 2026b], which analysed the existing data pipelines at Swiss Post with regard to operational reporting. Based on this groundwork, the next logical step was to adapt the existing reports to the new reporting platforms. However, in the meantime, the reporting data pipelines were updated. The old data pipelines remained available, but all further updates were performed only on the new pipelines. As a result of this development, the focus of this project shifted towards supporting these changes and helping the team to improve the monitoring and assessment of the existing data. Furthermore, an important objective was to establish the groundwork for the upcoming master's thesis by evaluating whether the available tools could serve as the basis for a use case to be explored in greater depth during the thesis.

Problem generalisation

This project bridges the groundwork established in the previous research towards the follow-up research in the master thesis. First, it has to perform the next step so that the stakeholders are able to continue checking their new data sources. Second, it must help to identify and test the tool for the master thesis. This is the most important part of the research, as it has to confirm that the selected tool is indeed a good match for the upcoming master thesis.

3.2. Research gap

Given this problem statement, the following questions are researched in more detail in the next two chapters. First, a theoretical introduction to the terms is provided; in later chapters, their specific meaning in the context of this project is discussed.

3.3. High-level research question

What is the next meaningful step to take in order to improve data availability and data quality within the DWH LS OPS team at Swiss Post? How can the business and IT be supported in achieving their objectives? While the solutions should be as simple as possible, they should be based on best practices and verified through scientific research. The research gap is addressed in the solution and implementation chapters. The main goal is to ensure that the conducted research provides tangible benefits to the team and contributes to future improvements. It is particularly important for the team that the research results are presented continuously so that progress, as well as potential deviations and changes, can be discussed.

3.4. Success definition

The success criteria of this project are threefold and are defined based on the research gap as well as the initial project hand-in, which describes the business requirements.

First, the primary business objective is to achieve improvements through report updates. The goal of this project is to replace the existing report immediately and to establish concrete steps towards easier daily usage of the new report.

Second, communication with the business is another important success factor. The results from the research and implementation must be presented to the product owner (PO) on a regular basis.

Third, the testing and documentation of the basic setup required for the new DQM tool, as a theoretical and practical foundation for the master thesis, are the main achievements to be reached.

3.5. S.M.A.R.T. Goals

The goals of this project are now defined here as "SMART goals", as shown in the following list. They are all **S**pecific: the goal is clearly formulated; **M**easurable: there are measurable criteria defining when the goal is reached; **A**chievable: they are realistic given the available resources; **R**elevant: the goal is directly linked to the project's success; and **T**ime-bound: there is a defined timeframe for achievement.

- **1) Setup of an Improved Data Quality Report**

  WHY: This will help the team to run the daily checks faster.

  By the end of the implementation phase, the existing data quality report is to be updated to use the new infrastructure. This includes new reporting data pipelines and the new structure using a star schema. Also, Power BI is to be used.

  Success is measured by user feedback, which has to be obtained by the ASTV team. The feedback is to be implemented, and the report must be on the production systems.

- **2) Evaluation and Selection of DQM Platform for the Master Thesis**

  WHY: To define the DQM platform for the master thesis.

  During the implementation phase, the tool for the master thesis has to be defined. This requires a structured approach for the selection of the tool, including the setup of requirement lists. The team is to follow the DQM team during the evaluation of the tools.

  Success is measured by having the DQM tool defined. The selected tool has to be discussed with the architect of the DWH LS OPS team to make sure it fits the team's needs.

- **3) Validation of the Selected DQM Tool**

  WHY: The DQM platform must be tested for applicability to the master thesis.

  By the end of the implementation phase, the DQM tool has to be tested for applicability to the master thesis. It must be proven that the required infrastructure can be set up for a test. Furthermore, a simplified test has to be run to confirm that the tool can handle the data sources.

  Success is measured by having the DQM tool tested. The tool must return the test values to prove that it can be used for the follow-up master thesis.

3.6. This project as a key step towards the "future state" reporting solution

As stated earlier the solutional goal of this project and its follow up is to provide an MVP for DQM, from which the key learnings will enable the full "future state" reporting solution. This MVP should implement an automatic data quality check, replacing an occasional, non automated check performed by the business. This is important, as the DQM check is currently executed by the business instead of IT, which creates challenges with regard to the desired target state. The expected situation would be that IT executes data quality checks and defines thresholds together with the business. At present, however, the knowledge about thresholds and other relevant information resides solely with the business.

In summary, having IT run the data checks would enable continuous follow-up of data quality. For example, a report based on automatically executed checks could show system uptime over a longer period of time. This information would later allow IT to extrapolate standard value ranges, which in turn would enable the definition of alerting thresholds. In short, having DQM checks executed automatically rather than occasionally by the business provides multiple benefits for both IT and the business.



Figure 3.1.: DQM manual checks against automated checks by IT

3.7. Expected benefits of this reporting solution

A single visualisation does not explain the full benefit of the intended solution. For instance, the question arises as to why the business should stop performing data quality checks when it has the most detailed knowledge of the data. This is quickly clarified by taking a step back. The question should not be why checks are removed from the business, but why the reports are performed by the business in the first place. The benefits of this shift are therefore outlined in more detail in the following.

Benefit 1: Automation over manual processes to reduce effort

The business currently spends 30 to 45 minutes per day on a single data quality check, according to the initial project hand in description. From the duration of the check, it can be inferred that anyone requiring the data quality check before a report based on it is approved must wait until it is finished, resulting in a waiting time of 30 to 45 minutes. This fact provides the basis for understanding why manual processes are not suitable at scale. When scaling up to ten data quality checks, five to seven and a half hours are spent by the business on data quality checks. The waiting time for the last reports to be approved increases accordingly to five to seven and a half hours. When increasing to 20 checks per day, ten to 15 hours are spent, and the waiting time rises accordingly, accounting for more than one full-time position and long delays in report data release. It should be noted that this amount of time is spent solely to determine whether the defined check passes or not. If the same initial check is executed by an automated data quality process, the expectation is that execution takes less than one minute; for simplicity, an execution time of one minute is assumed. As no manual initiation is required, no human time is spent performing the check. For comparison, the one minute waiting time for report approval and release is still considered. The business can then be informed whether the check returned an OK or not OK result, and the outcome can be stored for future reference.

When scaling this approach to ten checks, the total execution time remains ten times one minute. However, as the IT system can execute these checks in parallel, the waiting time for results remains at one minute. When increasing to 20 checks per day, the IT systems likewise execute the checks in parallel, and the execution and waiting time remain unchanged.

Benefit 2: Scalability to further reduce overall time required

In summary, manual data quality checks cannot be scaled and lead to excessive use of human resources for repetitive tasks, as well as long waiting times for approved datasets. Automated data quality checks, by contrast, allow many checks to be executed in parallel within the same overall time frame. It should be noted that this comparison assumes all checks have similar complexity.

Benefit 3: Rule changes without query changes

The current manual reports, which are business driven, have their logic built directly into the queries executed on the data. If any rule changes or a part of the data processing is modified, these queries need to be adapted. Thus, the check is deeply tied to the data source. To illustrate this with an example, a data quality check may confirm that 30% of the data is category A and 70% is category B, as this is considered the expected and valid distribution. If changes to processing pipelines or client behaviour cause the share of A to increase to 45% and B to decrease to 55%, the query performing the check must be adapted. However, when the check is performed by an application that injects these thresholds into a query, the query itself does not need to change, and only the parameters must be adjusted.

Benefit 4: Reusability to improve stability and performance

By moving the responsibility for checks from manual, business-driven processes to automated checks executed by IT, the checks can be designed to be more generic. Generic checks can be reused across different use cases. This separation of responsibilities and concerns leads to improved stability and performance when applying the checks, which in turn enables strong reusability. The same check type can be applied to different data sources by adapting parameters such as the query and thresholds, while the underlying check logic remains unchanged.

3.8. Initial use case

The initial use case was defined early in the project, on 18.10.2026, to serve as a foundation for future research. First, a definition of what the use case must describe is provided. Then, the initial use case itself is described, which may be refined during the upcoming research process.

- **Context:** In which company and industry sector does the use case take place? More specifically, which team or business process is involved?
- **Problem:** What problem is addressed by the use case, and what are its effects on the actors affected by it?
- **Actors:** Who is directly addressing the problem? Who contributes to the use case by providing definitions, testing, or needing to adapt their way of working?
- **Goal:** What is the ultimate goal of the use case, and what should be achieved?
- **Boundaries:** What is in and out of the scope of the use case? What may be strictly excluded?
- **High-level scenario:** A high-level description of the scenario in which the use case is situated. What triggers the use case, and what results are expected?

Context

The use case is situated within Swiss Post, in the area of its postal services. More specifically, it concerns the centralised sorting infrastructure and, looking more closely, only the data that is retrieved and stored for analytical reporting by the DWH LS OPS team.

Problem

The problem addressed is that there is no automated testing of data quality and availability for analytical reporting. The checks are currently performed ad hoc by business stakeholders rather than being performed automatically by the IT team that provides the data to them. This situation leads to delayed response times when data is missing or incorrect.

Actors

The main actors are the members of the academic project team working on the use case. Furthermore, the business stakeholders will use and test the solution for correctness, while the IT stakeholders will run the solution and benefit from being more responsive to data quality and availability problems. The IT architects will define the architecture that must be respected.

Goal

The goal is to provide an improvement for the business stakeholders and IT, or to show how their current problem can be addressed and alleviated in a follow-up to this project.

Boundaries

This project is strictly limited to the context and problem described within this use case. Its solution is not planned for the whole of Swiss Post. Also, the solution provided will be limited to one use case, which may be solved in different ways in order to compare the viability of different approaches.

High-level scenario

Swiss Post currently checks the data quality of its analytical reporting data by performing daily checks after each postal day. A postal day starts at 06:00 in the morning and ends at 05:59 the next day. As the sorting equipment should be free of parcels at the end of each postal day, the number of parcels that have been put onto the machines should be equal to the number removed from them. Thus, the number of parcels within the analytical reporting data should also be the same. If the numbers differ by more than a defined threshold of 1 percent when at least 10'000 parcels have been sorted, the data is considered incorrect and to have data quality or availability issues. These issues are then reported to IT, which has to run checks to determine why the data for that day is incorrect.

3.9. Research Gap, Todo

Todo Work for future Roland

3.10. Research Objective, Todo

Todo Work for future Roland

3.11. Research Question, Todo

Todo, Future Roland will have to write here... The Research Chapter will need rework... Read what is to be kept... What should be removed... etc...

3.12. Research Hypotheses, Todo

Todo, Future Roland will have to write here... The Research Chapter will need rework... Read what is to be kept... What should be removed... etc...

3.13. Research Methodology, Todo

Todo Work for future Roland

3.13.1. Research Approach, Todo

3.13.2. Comparison of Detection Approaches, Todo

3.13.3. Experimental Design, Todo

3.13.4. Evaluation Criteria and Metrics, Todo

3.14. Success Definition, Todo

Todo, Future Roland will have to write here... The Research Chapter will need rework... Read what is to be kept... What should be removed... etc...

3.15. S.M.A.R.T. Goals, Todo

Todo Work for future Roland

3.16. Expected Contributions, Todo

Todo Work for future Roland

3.16.1. Practical Contribution, Todo

3.16.2. Scientific Contribution, todo

3.17. Expected Benefits for the Reporting Solution, Todo

Todo Work for future Roland

4. Solution landscape

Todo, Future Roland will have to write here... The Solution Landscape needs rework and eventual restructure However keep a lot of content... and add further content :=) etc...

In order to enable the reader to understand the complexity of the Swiss Post DWH LS OPS, the data lineage must be presented at a high level. This chapter provides an overview of this lineage while deliberately omitting details that are not relevant to this project. It further explains the scope of this project within the data lineage and describes the reporting lineage with relevant tools. Furthermore, it is important to note that the solution landscape reflects the current state of knowledge and is limited to the project scope. Any additional knowledge acquired during implementation is discussed in the Results and Target Analysis chapters.

4.1. The Swiss Post addresses data quality in multiple ways

At the start of this project, the objective was to develop a minimal viable product (MVP) to replace a manual data quality check performed by the business with a more robust solution, including automation, the delivery of reports, as well as an appropriate notification and alerting. However, after beginning this project, Swiss Post management assigned a new priority to the topic of data quality management (DQM) overall; therefore, quite unexpectedly, this project acquired an additional set of key stakeholders, located in different teams as discussed in chapter 2.2.

Fortunately, this project as originally conceived fit very well into the larger DQM picture. Coordination with these teams although not originally envisioned (such as the DQM team described in chapter 2.2) therefore became an important task. This coordination was essential for several reasons: first and most obviously, the project must avoid reinventing existing solutions or overlapping in an unplanned way with other new initiatives. Second, it must ensure that methods and deliverables of this project conform to any new data architecture standards; for example, data quality dimensions have already been defined and documented by other teams. Finally, it must ensure that ongoing work by other teams is taken into account.

4.2. Rule-based DQM initiative

The current architectural goal pursued by the DQM team is shown in the following visualisation. The rule evaluation engine, shown at the top centre, executes defined rules on target data and stores the corresponding rule results in a database. In this context, a rule represents a data quality check for a defined dataset, and many different rules can be defined. These rules may, for example, specify allowed values, such as restricting the gender attribute to the three values *male*, *female*, and *not stated*. Alternatively, they may define expected data types, such as requiring dates to be stored in UTC format, or specify valid value ranges depending on the weekday.

Although the rule evaluation engine was a major topic during the weekly exchanges with the DQM team, it was not implemented by the project team. Nevertheless, its development progress was regularly monitored and discussed during these meetings. Furthermore, the ASTV team used this tool for its intended purpose. However, this usage was outside the scope of this project.

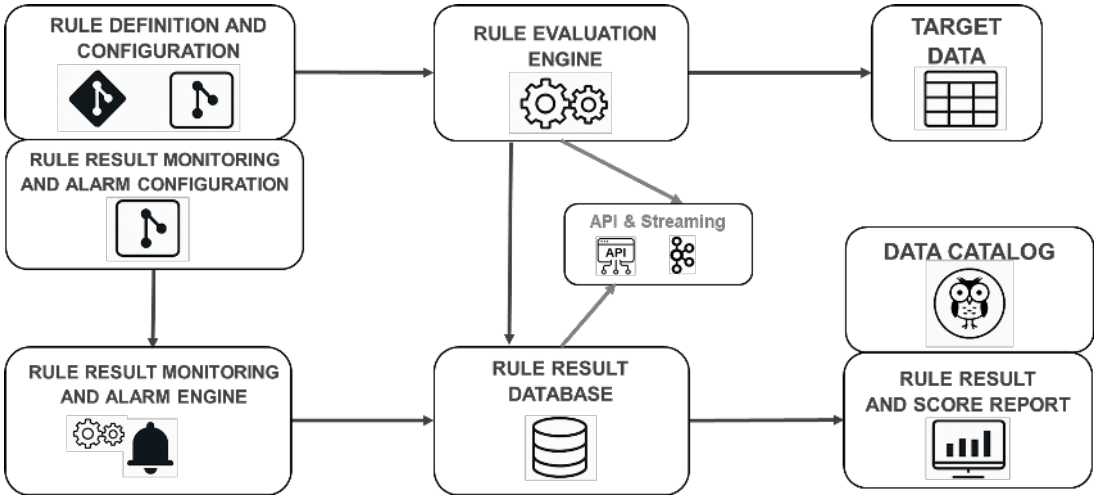


Figure 4.1.: The rule check engine architecture for future DQM

High-level target

From the complete target solution architecture, the first component to be implemented is a rule-based DQM suite. Its objective is to define generic data quality checks that can be applied across different datasets. This is achieved either by abstracting the check from the underlying data or by providing an SQL query that returns the data in a format allowing the same check to be applied to different datasets. The technology used to implement this solution is not discussed in this document, as the implementation was carried out by another team and is therefore outside the scope of this project. This approach consists of four main steps:

- **Configuration**: Configuration defines the tables to be checked, the execution frequency, and the type of check.
- **Data preparation**: An SQL query prepares the data in a format suitable for the check.
- **Rule execution**: An application executes the defined checks on the prepared data.
- **Result persistence**: The results are stored in a database, which enables reporting.

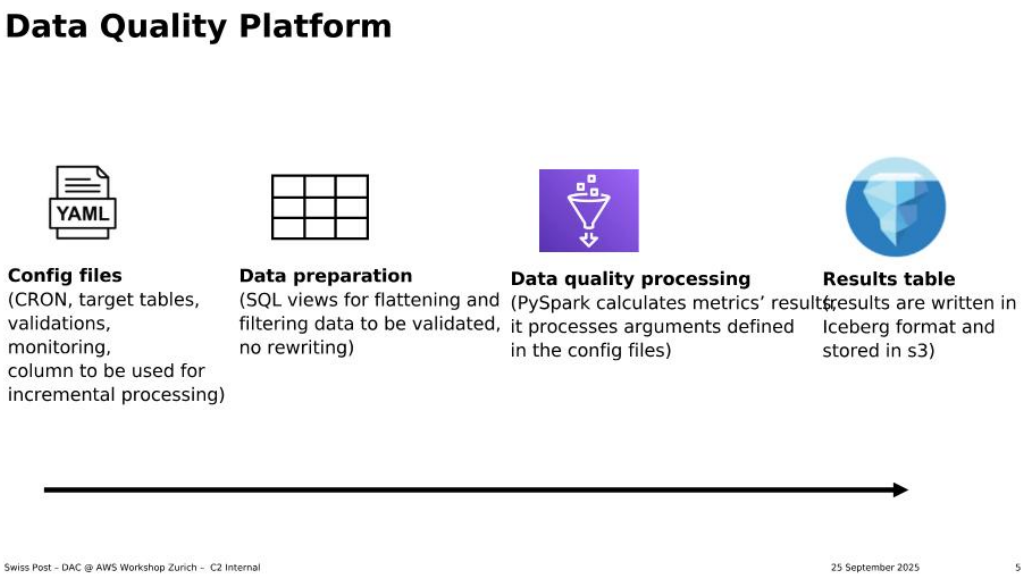


Figure 4.2.: Intention of a DQM platform, high level

Discarded solutions

This initiative is of high importance to the project, as it creates a major organisation-wide constraint to follow the DQM team's choice of tool. The rationale for developing a rule-based DQM solution is that existing tools, such as AWS Glue Data Quality, do not support views with complex data types, while PyDeequ does not provide row-level results. These limitations in the evaluated tools led to the decision to develop a dedicated solution at Swiss Post, as described above. The following section provides a brief overview of the tools and approaches that were evaluated but ultimately discarded as unsuitable. The following screenshot shows the presentation in which the DQM team summarised the evaluated tools and explained the reasoning behind their decision.

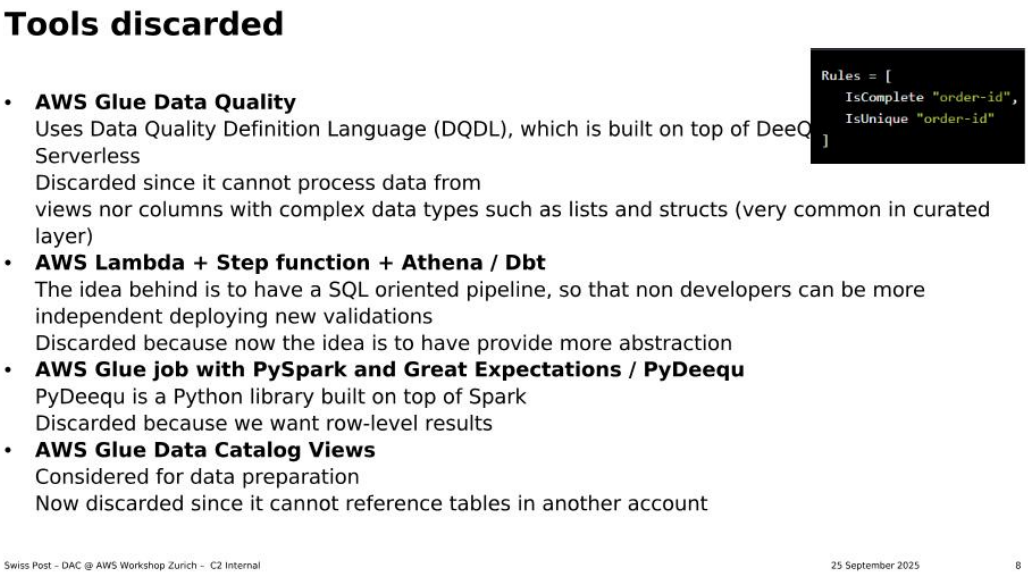


Figure 4.3.: The list of tools discarded and the reasons for their exclusion

State at the end of this project

By the end of 2025, the DQM team set up a PoC of the rule check engine based on the design shown at the start of this section 4.1. It allows running a check on a database view for specific target data source and storing the rule results in a designated rule result database. However, many details were still not defined or implemented. Such as the location where the results check database is to be installed which has not been defined. Also the alerting and the setup of dashboards are currently out of scope, as are later additions such as deriving expected value ranges from historical data or holiday calendars.

4.3. Heterogeneous Sorting Centre Environments

Before turning to the data lineage, an important constraint must be highlighted. The hardware used by Swiss Post in its high-tech "cyber" postal sorting centres differs in two main ways: across different providers and across different sizes and organisational structures of sorting centres. This, in turn, also led to a diversity of processes for performing similar actions. This diversity at different levels brings certain benefits but also introduces additional synchronisation effort in other areas.

Different sorting hardware providers

The hardware used by Swiss Post in its sorting centers is sourced from different providers. This is an essential risk management measure to reduce dependency on a single vendor. To mitigate this risk, Swiss Post decided to rely on at least two providers for essential hardware, not only to avoid dependency but also to protect against delivery issues or even the bankruptcy of individual providers. From a risk management perspective, this approach supports business continuity. However, from an IT perspective, it introduces challenges, as different hardware providers deliver data in different formats or with varying characteristics.

Different sorting center sizes and structure

The Swiss Post parcel sorting centers are distinguished into two categories. There is one large PC parcel center per region, which is supplemented by smaller RPC regional parcel centers. Regions encompass multiple cantons and may split cantons into the four current regions: west, middle, east, and south. The organisation into larger parcel centers and smaller regional parcel centers helps to optimise processes and reduce the carbon footprint of postal services. However, each PC and RPC differs from the others.

First, due to differences in size, processes differ between small and large centers, and the way of work also varies as organisational structures are different. Second, beyond the distinction between small and large centers, each center has a unique layout. Third, tasks can differ between centers; for example, some also handle letters. While these differences are reasonable from an operational perspective, they introduce challenges for centralised reporting, where all variations need to be harmonised. The following figure shows selected PC and RPC locations distributed across Switzerland.



Figure 4.4.: Selection of parcel sorting centers distributed over Switzerland

Different sorting processes

To go one step further, the different size and equipment at the sorting centers lead to different actions performed in the sorting centers. The same event in a sorting center can be caused by different previous actions making reporting on them difficult. As an example, the submitted and discharged events are described further.

Submitted event

The submitted event is the first automatic scan that occurs after a mail piece is placed onto the automatic sorting machine. In the context of this project, a mail piece refers to a parcel. However, parcels are not always placed onto the sorting machine in the same way. This can occur via roll boxes (RX), which are automatically emptied by robots. An RX is a container, as depicted in the following figures, that is used to transport different mail pieces through sorting and delivery facilities. Alternatively, parcels can be placed onto the automatic machine manually by staff, either by removing them from RX units or by emptying truck containers onto conveyor belts, as shown in the subsequent figures.

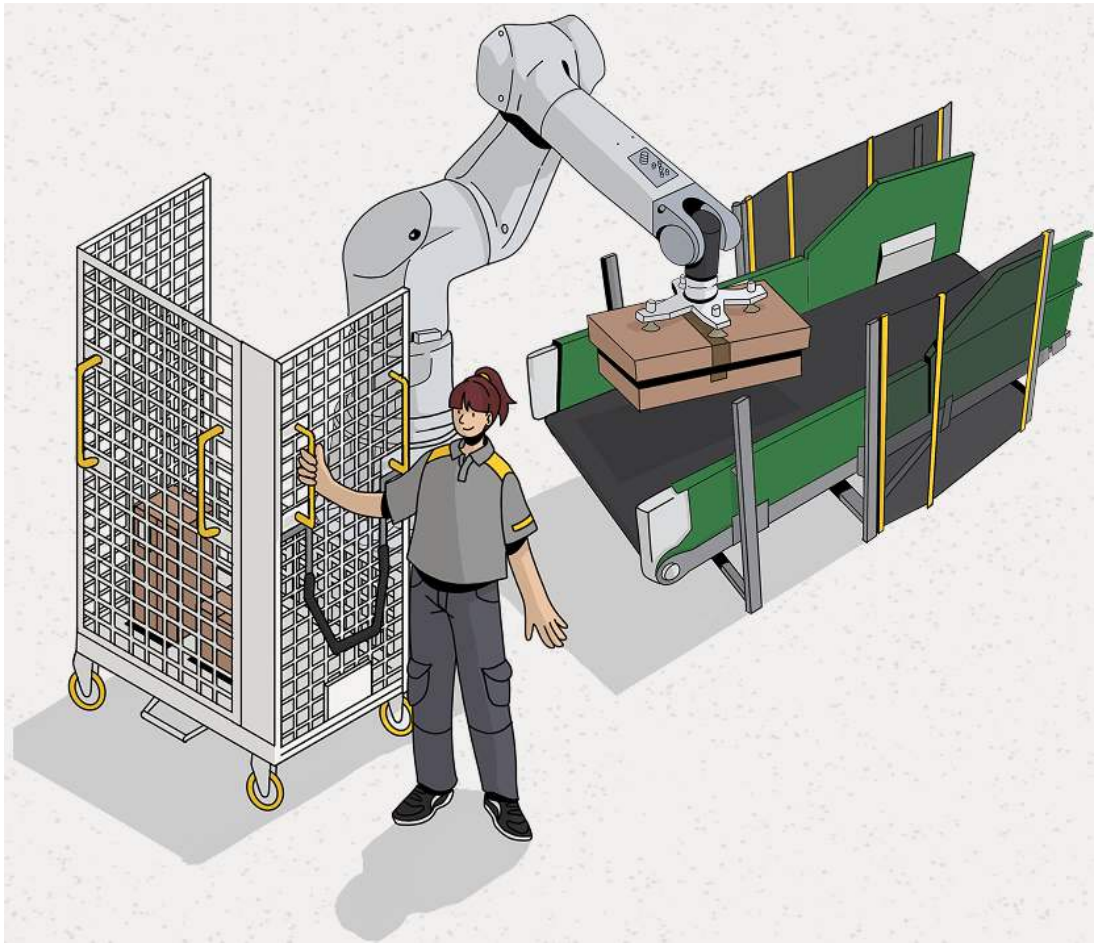


Figure 4.5.: Parcel sorting, parcels submitted by robot

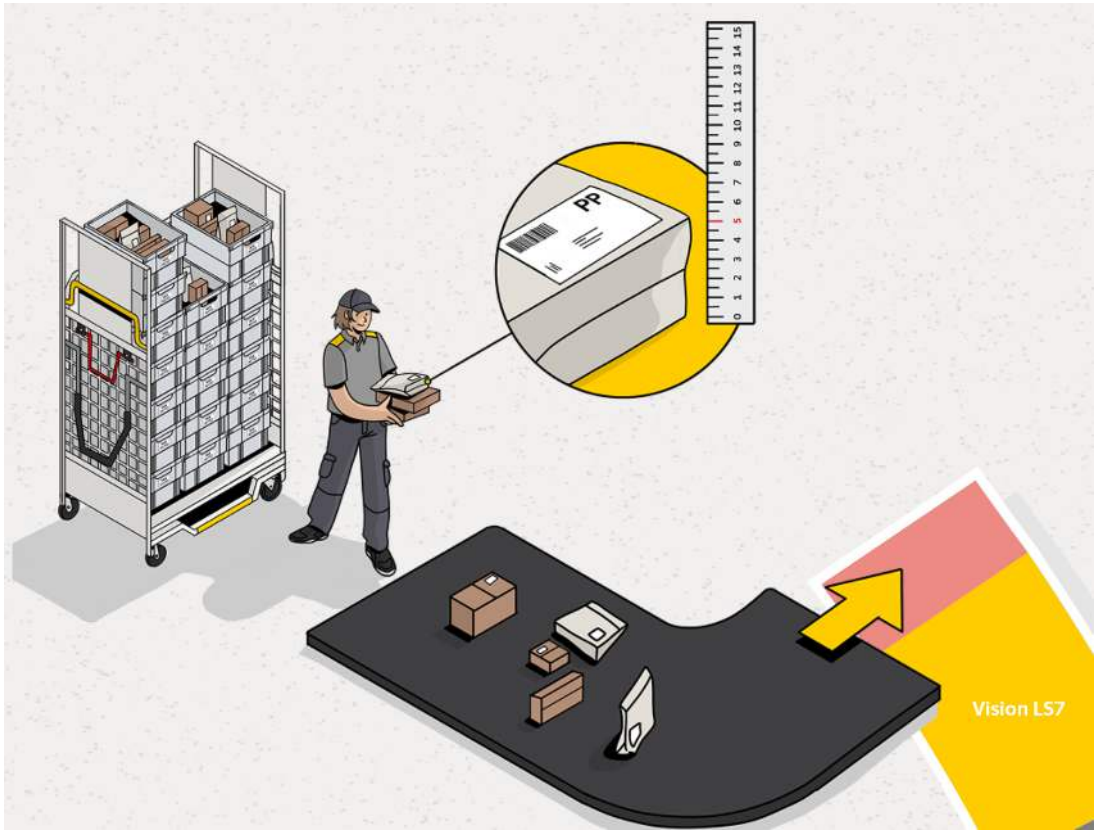


Figure 4.6.: Parcel sorting, smaller items submitted manually



Figure 4.7.: Manual parcel onloading from a truck

Discharged event

The discharged event is complementary to the submitted event. It is the last automatic scan event of a parcel when it leaves the automated sorting machine. This can happen in different ways, such as when the parcel is discharged via a chute where staff load it into an RX, when it is discharged directly into an RX automatically by the machine, or when it is automatically ejected after completing a defined number of cycles through the machine. All of these possibilities trigger different follow-up actions. However, in all mentioned cases, a discharged scan is performed after the parcel is removed from the automatic sorting machine. This is the important aspect for reporting, as these two events (submitted and discharged) occur once per complete cycle that a parcel runs through the automatic sorters.

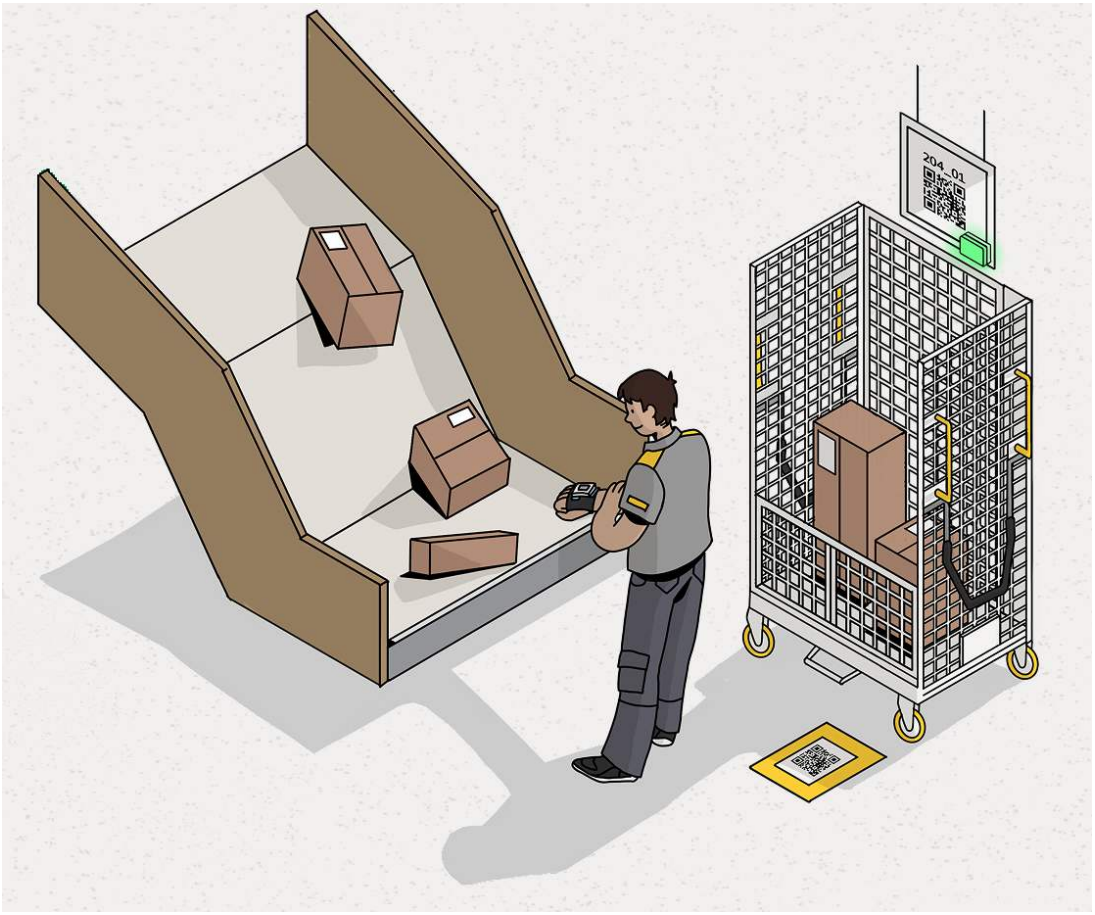


Figure 4.8.: Parcel discharged, manual loading into RX

4.4. Data Lineage

The data relevant to this project are those used by the DWH LS OPS team. They consist of cleaned and deduplicated data stored in the curated database layer. However, these data originate from sensor data and arrive in the data warehouse (DWH) in a raw format. Before reaching the curated layer, the data pass through four main processing steps, which are also shown in the following figure:

- **Physical data source**: Parcels are scanned by different scanners in the sorting centres. As described above, Swiss Post uses a variety of devices from different hardware vendors, which naturally results in a large number of data entities that must be stored. Each parcel passes through multiple scanners during its journey across the Swiss Post's 13 sorting centres, generating multiple data entities that are

transferred to the data warehouse. Determining the exact number of generated events was outside the scope of this project.

- **Initial data storage**: The scan events are first saved in AWS S3. At first glance this might be a very unusual choice, but the reasons make sense. First, the Swiss Post has a cloud strategy which includes AWS as a service provider; hence the choice of AWS is no surprise. But more importantly, Amazon S3 (Simple Storage Service) is the object storage service of AWS. Exactly which data are stored here and how is addressed in a following section.
- **Kafka ingestion**: Apache Kafka is an open-source distributed event streaming platform designed for high-throughput, fault-tolerant, real-time data pipelines and event-driven architectures.The Swiss Post [operates a Kafka environment in its down data centers] [uses the Amazon MSK Managed Streaming for Kafka]. With this, so-called "producers" read the data from the initial storage and ingest it into Kafka "topics," to which other systems can "subscribe" and obtain the data.
- **Data delivery to DWH**: Kafka consumers ingest the events into the DWH LS OPS raw data layer. Which is the first storage layer in this teams AWS S3 account, saving the data as recieved from Apache Kafka.

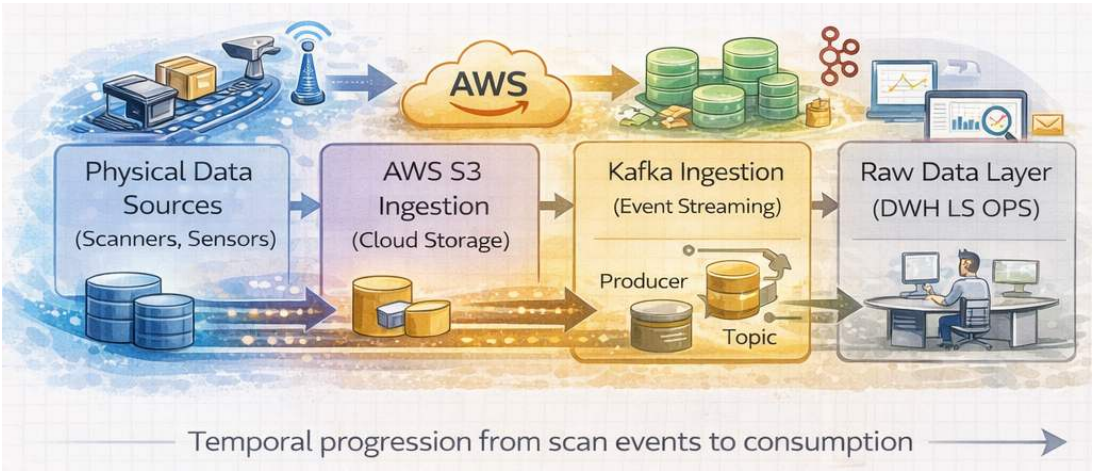


Figure 4.9.: An artistic high-level view of the DWH LS OPS data lineage

KEY POINT: From the data creating stage until the data arrives at the DWH, it passes through four IT system steps discussed here. This results in four potential sources of failure, in which each system is counted only once and by ignoring different possible errors per step. This landscape makes it difficult to trace errors that can occur at any step. Additionally, the variety of errors is large, as they may alter data values or result in no data being transmitted at a given step, even though the operation of the sorting center itself continues without issues. Therefore, at a technical level, providing clarity especially during the operational aspect is an important key aspect of this project.

Finally, to complete the picture of the data lineage, one more step is relevant. After the data has arrived in the raw layer at DWH LS OPS, it must be cleaned and deduplicated. At this step, differences in hardware providers and the structural differences of sorting centers come into effect, and the unification of the data takes place. All of this is performed in the step towards the curated database layer, after which a reporting layer serves as the final layer from which reporting tools retrieve data as introduced in chapter **??**. This is illustrated here with a focus on the central step from the raw layer to the curated layer. Even after the curated layer, there is one additional layer, namely the reporting layer.

The different layers in the cloud are set up according to a structure similar to that used in the classical on-premises DWH at Swiss Post. This similarity to the on-premises DWH stems from deliberate architectural

alignment. The cloud infrastructure was built with the guidance of on-premises developers using their domain knowledge. In this way, functional and structural consistency was maintained, as the architecture continues to follow the Kimball star schema and the layered architecture as defined in the Kimball technical architecture [Kimball Group n.d.].

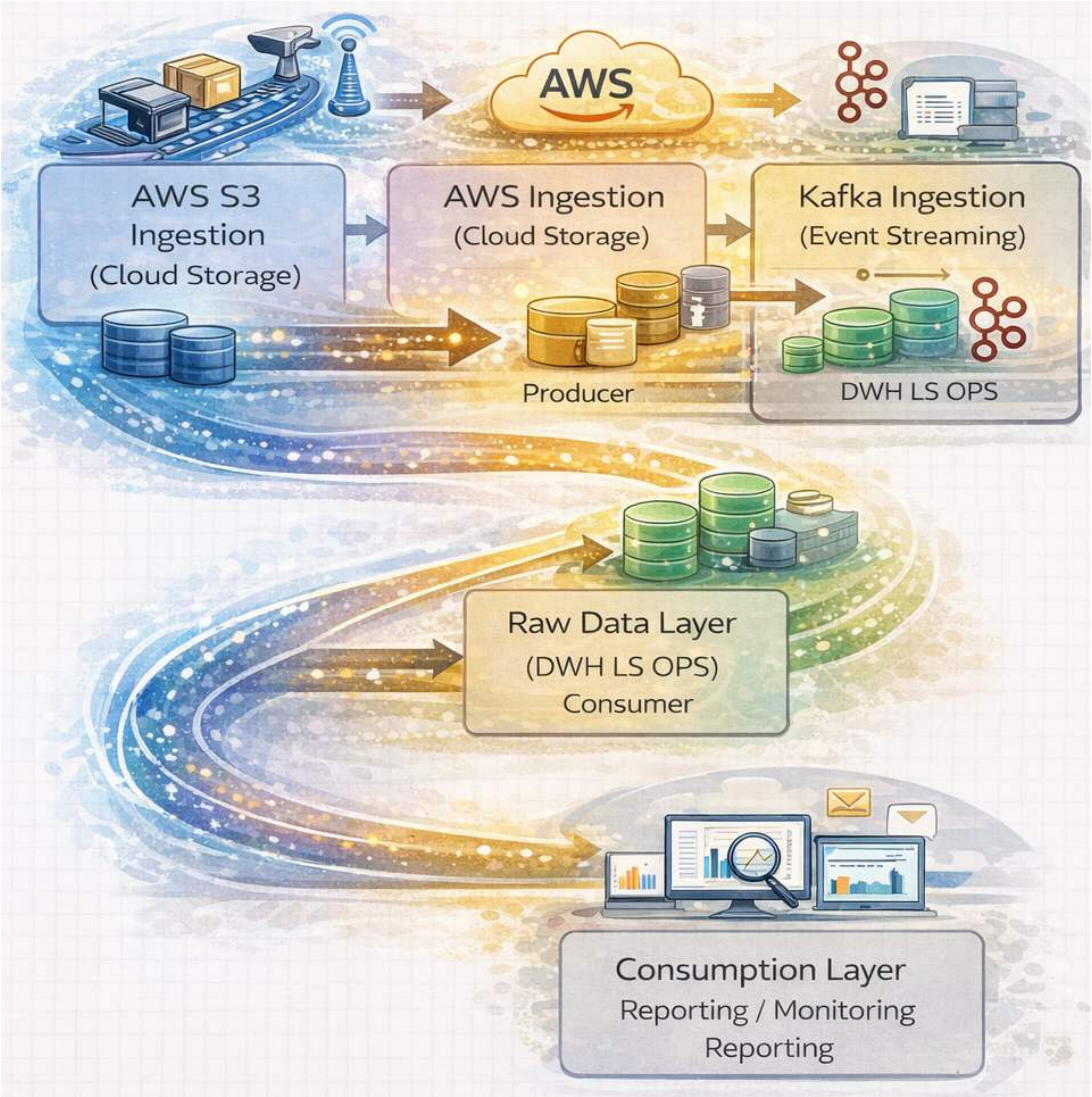


Figure 4.10.: An artistic high-level view of the full DWH LS OPS data lineage up to reporting

4.5. Existing reporting pipeline

The current reporting pipeline at DWH LS OPS team consists of four main components as shown in following figure and introduced in chapter **??**. First, the storage layer (in figure : **Datenspeicherung**), where data is retrieved, cleaned, and saved to Amazon S3. Second, the query and processing layer, where cleaned data is queried using Amazon Athena and DSP(in figure : **Datenselektierung, Datenpreparation Datenmodelle Data Federation**), and where data is passed and persisted in data models for reports. Third, the reporting and consumption layer, in which data from DSP is used in SAC reports (in figure : **Business Layer**).

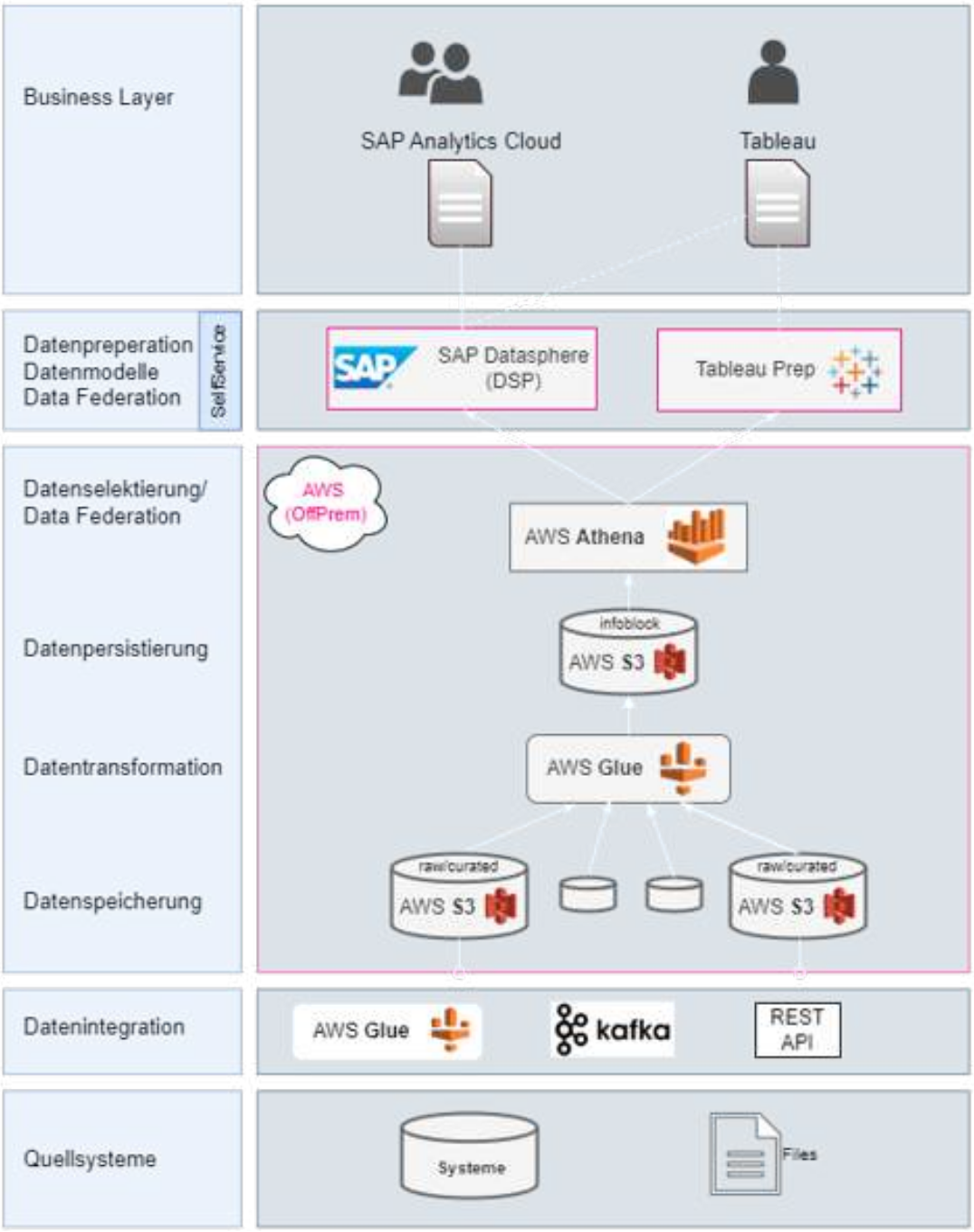


Figure 4.11.: The DWH LS OPS reporting layers

AWS S3

As described by Amazon: *"Amazon Simple Storage Service (Amazon S3) is an object storage service that offers industry-leading scalability, data availability, security, and performance. Customers of all sizes and industries can use Amazon S3 to store and protect any amount of data for a range of use cases, such as data lakes, websites, mobile applications, backup and restore, archive, enterprise applications, IoT devices, and big data analytics. Amazon S3 provides management features so that you can optimize, organize, and configure access to your data to meet your specific business, organizational, and compliance requirements." [Amazon Web Services 2025b].*

This is relevant for Swiss Post, as Amazon S3 serves as the central storage for data from mail piece events,

which include all scans of parcels and mail in sorting centers, as well as later stages up to the delivery process. This storage is used by the DWH LS OPS team. All data entering the data lineage pipeline, for example through Kafka or through direct database connections, is stored in Amazon S3. Thus, it represents the first persistence point in the reporting pipeline.

Layered architecture It is important to note that the data is not only initially saved to Amazon S3 in a raw layer, but is also cleaned and prepared for reporting in two additional layers. To separate the storage of raw data from the cleaning process, multiple layers are used. This ensures that all data in its original format is saved first before processing, modification, augmentation, and checks for completeness or correctness are performed. Each step can therefore be reproduced and repeated, which is useful whenever data needs to be reloaded, for example when data was incomplete for a day or when new processing methods are developed. After the raw layer, data is deduplicated and cleaned, and then saved into the curated layer. The final top layer, the reporting layer, mainly adds translations of fields into business terminology, for example by changing column labels. This is useful for the business, as it enables consistent naming of columns across tables or data products.

Infoblock On this level, infoblocks are generated. This is a Swiss Post internal term used for report-ready data products. An infoblock consists of cleaned, deduplicated, curated data featuring joined datasets that use business naming, e.g. post day and processing priority types. Basically, the infoblocks are the final layer employed in the reporting layer, making them accessible to reporting tools such as DSP and SAC, which are described in detail below. A more common term for an infoblock known throughout the industry is *data product* [*What is a Data Product?* N.d.(a); *What is a Data Product?* N.d.(b)].

AWS Athena Query

The next step is Amazon Athena as the query layer, which Amazon describes as follows: *"Amazon Athena is an interactive query service that makes it easy to analyze data directly in Amazon Simple Storage Service (Amazon S3) using standard SQL. With a few actions in the AWS Management Console, you can point Athena at your data stored in Amazon S3 and begin using standard SQL to run ad-hoc queries and get results in seconds." [Amazon Web Services 2025a]*

Amazon Athena is used by Swiss Post DWH LS OPS for all queries on Amazon S3, which is the next step in the reporting pipeline. Athena is the tool designed by AWS for use with Amazon S3, making it the logical choice for querying the stored data. The data is queried for reporting purposes by selecting the relevant subsets.

DQM relevance The querying layer is relevant to DQM, since data quality issues may become apparent first at this earliest checkpoint, for example when errors occur in the data itself and counts or joins become invalid. While the data is queried at this layer for reporting purposes, it is also queried here for data quality checks. Such checks typically address missing data and data completeness, but can also extend to data plausibility, for example when postal codes are compared to cities. In cases where issues occur at this layer, they are passed downstream towards DSP and SAC, provided they do not already break the pipeline at this stage.

SAP Datasphere (DSP) and SAP Analytics Cloud (SAC)

At Swiss Post, the next step in the reporting pipeline uses the two dependent tools SAP Datasphere (DSP) and SAP Analytics Cloud (SAC). These are described as follows by SAP:

SAP Datasphere enables organizations to combine, harmonize, and model data from multiple SAP and non-SAP sources within a semantically rich, cloud-based data environment. SAP Analytics Cloud can directly consume these governed and business-ready data models via live connections, enabling interactive reporting, analytics, planning, and forecasting. In addition, planning and forecast data created in SAP Analytics Cloud can be written back to SAP Datasphere, allowing analytical and planning data to coexist in a single, integrated data foundation [SAP SE 2024].

SAP Datasphere (DSP) is SAP's cloud-based data platform for integrating, modeling, and analyzing data across SAP and non-SAP systems while preserving business semantics. It enables federated access to distributed data without full replication and supports end-to-end data lineage and governance. DSP at Swiss Post is used as the analytical backbone for reporting, analytics, and data-driven decision-making in SAP landscapes.

SAP Analytics Cloud (SAC), in turn, uses the data model provided by DSP to present this information on the business layer to end users. More information on this is provided in the separate section introducing the tool.

DSP data model

After querying, the data is persisted in DSP, where it is made available in data models. This persistence and processing layer is essential in the current pipeline, as data can be made available to SAC only through DSP. At Swiss Post, data persistence has proven essential to improving the response time of the reporting pipeline as a whole. With this persistence, DSP does not pass queries back to Amazon Athena, creating a clear separation between the two layers. Due to this separation, DSP is not responsible for DQM, which is handled in lower layers. Finally, DSP is the layer in which business logic is applied in a semantic form, defining what the data represents. At Swiss Post, this is done by applying business oriented naming and calculations. Different report consumers may require the same values to be structured in different ways; for example, productivity in a sorting center may be measured using total working hours against the number of mail pieces handled, or calculated using only time spent on sorting-related tasks, excluding breaks, training, and other activities.

DSP use case The data models in DSP are useful not only as a bridge to SAC (discussed below). They allow different data sources to be modelled and combined into unified data models. This is also relevant to Swiss Post, as it operates not only Amazon Athena data sources but also other database- or file-based sources. This combination of data inevitably inherits data quality issues from the underlying sources, which can affect joins, data loading, and further processing. Nonetheless, DSP is the sole source of data provided to SAC, which is the next logical step in the reporting pipeline. This also means that any unhandled data quality issues are passed directly to users consuming reports in SAC and may therefore affect confidence in the reporting pipeline depending on data quality.

SAP Analytics Cloud (SAC) report

SAC is the final step in the reporting pipeline and represents the reporting and consumption layer. As the last layer, it inherits the data from previous layers and is responsible only for its visualisation. The data is displayed passively, and no further transformations are performed at this stage. Consequently, any errors in data preparation or data quality issues from upstream layers are propagated to this layer together with the data itself.

As described earlier, SAC is dependent on DSP and cannot operate independently, nor does it provide native DQM capabilities. The reports generated by SAC are therefore fully dependent on the preceding layers, and

any issues or delays are reflected directly in the reports. This is particularly relevant for Swiss Post, as one report provided by DWH LS OPS is used daily at 07:00 for shift handover, which takes place one hour after the end of the post day. Some data sources required for this report become available only after 06:00, the end of the postday. If any upstream step is delayed, the report delivery is delayed accordingly.

Consumption layer As the consumption layer for reports, this is the point at which users with read access have their first and only interaction with the reporting pipeline. It is at this stage that user confidence is either established or lost, making it particularly critical. Users do not see the upstream processing steps and typically do not differentiate whether missing or delayed information is caused by AWS S3, Athena, or DSP.

4.6. Report visualisations and evaluations

After introducing the basic solution landscape, this project now examines the more specific context of report visualisations and the evaluation of automation.

Power BI

At Swiss Post, within the DWH LS OPS environment, Power BI is the designated business intelligence tool. It is intended to replace the currently used tools over time. The main reasons for this shift are the lower licence costs compared with the previous SAC; furthermore, it performed faster in direct comparison tests with its predecessor. Therefore, Power BI served as the BI tool in this project, as it is specifically replacing an old report in SAC.

Architectural position

When considering the data lineage, Power BI is replacing both DSP on the data model side and the front-end part of SAC. In total, it is replacing the work areas of the business stakeholders working together with the DWH LS OPS team. This is because they are involved in both the generation of tests and the updating of reports within the reporting and monitoring layer.

Star schema

During the speed tests and comparisons with the old business intelligence tool, the star schema as a data architecture was tested. It showed that Power BI performed better with this new structure. Therefore, as Power BI was designated as the new business intelligence tool, the architecture was changed accordingly.

Basic architecture of a star schema

When looking at database architecture, a star schema is basically a partly denormalised database that has been optimised for fast OLAP processing. The data are split into repeated dimension informations, which are linked with the factual data in the fact table in the middle. In such a architecture there is only one fact table holding the information the model is providing and multiple dimensions which are repeating informations. In the case of the message processing star shema, the fact table are the messages, who come with their specific informations such as an unique ID to identify them. They are then linked to the different dimensions trough other id's which can be linked to entries in the dimension tables. As a toy sample the message "ABC134" may have the desination 4, which is in the destination table the id 4, and can represent

a whole adress or a person. More information can be found at this source [Wikipedia contributors 2025]. Furthermore, the following figure shows a sample star shema with a fact table in the middle and many connected dimensions around it.

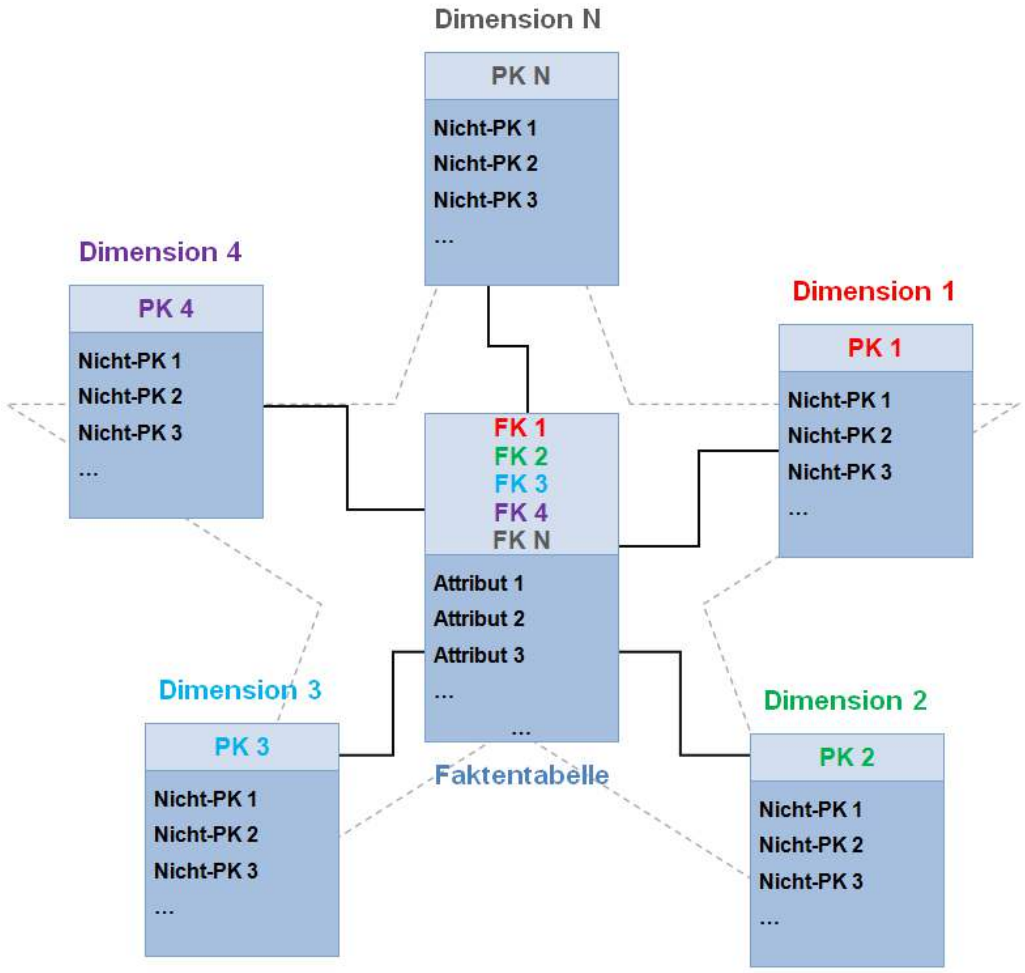


Figure 4.12.: Illustration of a star schema showing the relationship between a central fact table and multiple dimension tables. Image adapted from Wikipedia.

Basic architecture of a wide table

In contrast to a star schema, a wide table stores all the data in one large, very wide table. This concept of the wide table was introduced at Swiss Post together with SAC. In a wide table, the data are not split into different tables; instead, all existing measures are repeated for every dataset. For example, every data point measured at the same factory would have the same identifying address information repeated, rather than a single key pointing to an addresses table [QuestDB 2026]. Such a table makes it easier to create reports which are already known at the moment of setting up the table, as every data required is available. However if additional data is required a wide table has to be extended.

Reporting idea

When considering the previous report generated through SAC, it can be seen that it is not very visual. The data are shown only as a table without colour highlights or additional indicators. Without sufficient

business knowledge, it did not allow a user to determine whether the data quality was acceptable for a specific day. The current report is shown below.



Figure 4.13.: Existing plain DQM dashboard used as the baseline for the redesigned Power BI report

To improve this situation, a report that had to replace it needed to be more visual and apply some of the most basic business knowledge, allowing users to interpret its results more quickly. As the previous research had already defined which rules had to be applied in an automatic check, those rules formed the basis for the report's visualisation.

Business stakeholder feedback

During meetings with the business, their requirements were gathered. These were simple: the report had to be upgraded to use the new data source. Since the adoption of Power BI also enabled a more efficient reporting data structure, this had to be measurable again. As the data themselves had only been rearranged and were not fundamentally different, the checks and the way they had to be shown in a table remained the same. Regarding additional improvements to the report, no direction was given; however, approval was provided to improve the report with useful information.

DevOps

At Swiss Post, deployments are carried out by software engineers, effectively making their field of operations DevOps.

The field of DevOps is described by Wikipedia as follows [Wikipedia contributors 2026]:

DevOps is the integration and automation of software development and information technology operations. DevOps encompasses the tasks necessary for software development and can lead to both shortening development time and improving the development life cycle. According to American software architect Neal Ford, DevOps, particularly through continuous delivery, employs the "bring the pain forward" principle by tackling challenging tasks early, fostering automation, and enabling swift issue detection.

Although debated, DevOps is generally characterized by three key principles: shared ownership, workflow automation, and rapid feedback. From an academic perspective, Len Bass, Ingo Weber, and Liming Zhu—three computer science researchers from the CSIRO and the Software Engineering Institute—defined DevOps as "a set of practices intended to reduce the time between committing a change to a system and the change being placed into normal production, while ensuring high quality". However, the term is used in multiple contexts. At its most successful, DevOps is a combination of specific practices, culture change, and tools.

DevOps at Swiss Post

At Swiss Post, the DevOps process is already defined and is not changed by this project. It applies all defined tools and rules. The general rules are as follows:

- GitHub is used as the code repository, using SSO with Swiss Post accounts.
- Access to repositories is limited and managed by a permission system.
- Separate repositories exist for development and production environments.
- Repositories are also separated by function, e.g. repositories for Terraform IaC or SQL scripts for Athena.
- Development is performed on branches.
- To merge branches, a pull request is required; this triggers GitHub Actions.
- GitHub Actions perform checks depending on the repository and its function.
- To merge code into production environments, a review by another developer is required in addition to successful GitHub Actions.
- In addition to this deployment pipeline, DevOps engineers continue maintaining and supporting their code.

Frontend deployment pipelines

Regarding frontend reports, a similar process is already defined and also applied; it is not adapted by this project. When deploying front-end reports using Power BI, the process takes place entirely within the Microsoft environment.

- Reports are initially created in Power BI Desktop.
- Development and testing happen in the desktop version first.
- The report is deployed from Power BI Desktop to the Microsoft Fabric development environment.
- The code is checked in using Git integration in Microsoft Fabric.
- The report is tested first in the development environment.
- It is then deployed through the defined pipeline to the test environment.
- Stakeholders perform tests in the test environment.
- After successful testing and any required fixes, approval for production deployment is given.
- Production deployment is performed again through the deployment pipeline.

The deployment pipeline is shown in the figure below, showing the selection of environments from which the three-step process starts.

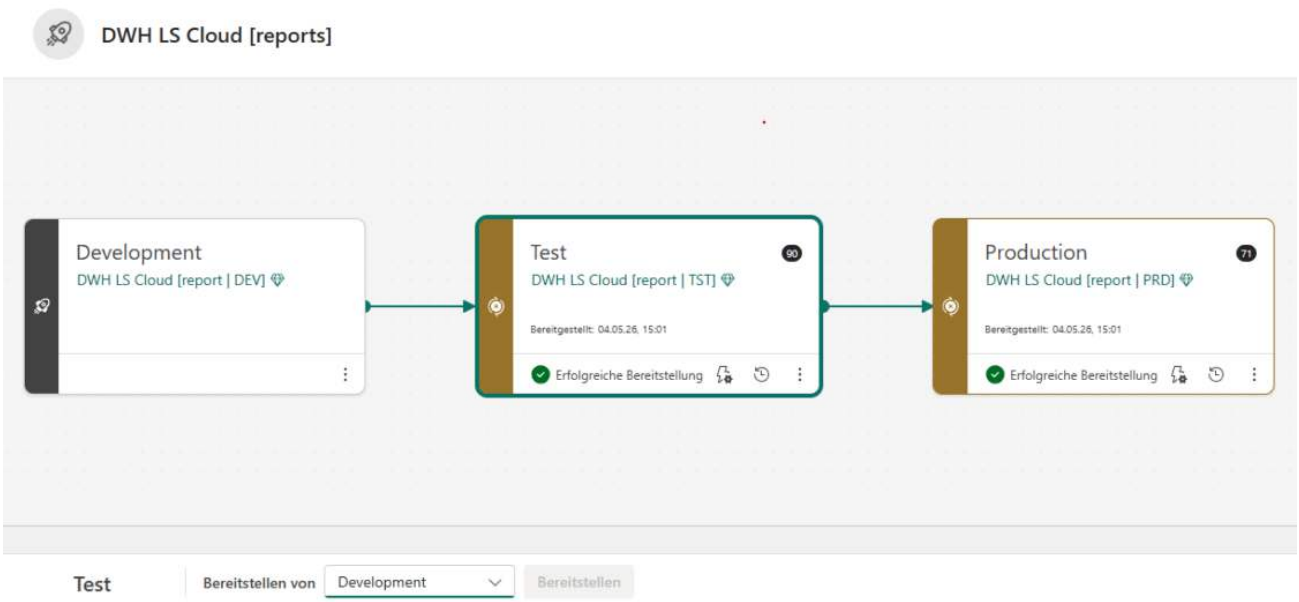


Figure 4.14.: Power BI deployment pipeline across the development, test, and production environments in Microsoft Fabric

4.7. DQM tool selection

The DQM tool selection required a structured approach. Such an approach required first setting up a list of requirements for the tool. Then an evaluation of how well the tools fulfilled these requirements. And finally the invitation of the providers that fulfilled the requirements to demonstrations.

Tool selection principles

The first step in evaluating a tool must be to determine what a tool actually needs to have. Commonly, such lists are called requirement lists. They have to incorporate the must-have features of a tool and the nice-to-have requirements. With such a list, the decision together with higher management is made easier, as it allows for scoring and provides an easy overview of how many points each tool scored. Based on such a points list, the best-performing or best-value tool can be selected.

Requirements list

At Swiss Post, the DQM team set up the requirements list. It incorporates the following main principles:

- **Data Coverage & Connectivity:** Integration into the existing tool landscape. The requirement to be cloud-first. The most important data sources that need to be handled.
- **Data Quality Capabilities:** The actual DQM requirements that the tool must fulfil. The expected usability, such as no-code environments.
- **Reporting & Monitoring:** The tool must allow exports to other tools for reporting purposes.
- **Performance and Reliability:** The tool needs to be responsive even on large datasets, as Swiss Post will want to run checks on very large transaction datasets.
- **Cost & Licensing:** The cost of a tool is relevant. How the cost is structured.

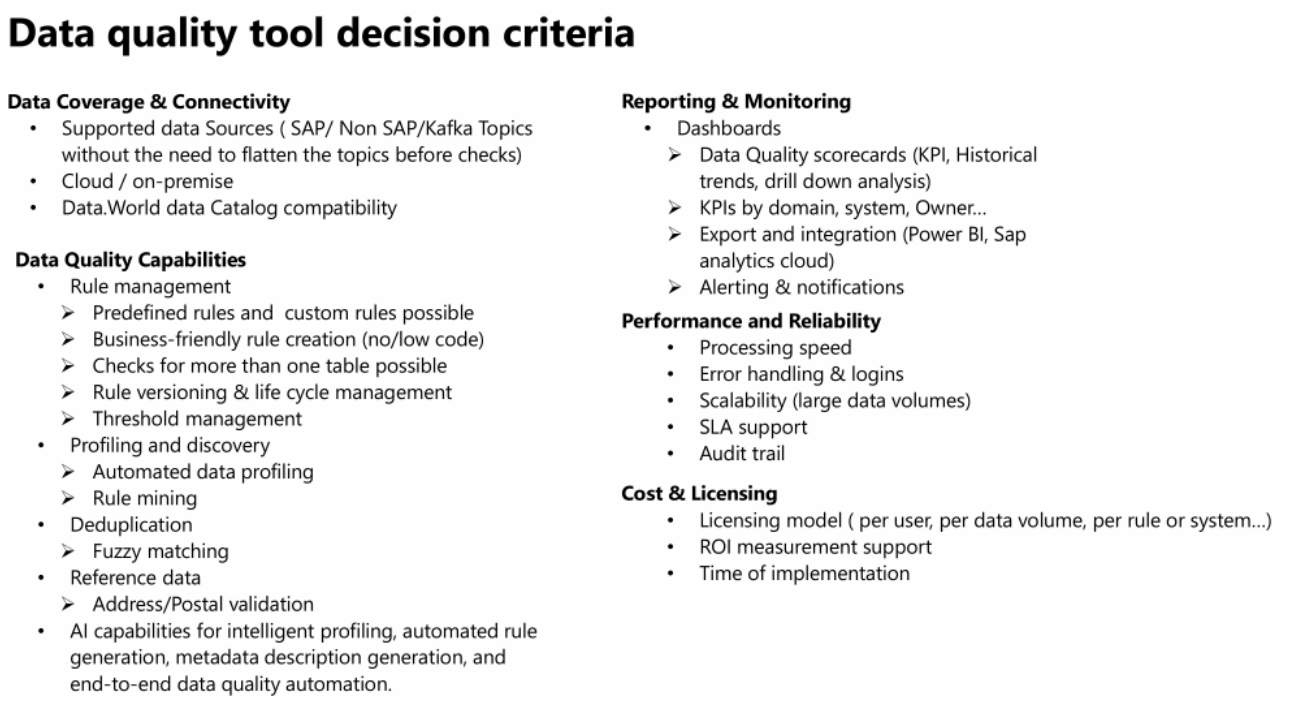


Figure 4.15.: Overview of the data quality management tool evaluation process and the assessed evaluation criteria

Additionally, there are the more soft criteria, which are nice to have. Soft criteria are criteria that are not mandatory but make the usage of an application easier for users, e.g. by providing easy management of access roles. The list as defined by Swiss Post is shown in the following screenshot:

- **Usability & User Experience:** User permission system allowing multiple different access levels and their provisioning through existing means.
- **Technical Requirements:** The security requirements that the cloud solution has to meet. Additionally, the connectivity options that it needs to support.

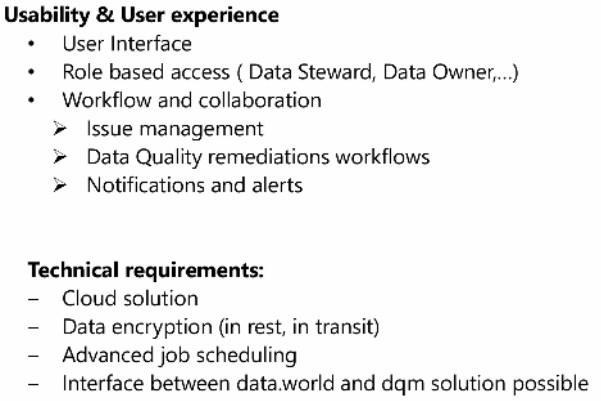


Figure 4.16.: Continuation of the data quality management tool evaluation, including the comparison of the evaluated solutions

DQM tooling selection

After defining the selection criteria, the tools are researched. The research on these tools was also carried out by the DQM team, so it is out of scope for this work. As a result of the tool research, the different

tools found are evaluated against the criteria. This finally results in the table shown below, highlighting the performance of the tools against the criteria. This table then serves as the basis for deciding which tools are invited for demonstrations or directly excluded. The following screenshot shows the five tools that were included in the evaluation, including the three in green that were invited for demonstrations.

- **Ataccama ONE**: Selected for demonstration.
- **Informatica IDMC (CDQ + optional Streaming)**: Selected for demonstration.
- **Soda (Core + Cloud)**: Selected for demonstration. The demonstration is documented later in this document.
- **Great Expectations**: Not selected for demonstration.
- **SAP Information Steward**: Not selected for demonstration.

Comparison of tools

Criterion	Ataccama ONE	Informatica IDMC (CDQ + optional Streaming)	Soda (Core + Cloud)	Great Expectations	SAP Information Steward
1) Business-rule DQ (custom rules + complex checks)	**Strong** – DQ evaluation rules + monitoring/reconciliation projects. (docs.ataccama.com)	**Strong** – profiling + rules + monitoring/scorecards. (Informatica)	**Strong** – flexible checks language; runs checks as code/config. (docs.soda.io)	**Strong** — Expectations are expressive/extensible tests for business logic; works with SQL-backed data, pandas, Spark. (GitHub)	**Medium** – rules + profiling/monitoring, but SAP-centric. (SAP)
2) Kafka "close to source" fit	**Partial** – possible via running processing engines in your environment; Kafka specifics must be proven in PoC. (docs.ataccama.com)	**Strong (if you include Streaming)** – Data Engineering Streaming provides prebuilt connectors incl. Kafka. (docs.informatica.com)	**Partial** – can work if your pipelines can query/scan where data lands; "close to Kafka" usually means embedded into streaming/pipeline patterns rather than native Kafka engine. (Needs design.) (docs.soda.io)	**Partial** — GX doesn't "inspect Kafka topics" by itself; it fits Kafka via your processing/landing pattern (e.g., validate in the streaming job/micro-batch, or after landing in a lake/DB). GX is designed to validate where data lives. (Great Expectations)	**Weak** – not designed as a Kafka-first enterprise DQ tool.
3) SAP finance/master data fit	**Strong/Partial** – enterprise platforms typically cover SAP via connectivity, but confirm specifics with your SAP landscape.	**Strong/Partial** – strong enterprise connectivity model via Secure Agents; confirm SAP specifics/modules. (docs.informatica.com)	**Partial** – depends how you access SAP data (extract/replication layer, connectors).	**Partial** — good if SAP data is extracted/replicated to a database/lake/analytics platform that GX can query; not SAP-native stewardship/DQ inside SAP. (PoC needed for your SAP access pattern.)	**Strong (within SAP landscape)** – profiling, validation rules, monitoring in SAP contexts. (SAP)
4) Hybrid deployment & "process data locally"	**Strong** – hybrid model with Data Processing Engines running in your network. (docs.ataccama.com)	**Strong** – Secure Agent group runs within your network/cloud and is used to access on-prem data. (docs.informatica.com)	**Strong** – Soda Core runs where you execute it; Soda Agent can be self-hosted (K8s/Helm). (docs.soda.io)	**Strong** — runs wherever you can run Python (on-prem, cloud, hybrid) and validates data "where it lives." (Great Expectations)	**On-prem oriented** – good if you want on-prem, less flexible for broader hybrid patterns. (SAP)
5) Governance + low-code rule authoring	**Strong** – platform UI for rules/monitoring and governance concepts. (docs.ataccama.com)	**Strong** – visual/no-code positioning; training materials emphasize building rules and scorecards. (Informatica)	**Medium** – can be no-code in Soda Cloud, but often starts engineer-led; agent model supports UI checks. (docs.soda.io)	**Medium** — GX Core is code-first; GX Cloud emphasizes point-and-click UI to involve nontechnical stakeholders. (Great Expectations)	**Medium** – stewardship dashboards/scorecards; governance is SAP-centric. (SAP)
6) Workflow: alert routing, case mgmt, exception handling	**Partial** – platform capabilities exist, but your wish is "as much as possible in data.world"; you'll likely still integrate.	**Partial** – similar: tooling exists, but aligning workflow into data.world will be integration work.	**Partial** – Soda Cloud has collaboration/alerts; "case mgmt/waivers" may still be better in data.world processes.	**Partial** — GX integrates well with orchestration/alerting patterns (e.g., Airflow operator examples), but it's not a full case-management/waiver system out of the box. (astronomer.io)	**Partial** – some review workflows exist, but likely not your enterprise-wide workflow hub.
7) data.world integration	**Partial (via API)** – likely publish results through data.world DQ API. (Data.world)	**Partial (via API)** – same pattern. (Data.world)	**Partial (via API)** – same pattern; technically straightforward because results are already "check run" oriented. (Data.world)	**Partial (via API)** — you'd publish GX validation results into data.world using the Data Quality API (mapping GX results → check runs). (Data World Docs)	**Partial (via API)** – possible but usually more custom. (Data.world)
8) Power BI consumption (aggregated metrics)	**Strong/Partial** – most enterprise platforms expose scorecards/metrics; may still export to Power BI.	**Strong/Partial** – scorecarding and exporting exception records are part of CDQ training/usage. (Informatica)	**Strong/Partial** – checks produce aggregated results; you can store/export for Power BI. (GitHub)	**Strong/Partial** — GX produces structured results you can store and visualize; if you use data.world as the hub, it supports Power BI integration. (Data World Docs)	**Medium** – has dashboards/scorecards, but broader Power BI strategy likely requires exports. (SAP)
9) Start small (few checks) with attractive entry cost	**Medium** – typically enterprise licensing/implementation.	**Medium** – enterprise platform; Secure Agent + services can be heavier.	**Strong** – Soda Core is free/open-source; scale into Soda Cloud/Agent as needed. (docs.soda.io)	**Strong** — GX Core is open source; GX Cloud has a free Developer option (data-asset based) and tiers based on # assets under test. (Great Expectations)	**Medium/Weak** – SAP tool (licensing/stack dependency); also lifecycle signal. (community.sap.com)
10) Strategic risk / roadmap	**Low/Medium** – active platform evolution (verify procurement/roadmap).	**Low/Medium** – major vendor; broad cloud platform.	**Low/Medium** – solid modern approach; depends on your appetite for "DQ as code" model.	**Low/Medium** — positioned as the "open-source standard" for data quality testing, which typically reduces lock-in risk. (Great Expectations)	**Higher** – SAP indicates maintenance timelines (mainstream to 2026, CSM to 2028). (community.sap.com)

Figure 4.17.: Example of the comparative evaluation matrix used during the assessment of different data quality management tools

Design Decisions

The next step after the evaluation table is to invite the selected tool vendors or providers to a demonstration. During this demonstration, the providers are asked to show the performance of their tool against the defined criteria, but they can also demonstrate any specific benefits that they provide. The demonstration gives the selecting company a more detailed view of the capabilities of the tool when applied to its dataset and whether any shortcomings would prevent a positive decision. Furthermore, it allows the selecting company to ask questions directly about any open items. After a demonstration, the scoring can be adjusted and, finally, a decision made on whether to proceed with a provider into a PoC or even contract negotiations, or to inform them that they have been excluded from the selection.

Alternative Outcome

The previous subsection assumes that a tool can be found for the defined use case. However, in case such a tool cannot be found, alternative ways need to be followed. In this work, the focus is on the solution; the why question will therefore also be excluded from the solution architecture. In case no solution is found:

1. **Reduce and adapt:** If the full use case cannot be satisfied by the tools at hand, the original use case can be adapted. The implementation can be adapted to a smaller use case while waiting for a suitable technology.
2. **Reconsider the discarded:** The previously discarded solutions or approaches can be reconsidered. Engage in discussions with the tools being developed internally. Review the tools which were excluded from selection to find out if they are fit for the use case of this project.
3. **Consult and discuss:** In large organisations, multiple initiatives can happen at the same time which are not seen on the team level. Therefore, engaging with a solution architect can open up the view towards other options. It allows identifying possible repurposing of existing tools which can fulfil the project's needs.

4.8. DQM tool testing

After the tool evaluation and selection, the testing should confirm whether it can execute the required tests in the following master thesis. Testing requires knowledge of how to use the tool and a simplification of the overall topic of the thesis.

Knowledge buildup

In order to understand the basic function of a tool, the required knowledge had to be acquired first. To do so, a variety of options are available. These are either based on documentation and how-to guides or direct knowledge transfer between developers who developed the tool or have already used it. In addition to the tool, additional technologies may be required to run tests, which in turn have to be studied as well. As, at this stage, the decision is pending, further details are not available as to what needs to be part of the knowledge buildup.

Tool testing

The tool had to be tested in order to confirm its applicability to the master thesis. The test of the application requires the use case to be broken down into a smaller test, with which the team can decide whether the tool can be used for further development. If such a tool were to fail the test, a return to the previous step and a continuation of the tool search would be necessary. In addition to the tool itself being tested, the setup required for it to run is equally part of the test. In the case of Swiss Post, this can include steps for permissions and infrastructure, which are commonly set up using Terraform.

Test design

The test design is important so that the selected case is both a step towards the complete use case and easier to execute. It must also include the infrastructure to be built for the use case. For the use case at hand, the test must prove that data can be queried with the tool, returning results that can be further processed. The processing of the data is, however, not required as part of the test design, as this may require multiple data points to be included in a check.

For this use case, which is the counting of multiple events per day, the count of one event is sufficient for a test. Therefore, the query at hand has to be reduced so that it only returns one event. However, even one event still requires identifying data, namely the date and the sorting centres. In addition to the idea of a final check, the test also has to be modular. First, the test will check whether any data are returned at all, then it will go deeper by checking the more specific event.

4.9. Iteration 1, Topic to be defined, TODO

Todo, Future Roland will have to write here... The Solution Landscape needs rework and eventual restructure However keep a lot of content... and add further content :=) etc...

4.10. Iteration 2, Topic to be defined, TODO

Todo, Future Roland will have to write here... The Solution Landscape needs rework and eventual restructure However keep a lot of content... and add further content :=) etc...

5. Implementation, Todo

Work for Future Roland, Todo Will have to write this chapter again... obviously because here we only write new work

This chapter uses the research findings and the solution landscape developed in the previous chapters as a basis for the implementation. With the defined SMART goals, the analysed reporting pipeline and data lineage of the previous research [Roccaro 2026b], being the basis for the successful completion of the SMART goals set in this project. The three goals are handled separately, as their completion is not dependent on each other. Those three goals are:

- To do the update of the existing data quality report, while using the new data products set up by the ASTV team, thereby also requiring the new reporting software Power BI and the new setup of data sources in star schema.
- Following up with the platform and data quality group regarding the tool decision for data quality management at Swiss Post in the future. This task started as the evaluation of tools, as the DQM platform described in the previous research [Roccaro 2026b] had shown that it would not be fit as a tool for the entire Swiss Post.
- As a preparation for the upcoming Master Thesis, the data quality tools to be used were checked initially. This included performing a concise test of counting data points with the defined monitoring setup.

The validation of the achieved results is performed in the next chapters.

5.1. Implementation Iteration1, Todo

Work for Future Roland, Todo Will have to write this chapter again... obviously because here we only write new work

5.1.1. Design

5.1.2. Implementation

5.1.3. Technical Validation

5.2. Implementation Iteration 2, Todo

Work for Future Roland, Todo Will have to write this chapter again... obviously because here we only write new work

5.2.1. Analysis of Iteration 1, Todo

Write what changed after iteration 1, What will Iteration 2 improve or research

5.2.2. Design

5.2.3. Implementation

5.2.4. Technical Validation

6. Results, Todo

Work for Future Roland, Todo Will have to write this chapter again... obviously because here we only write new work

This is the key chapter that reflects on the initial goals, the work carried out to achieve the goals, and the degree to which they were achieved. This chapter is structured around each goal, and special attention is paid to the key learnings that will be useful for subsequent phases of this work.

6.1. Introduction

The results presented here are part of a structured process that began step by step with the research question and led to the solution architecture and its implementation. Before delving further into the results of this project, a brief reminder of its scope helps to maintain focus. The project supports the DWH LS OPS with the goal of improving data availability and data quality management by replacing manual checks performed by the business with automated checks executed by IT. Through this change, data about these checks can become more consistent and enable the business to monitor them more easily. To achieve this, the existing reports need to be maintained to work with updated data sources, and the search for a new DQM tool and its validation for use in the upcoming master's thesis were major milestones.

6.2. Evaluation criteria Experimental Conditions, TODO

First, a result is defined as every piece of information that helps the reader become familiar with the context of the work. Second, it includes all information that supports understanding of the methods used during the project. Third, it comprises the main goal of this work, the SMART goals and their results. While the supporting results help to understand the environment, only the SMART goal results are the effectively relevant results of this work.

These SMART goals have been defined in the research chapter at 3.5:

- **1) Setup of an improved data quality report**
- **2) Evaluation and Selection of DQM platform for master thesis**
- **3) Validation of selected DQM tool**

6.3. SMART goal 1: Setup of an improved data quality report

Goal definition

1) Setup of an improved data quality report

WHY: This will help the team to run the daily checks faster

By the end of the implementation phase, the existing data quality report is to be updated to use the new infrastructure. This includes new reporting data pipelines and the new structure using star shema. Also, Power BI is to be used.

Success is measured by user feedback which has to be retrieved by ASTV team. The feedback is to be implemented and the report must be on the productive systems.

Core results

- **Update of data source:** In the implementation chapter, the data source for the third-party checks was moved from being stored on a shared drive to being stored on SharePoint, with synchronisation to Amazon S3. As a result, the data are versioned and available for other uses.
- **Power BI Setup:** At the beginning of the implementation chapter, the setup of Power BI was performed. It used the existing guides of Swiss Post and, in turn, allowed the development of the new improved DQM report to begin.
- **Model data setup:** The data setup carried out throughout the first section of the implementation chapter established a model that was extendable and allowed filters to be applied to all the data.
- **New visuals to support faster interpretation by stakeholders:** The new visuals developed allowed three things. They enabled a fast visual check that the data were correct across the four comparisons performed. They also allowed changes to be viewed over time using the line chart, and finally, the table containing the data allowed the detailed checks per day and location to be confirmed.
- **Scrum review and post-review feedback:** The Scrum review presented the report to the whole ASTV team and was a valuable means of obtaining feedback. Further feedback was gathered in separate meetings and subsequently applied, resulting in the final version of the report being brought into the production environment.
- **New Athena view and applied DevOps:** During the implementation of the report in Power BI, a separate view was set up in Athena. This new view allowed the new data sources to be used while the old view remained unchanged. This improved not only the performance of the view but also ensured that the two were not entangled. In addition, the DevOps deployment process at Swiss Post was described and applied for the deployment of this new view.

Supporting Results

The following results support the later implementation steps:

- In the solution landscape chapter, the definition of Power BI as the new reporting environment was established. Furthermore, the change from a wide table to a star schema was established.
- In the solution architecture, the metrics are reflected in the definitions of the relevant submitted and discharged events, as described in Sections 4.3 and 4.3.

What Was Not Achieved?

All objectives were fulfilled.

Goal Verdict

This goal was completely fulfilled. The report has been implemented, including the application of stakeholder feedback.

Learning and Outlook for Future Work

The importance of running checks between different environments from the first day of development became apparent. This was due to the issues that occurred with the different Power BI environments. Furthermore, it is important to obtain stakeholder feedback as early as possible and to plan enough time for implementing their requests.

6.4. SMART goal 2: Evaluation and Selection of DQM platform for master thesis

Goal definition

2) Evaluation and Selection of DQM platform for master thesis

WHY: To define the DQM platform for master thesis

During the implementation phase, the tool for the master thesis has to be defined. This requires a structured approach for the selection of the tool, including the setup of requirements lists. The team is to follow the DQM team during the evaluation of tools.

Success is measured by having the DQM tool defined. The tool selected has to be discussed with the architect of DWH LS OPS team to make sure it fits the teams needs.

Core Results

- **Tool selection for Swiss Post:** The DQM team decided that the tool which was part of the evaluation in the previous project, using a rule-based approach, was not suitable for the whole of Swiss Post. Therefore, the evaluation of other tools was launched.
- **Demonstration and decision:** The DQM tool Soda was invited to an online demonstration at Swiss Post. During this demonstration, its key features were presented. The team's decision was that, based on the demonstrated features, the missing connection to SAP data was the main deciding factor for not continuing with Soda.
- **DQM platform testing:** Towards the end of this project's timeline, the DQM platform was presented as being ready for testing.
- **Discussion with solution architect:** During a meeting with the solution architect, an alternative to waiting for the tool decision and the evaluation of the rule-based DQM tool was discussed and agreed upon. The job monitoring that was set up during the AWS Modernisation initiative could be used for data monitoring. This led to the third SMART Goal.

Supporting Results

The supporting results, while important achievements, were not researched or generated by the project team. However, these results are an important step towards the goal at hand; therefore, they are still presented.

- **Requirements list:** The list of requirements established by the DQM team defined which features are required in a requirements list. This list defines which requirements are must-have requirements and which are soft criteria.
- **DQM tooling selection:** The DQM tooling selection was performed by the DQM team based on the researched tools and their performance against the requirements list. The three best-performing tools were selected for continued evaluation.

What Was Not Achieved?

- **Final tool selection:** The DQM tool for the whole of Swiss Post had not yet been selected by the end of this project.

Goal Verdict

This SMART Goal was achieved. However, as discussed in the text, not all of the results presented here were developed by the project team

Learning and Outlook for Future Work

This goal was important for the following goal and was successful in providing that decision. However, the goal, as it was set up, was out of scope regarding the project team's function and responsibilities, as the setting up of the requirements list and the selection of the tools were performed by the DQM team. Goals must take into account the responsibilities and functions at Swiss Post.

6.5. SMART goal 3: Validation of the Selected DQM Tool

Goal definition

3) Validation of the Selected DQM Tool

WHY: The DQM platform must be tested for applicability to the master thesis.

By the end of the implementation phase, the DQM tool has to be tested for applicability to the master thesis. It must be proven that the required infrastructure can be set up for a test. Furthermore, a simplified test has to be run to confirm that the tool can handle the data sources.

Success is measured by having the DQM tool tested. The tool must return the test values to prove that it can be used for the follow-up master thesis.

Core Results

- **Knowledge transfer sessions:** The knowledge transfer sessions took place through extensive documentation in Confluence. Additionally, the consultants who developed the solution provided online sessions during which the functioning of the monitoring tool was described.
- **Design of a measurable use case:** The setup of a measurable use case was a key step towards testing the tool. The test was set up based on the existing query and therefore only had to be changed to fit the requirements.

- **Testing and creating a view for testing:** The new view was designed to have a good response time. This required multiple redesigns of the query, which was ultimately shortened so that it only contained one event being checked for one sorting centre.
- **Creating new resources:** While testing the tool, new resources had to be created. These were created using Terraform scripts, which are the Swiss Post standard for managing infrastructure as code.
- **Code execution testing:** The code testing led to the discovery of an error in the tool and how it handled date conditions. This was reported to the AWS Modernisation team, which in turn fixed the bug. This led to the successful execution of the code and the retrieval of results by the monitoring software setup.

Supporting Results

- **DQM tool testing:** The basis for planning the test was described in the solution architecture. The main steps are knowledge buildup, tool testing, and test design.

What Was Not Achieved?

The goal was completely fulfilled; nothing was left unachieved.

Goal Verdict

This goal was successfully achieved using the given use case. Thus, the master thesis can use this tool as a basis for entering the topic.

Learning and Outlook for Future Work

An important lesson learned here is the importance of early meetings with the developers of new tools. In the case of AWS Modernisation, the developers were only available until before the end of this project's timeline, so meetings had to be reserved early. Also important was early testing, which led to the discovery of a bug that was in turn fixed by the consultants rather than having to be debugged and fixed later by the project team. Furthermore, the importance of allowing enough time for the setup of the infrastructure must be taken into account during planning.

6.6. Stakeholder Evaluation, How many errors, Todo

Work for Future Roland, Todo Will have to write this chapter again... obviously because here we only write new work

6.7. Comparative Evaluation, Different solutions, Todo

Work for Future Roland, Todo Will have to write this chapter again... obviously because here we only write new work

6.8. Practical Stakeholder Evaluation, ReviewInputs, Usability solutions, Todo

Work for Future Roland, Todo Will have to write this chapter again... obviously because here we only write new work

6.9. Summary

With this chapter, the SMART goals defined in the research chapter have been analysed for completion. Two goals were completely fulfilled, one partially. The partially fulfilled second goal was limited due to responsibility constraints related to personal and team responsibility. The goal with partial fulfilled results points towards the need for goals planed along the lines of responisbilities, in addition to realistic time frames. However, it is important to note that even tough one goal was not reached completely, the overall results of the research helped to provide a solid for subsequent master thesis.

7. Discussion, Todo

Work for Future Roland, Todo Will have to write this chapter again... obviously because here we only write new work

7.1. Interpretation of Results, Todo

Work for Future Roland, Todo Will have to write this chapter again... obviously because here we only write new work

7.2. Evaluation of Research Hypotheses, Todo

Work for Future Roland, Todo Will have to write this chapter again... obviously because here we only write new work

7.3. Answer to the Research Question, Todo

Work for Future Roland, Todo Will have to write this chapter again... obviously because here we only write new work

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Work for Future Roland, Todo Will have to write this chapter again... obviously because here we only write new work

7.5. Practical Implications , Todo

Work for Future Roland, Todo Will have to write this chapter again... obviously because here we only write new work

7.6. Generalisability of the Results , Todo

Work for Future Roland, Todo Will have to write this chapter again... obviously because here we only write new work

7.7. Threats to Validity and Limitations, Todo

Work for Future Roland, Todo Will have to write this chapter again... obviously because here we only write new work

8. Reflection and outlook, Todo

Todo Work for future Roland Chapter kept but content has to be reworked

This reflection chapter concludes the second step in the epic quest of the academic team's towards improving data quality and data availability at Swiss Post. It evaluates the project with regard to methodology, content, and personal learnings, and concludes with an outlook on how the project can be followed up.



Figure 8.1.: Reflections with the lords of DQM

8.1. Summary of Findings, Todo

Todo Work for future Roland Chapter kept but content has to be reworked

8.1. Summary of Findings, X 8.2. Scientific Contribution , X 8.3. Practical Contribution , X 8.4. Future Work , X

8.2. Scientific Contribution, Todo

Todo Work for future Roland Chapter kept but content has to be reworked

8.3. Practical Contribution, Todo

Todo Work for future Roland Chapter kept but content has to be reworked

8.4. Personal Reflection , Todo

Todo Work for future Roland Chapter kept but content has to be reworked

8.5. Future Work, Todo

Todo Work for future Roland Chapter kept but content has to be reworked

9. Statement of Authorship

The author hereby declares that this work is the result of their own independent effort and that no sources other than those listed in the bibliography and identified as references have been used. Glossary entries have been partially written by the author using public descriptions (e.g., product homepages), copied from Wikipedia, or generated by ChatGPT based on provided bullet points. Further details regarding the use of LLM is decribed within the project management chapter.

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Glossary

Amazon Athena Amazon Athena runs interactive queries directly against data in Amazon S3. 52, 59

Amazon S3 Amazon Simple Storage Service (S3) is a service offered by Amazon Web Services (AWS) that provides object storage through a web service interface. Amazon S3 uses the same scalable storage infrastructure that Amazon.com uses to run its e-commerce network. 52, 68

Analytical Reporting Analytical Reporting is focusing on interpreting data. To be able to make management decisions over multiple days, weeks or months. It is fundamentally different to operational reporting which monitors the current status e.g. to check if machines are operating without errors. 1–3, 12

ASTV An abbreviation used at Swiss Post within the postal services. It is an acronym for the German terms: "Aufgabe" –> Receiving, "Sortierung" –> Sorting, "Transport" –> Transport, "Verzollung" –> Customs clearance. This abbreviation encompasses the entire postal process, except for the delivery step. 9, 12, 38, 43, 68

ASTVZ An abbreviation used at Swiss Post within the postal services. It is an acronym for the German terms: "Aufgabe" → Receiving, "Sortierung" → Sorting, "Transport" → Transport, "Verzollung" → Customs clearance, and "Zustellung" → Delivery. This abbreviation encompasses the entire postal process, from receiving a mail piece until it is delivered. 3

AWS Glue AWS Glue is a serverless data integration service that makes it easy for analytics users to discover, prepare, move, and integrate data from multiple sources. You can use it for analytics, machine learning, and application development. It also includes additional productivity and data ops tooling for authoring, running jobs, and implementing business workflows. https://docs.aws.amazon.com/glue/latest/is-glue.html. 13

AWS Glue Data Quality AWS Glue Data Quality allows you to measure and monitor the quality of your data so that you can make good business decisions. Built on top of the open-source DeeQu framework, AWS Glue Data Quality provides a managed, serverless experience. AWS Glue Data Quality works with Data Quality Definition Language (DQDL), which is a domain specific language that you use to define data quality rules. 45

BizDevOps BizDevOps extends DevOps by including the business side. In this constellation, the business becomes part of the team. This setup improves communication between developers and the people requesting the work. 3, 9, 12, 17

Business analyst A business analyst (BA) is a person who processes, interprets and documents business processes, products, services and software through analysis of data. The role of a business analyst is to ensure business efficiency increases through their knowledge of both IT and business function. 12

Business intelligence Business intelligence (BI) consists of strategies, methodologies, and technologies used by enterprises for data analysis and management of business information to inform business strategies and business operations. Common functions of BI technologies include reporting, online analytical processing, analytics, dashboard development, data mining, process mining, com-

plex event processing, business performance management, benchmarking, text mining, predictive analytics, and prescriptive analytics. 12

Curated database layer The curated database layer is the first layer in the Swiss Post DWH setup in which data is cleaned and standardised and duplicates are removed. This occurs after the data is received in a raw format. 50

D & F, Delivery and Finance The Delivery and Finance team is a BizDevOps team in DWH LS OPS handling requirements related to delivery and finance, including regional controlling and process controlling. 12

Data Availability (DA) Data availability, in the context of this document, refers to whether the data is accessible e.g. whether a report can be retrieved and used when needed. 1, 7, 8, 38, 67

Data Quality (DQ) Data quality refers to the state of qualitative or quantitative pieces of information. There are many definitions of data quality, but data is generally considered high quality if it is "fit for [its] intended uses in operations, decision making and planning". Data is deemed of high quality if it correctly represents the real-world construct to which it refers. Apart from these definitions, as the number of data sources increases, the question of internal data consistency becomes significant, regardless of fitness for use for any particular external purpose. 1, 7, 8, 38, 67

Data Quality Management (DQM) Data quality management is the management of data quality topics. It describes how data quality is handled in a company. However, in this document, most references to DQM refer to the IT team working at Swiss Post on the DQM topic. This team is implementing defined software that is intended to promote improvements in data quality. 12, 13, 15, 18, 43, 45, 46, 60, 67, 68

DevOps DevOps is the integration and automation of software development and information technology operations. DevOps encompasses the tasks necessary for software development and can lead to both shortening development time and improving the development life cycle. 58

DSP, SAP Datasphere DSP (SAP Datasphere) enables the creation of data models that integrate SAP and non-SAP data sources. These sources can include databases as well as Excel files. The data sources can then be further joined and aggregated. 52, 54–56

DWH In computing, a data warehouse (DW or DWH), also known as an enterprise data warehouse (EDW), is a system used for reporting and data analysis and is a core component of business intelligence. Data warehouses are central repositories of data integrated from disparate sources. 12, 50

DWH LS OPS A Swiss post abreviation, DWH -> "Data Warehouse", LS -> "Logistic Services", OPS -> "Operations". Together, this refers to the Data Warehouse that supports and manages the operational processes of the postal logistics services. 3, 8, 9, 12, 17, 38, 43, 56, 67, 69

Epic Epics are large bodies of work in Agile that are broken down into smaller user stories for incremental delivery. They help teams organize, prioritize, and track progress toward bigger goals while maintaining agility. 19

ETL Extract, transform, load (ETL) is a three-phase computing process where data is extracted from an input source, transformed (including cleaning), and loaded into an output data container. 9, 13

Greenfield A greenfield refers to a system landscape where no prior system exists. In IT, a greenfield project means that a new system is built from scratch, without having to consider or integrate with any existing legacy systems. Because there is no predecessor system, development can proceed more freely, without compatibility constraints — which often makes the implementation simpler and more flexible compared to a migration or redesign of an existing system (a brownfield). 7

High Level High level refers to viewing a use case from a general or abstract perspective, rather than analyzing all details. It focuses on the overall picture instead of individual elements. Instead of examining every single event during a day, a high-level analysis would simply check whether most events fall within the expected range during that day. 8

IaC Infrastructure as Code (IaC) is the process of managing and provisioning computer data center resources through machine-readable definition files, rather than via physical hardware configuration or interactive configuration tools. 59

Infoblock Infoblock (at Swiss Post in the context of AWS data storage) An Infoblock is a standardized data product that is provided within the data architecture of Swiss Post for reporting and analysis purposes. It serves as a logical unit of processed and quality-assured data that is extracted from different source systems, transformed, and stored in a consistent structure in the AWS cloud. This concept is similar to data products as defined in modern data management practices. 54

Jour Fixe A Jour Fixe is a weekly meeting used in project management to provide updates on progress, issues encountered during work, and problems that remain unresolved and require assistance. It serves to maintain communication and alignment with stakeholders. 18

Kafka Topics A Kafka topic is a core component of Apache Kafka. It is a named storage structure where producers write messages or data streams. The content of the topic can then be read and processed by consumers. 9

KPI, key performance indicator A performance indicator or key performance indicator (KPI) is a type of performance measurement. KPIs evaluate the success of an organization or of a particular activity (such as projects, programs, products and other initiatives) in which it engages. KPIs provide a focus for strategic and operational improvement, create an analytical basis for decision making and help focus attention on what matters most. 13

LS, Logistics Services Logistics Services (LS) is responsible for efficient and secure logistics of letters, printed matter and newspapers, as well as parcels, goods and merchandise in Switzerland and internationally. Logistics Services is the largest business unit of Swiss Post and makes a substantial contribution to the company's success with over 21 000 employees. 12

MIAR MIAR is the internal reporting and project effort tracking system used at Swiss Post. It is integrated with applications for working time recording, such as check-in and check-out. It allows employees to allocate worked hours to projects, which in turn enables monitoring of project efforts. 30

Microsoft Power BI (PBI) Microsoft Power BI (PBI) is an interactive data visualization software product developed by Microsoft with a primary focus on business intelligence (BI). It is part of the Microsoft Power Platform. Power BI is a collection of software services, apps, and connectors that work together to turn various sources of data into static and interactive data visualizations. 65

Microsoft Power BI (PBI) Microsoft Power BI (PBI) is an interactive data visualization, database management, and data modeling software product developed by Microsoft with a primary focus on business intelligence (BI). 56

MPP Mengenplanung und Prognose Pakete MPP Parcel Volume Planning and Forecasting is a team within Swiss Post. It is responsible for improving current planning and forecasting methods. Based on this information, important decisions are made, such as determining the personnel required for peak periods like the Christmas season with increased parcel volumes. The team analyses current data sources and aims to provide more precise methods and data sources for the process. 12

MVP A Minimal Viable Product (MVP) is the simplest version of a system that successfully fulfils a specific use case. It is developed with just the core functionality needed to provide value to users. An MVP can serve two main purposes: to demonstrate that a concept or method is feasible (proof of concept), or as an initial implementation that will later be improved through further development and updates. 7, 43

OLAP In computing, online analytical processing (OLAP) , is an approach to quickly answer multi-dimensional analytical (MDA) queries. The term OLAP was created as a slight modification of the traditional database term online transaction processing (OLTP). OLAP is part of the broader category of business intelligence, which also encompasses relational databases, report writing and data mining. 56

One to One Migrations In the context of reporting, a one-to-one migration refers to rebuilding an existing report exactly as it is, but on a different software or hardware platform. The structure, logic, and content of the report remain unchanged only the underlying technology is replaced. 7

Parcel center A parcel center is a large sorting center for parcels at Swiss Post. Each region has one parcel center, which serves as the main hub for parcel sorting. Every parcel is first gathered at a parcel center and later distributed to smaller regional parcel centers if its delivery is not scheduled for the immediate vicinity of the center. 46

Post day The post day starts at 06:00 AM and lasts until 05:59 AM the next day. This is the essential date format used by Swiss Post for reporting. 54, 56

Product Owner (PO) Each scrum team has one product owner. The product owner focuses on the business side of product development and spends the majority of time liaising with stakeholders and the team. The role is intended to primarily represent the product's stakeholders, the voice of the customer, or the desires of a committee, and bears responsibility for the delivery of business results. 38

Proof of concept (POC) Proof of concept (POC), also known as proof of principle, is a realization of a certain method or idea in order to demonstrate its feasibility, or a demonstration in principle with the aim of verifying that some concept or theory has practical potential. A proof of concept is usually small and may or may not be complete. 62

PyDeequ PyDeequ is a Python API for Deequ, a library built on top of Apache Spark for defining "unit tests for data", which measure data quality in large datasets. PyDeequ is written to support usage of Deequ in Python. 45

Region Swiss Post organises its postal services into four regions covering the whole of Switzerland. These regions are West, Middle, East, and South. The regions are not based on cantonal borders and may split cantons between regions. 46

Regional parcel center A regional parcel center is a smaller sorting center for parcels at Swiss Post. Multiple regional parcel centers are located within one region and handle parcels for smaller geographic areas. 46

Risk Appetite Risk appetite is a term used in risk management. It describes an individual's perception and tolerance of risk. Even when two people are facing the same risk, they may perceive its severity and probability differently, based on their personal risk appetite. 31, 33

Rollbox (RX) The rollbox (RX) is used to temporarily transport parcels throughout the Swiss Post workflow. It can hold many mail pieces and is used to move them between different processing areas, making it an essential component of parcel handling at Swiss Post. 47

SAC, SAP Analytics Cloud SAC, SAP Analytics Cloud is a plattform for online analytical reporting. It allows to show data in tables and dashboards and allows Reporting and predictive analysis. 52, 54–57

Scrum Scrum is an agile team collaboration framework commonly used in software development and other industries. Scrum prescribes for teams to break work into goals to be completed within timeboxed iterations, called sprints. 17

Smart Goals S.M.A.R.T. (or SMART) is an acronym used as a mnemonic device to establish criteria for effective goal-setting and objective development. The S.M.A.R.T. criteria he proposes are as follows: Specific: Targeting a particular area for improvement, Measurable: Quantifying, or at least suggesting, an indicator of progress, Assignable: Defining responsibility clearly, Realistic: Outlining attainable results with available resources, Time-related: Including a timeline for expected results. 38

Soft Criteria Non mandatory evaluation criteria that do not determine whether a solution is technically suitable, but improve usability, maintainability, or the overall user experience. Examples include intuitive user interfaces, simplified administration, or easy management of user roles and permissions. 61

Sorting Center In the postal services of Swiss Post, sorting centers are an essential part of the logistics process for handling letters and parcels. Items are first collected and sent to regional centralized facilities, where they are either: routed to sorting centers in other regions, or processed locally and dispatched to the distribution network of the current region. These centers enable efficient processing and routing of mail within Switzerland and across regions. 4

Star Schema A dimensional data modelling approach following the Kimball methodology, consisting of a central fact table linked to multiple denormalised dimension tables. The denormalisation is performed to improve the performance of analytical queries. It is widely used in data warehouses and business intelligence systems. 65

Terraform An open-source infrastructure as code (IaC) tool developed by HashiCorp that enables infrastructure to be defined and managed using declarative configuration files. Terraform automates the provisioning, modification, and removal of cloud resources while maintaining the desired infrastructure state. It supports multiple cloud providers and promotes consistent, version-controlled infrastructure deployments. 63

Wide table A wide table is a database table structure characterized by a large number of columns, often dozens or hundreds, storing multiple attributes about each record in a single row. In time-series databases, wide tables are commonly used for storing sensor data, financial market data, and other scenarios where many metrics need to be captured at each timestamp. 57

WYSIWYG (what you see is what you get) In computing, WYSIWYG (what you see is what you get) is software that allows content to be edited in a form that resembles its appearance when printed or displayed as a finished product, such as a printed document, web page, or slide presentation. WYSIWYG implies a user interface that allows the user to view something very similar to the result while the document is being created. 27

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A. Amazon Athena Queries

This appendix provides the Amazon Athena queries. They are stored here since putting them into the document would disrupt normal flow.

A.1. Full DQM view query

The full query which creates the view beeing main data source of the DQM report.

```
CREATE OR REPLACE VIEW "sendungsverarbeitung_dqm_v" AS
WITH
key_to_plz AS (
    SELECT 'VET' AS locationKey, '196000' AS plz
    UNION ALL SELECT 'HRK', '462150'
    UNION ALL SELECT 'CAD', '659100'
    UNION ALL SELECT 'WAL', '830000'
    UNION ALL SELECT 'HAE', '462000'
    UNION ALL SELECT 'UTZ', '720000'
    UNION ALL SELECT 'PRA', '413400'
    UNION ALL SELECT 'FRA', '852000'
    UNION ALL SELECT 'DAI', '131000'
    UNION ALL SELECT 'URD', '892000'
    UNION ALL SELECT 'BCH', '503000'
    UNION ALL SELECT 'RUE', '815000'
    UNION ALL SELECT 'OMU', '307100'
),

plz_to_key AS (
    SELECT '196000' AS plz, 'VET' AS locationKey
    UNION ALL SELECT '462150', 'HRK'
    UNION ALL SELECT '659100', 'CAD'
    UNION ALL SELECT '830470', 'WAL'
    UNION ALL SELECT '830000', 'WAL'
    UNION ALL SELECT '462000', 'HAE'
    UNION ALL SELECT '720000', 'UTZ'
    UNION ALL SELECT '413400', 'PRA'
    UNION ALL SELECT '852000', 'FRA'
    UNION ALL SELECT '131000', 'DAI'
    UNION ALL SELECT '892000', 'URD'
    UNION ALL SELECT '503000', 'BCH'
    UNION ALL SELECT '815000', 'RUE'
    UNION ALL SELECT '307100', 'OMU'
),

mp_discharged_raw AS (
    SELECT
        CAST(
            DATE_ADD(
```

```
                'hour', -6,
                discharged.timestamp AT TIME ZONE 'Europe/Zurich'
            ) AS DATE
        ) AS Posttag,
        discharged.sortingunit.locationKey AS locationKey,
        discharged.sourceId AS sourceId
    FROM curated_lsops.mailpiece_discharged
),

mp_discharged AS (
    SELECT
        Posttag,
        locationKey,
        CASE
            WHEN locationKey IN ('CAD','HRK','PRA','UTZ','VET','WAL')
                 AND sourceId NOT LIKE 'w%'
                 AND sourceId NOT LIKE 'W%'
                 AND TRIM(sourceId) <> '' THEN 1
            WHEN locationKey IN ('BCH','RUE','OMU') THEN 1
            WHEN locationKey IN ('HAE','FRA','DAI','URD') THEN 1
            ELSE 0
        END AS relevant_discharged
    FROM mp_discharged_raw
),

discharged_agg AS (
    SELECT
        Posttag,
        locationKey,
        SUM(relevant_discharged) AS num_discharged
    FROM mp_discharged
    GROUP BY 1, 2
),

mp_submitted_raw AS (
    SELECT
        CAST(
            DATE_ADD(
                'hour', -6,
                submitted.timestamp AT TIME ZONE 'Europe/Zurich'
            ) AS DATE
        ) AS Posttag,
        submitted.sortingunit.locationKey AS locationKey,
        submitted.sortingProcess AS sortingProcess
    FROM curated_lsops.mailpiece_submitted
),

mp_submitted AS (
    SELECT
        Posttag,
        locationKey,
        CASE
            WHEN locationKey IN ('HRK','VET','CAD','WAL','UTZ','PRA')
                 AND sortingProcess = 'AUTOMATIC' THEN 1
            WHEN locationKey IN ('BCH','RUE','OMU') THEN 1
            WHEN locationKey IN ('HAE','FRA','DAI','URD') THEN 1
            ELSE 0
```

```
        END AS relevant_submitted
    FROM mp_submitted_raw
),

submitted_agg AS (
    SELECT
        Posttag,
        locationKey,
        SUM(relevant_submitted) AS num_submitted
    FROM mp_submitted
    GROUP BY 1, 2
),


mp_coded_raw AS (
    SELECT
        coded.sortingCycleId AS sortingCycleId,
        CAST(
            DATE_ADD(
                'hour', -6,
                coded.timestamp AT TIME ZONE 'Europe/Zurich'
            ) AS DATE
        ) AS Posttag,
        coded.sortingunit.locationKey AS locationKey,
        coded.sortingProcess AS sortingProcess
    FROM curated_lsops.mailpiece_coded
),

mp_coded AS (
    SELECT DISTINCT
        sortingCycleId,
        Posttag,
        locationKey,
        CASE
            WHEN locationKey IN ('HRK','VET','CAD','WAL','UTZ','PRA')
                 AND sortingProcess = 'AUTOMATIC' THEN 1
            WHEN locationKey IN ('BCH','RUE','OMU') THEN 1
            WHEN locationKey IN ('HAE','FRA','DAI','URD') THEN 1
            ELSE 0
        END AS relevant_coded
    FROM mp_coded_raw
),

coded_agg AS (
    SELECT
        Posttag,
        locationKey,
        SUM(relevant_coded) AS num_coded
    FROM mp_coded
    GROUP BY 1, 2
),

mp_destinationAssigned_raw AS (
    SELECT
        destinationAssigned.sortingCycleId AS sortingCycleId,
        CAST(
            DATE_ADD(
```

```
                'hour', -6,
                destinationAssigned.timestamp AT TIME ZONE 'Europe/Zurich'
            ) AS DATE
        ) AS Posttag,
        destinationAssigned.sortingunit.locationKey AS locationKey,
        destinationAssigned.sortingParameters.sortingProcess AS sortingProcess
    FROM curated_lsops.mailpiece_destinationassigned
),

mp_destinationAssigned AS (
    SELECT DISTINCT
        sortingCycleId,
        Posttag,
        locationKey,
        CASE
            WHEN locationKey IN ('HRK','VET','CAD','WAL','UTZ','PRA')
                 AND sortingProcess = 'AUTOMATIC' THEN 1
            WHEN locationKey IN ('BCH','RUE','OMU') THEN 1
            WHEN locationKey IN ('HAE','FRA','DAI','URD') THEN 1
            ELSE 0
        END AS relevant_destinationAssigned
    FROM mp_destinationAssigned_raw
),

destinationassigned_agg AS (
    SELECT
        Posttag,
        locationKey,
        SUM(relevant_destinationAssigned) AS num_destinationassigned
    FROM mp_destinationAssigned
    GROUP BY 1, 2
),

sendungsverarbeitung_raw AS (
    SELECT
        sva.datum_sort.datum AS Posttag,
        plz_to_key.locationKey AS locationKey,
        sva.sortierung.sortierprozess_typ AS sortierprozess_typ
    FROM infoblock_lsops.sendungsverarbeitung sva
    LEFT JOIN plz_to_key
        ON plz_to_key.plz = sva.zentrum.zentrum_plz
),

sendungsverarbeitung AS (
    SELECT
        Posttag,
        locationKey,
        CASE
            WHEN locationKey IN ('HRK','VET','CAD','WAL','UTZ','PRA')
                 AND sortierprozess_typ = 'AUTOMATIC' THEN 1
            WHEN locationKey IN ('BCH','RUE','OMU') THEN 1
            WHEN locationKey IN ('HAE','FRA','DAI','URD') THEN 1
            ELSE 0
        END AS relevant_sva
    FROM sendungsverarbeitung_raw
),
```

```
sva_agg AS (
    SELECT
        Posttag,
        locationKey,
        SUM(relevant_sva) AS num_sva
    FROM sendungsverarbeitung
    GROUP BY 1, 2
),

device_stats AS (
    SELECT
        Posttag,
        sorting_unit_location_key AS locationKey,
        SUM(content_value) AS amount
    FROM "reporting_lsops"."anlagestatistik_curated_v"
    WHERE model = 'BEUMER'
      AND content_type = 'INDUCTION'
      AND content_value_index = 1
    GROUP BY 1, 2
)

SELECT
    submitted.Posttag,
    submitted.locationKey,
    key_to_plz.plz,
    discharged.num_discharged,
    submitted.num_submitted,
    coded.num_coded,
    destinationassigned.num_destinationassigned,
    sva.num_sva,
    dvs.amount AS num_stats
FROM submitted_agg submitted
INNER JOIN key_to_plz
    ON submitted.locationKey = key_to_plz.locationKey
INNER JOIN discharged_agg discharged
    ON submitted.Posttag = discharged.Posttag
   AND submitted.locationKey = discharged.locationKey
INNER JOIN coded_agg coded
    ON submitted.Posttag = coded.Posttag
   AND submitted.locationKey = coded.locationKey
INNER JOIN destinationassigned_agg destinationassigned
    ON submitted.Posttag = destinationassigned.Posttag
   AND submitted.locationKey = destinationassigned.locationKey
INNER JOIN sva_agg sva
    ON submitted.Posttag = sva.Posttag
   AND submitted.locationKey = sva.locationKey
LEFT JOIN device_stats dvs
    ON submitted.Posttag = dvs.Posttag
   AND submitted.locationKey = dvs.locationKey
WHERE submitted.locationKey IN (
    'HRK','VET','HAE','CAD','WAL','UTZ',
    'PRA','FRA','DAI','URD','BCH','RUE','OMU'
);
```

A.2. Compact DQM view query

The query after removing the parts which are not needed for the MVP.

```
WITH
key_to_plz AS (
    SELECT 'VET' AS locationKey, '196000' AS plz
    UNION ALL SELECT 'HRK', '462150'
    UNION ALL SELECT 'CAD', '659100'
    UNION ALL SELECT 'WAL', '830000'
    UNION ALL SELECT 'HAE', '462000'
    UNION ALL SELECT 'UTZ', '720000'
    UNION ALL SELECT 'PRA', '413400'
    UNION ALL SELECT 'FRA', '852000'
    UNION ALL SELECT 'DAI', '131000'
    UNION ALL SELECT 'URD', '892000'
    UNION ALL SELECT 'BCH', '503000'
    UNION ALL SELECT 'RUE', '815000'
    UNION ALL SELECT 'OMU', '307100'
),

plz_to_key AS (
    SELECT '196000' AS plz, 'VET' AS locationKey
    UNION ALL SELECT '462150', 'HRK'
    UNION ALL SELECT '659100', 'CAD'
    UNION ALL SELECT '830470', 'WAL'
    UNION ALL SELECT '830000', 'WAL'
    UNION ALL SELECT '462000', 'HAE'
    UNION ALL SELECT '720000', 'UTZ'
    UNION ALL SELECT '413400', 'PRA'
    UNION ALL SELECT '852000', 'FRA'
    UNION ALL SELECT '131000', 'DAI'
    UNION ALL SELECT '892000', 'URD'
    UNION ALL SELECT '503000', 'BCH'
    UNION ALL SELECT '815000', 'RUE'
    UNION ALL SELECT '307100', 'OMU'
),

mp_discharged_raw AS (
    SELECT
        CAST(
            DATE_ADD(
                'hour', -6,
                discharged.timestamp AT TIME ZONE 'Europe/Zurich'
            ) AS DATE
        ) AS Posttag,
        discharged.sortingunit.locationKey AS locationKey,
        discharged.sourceId AS sourceId
    FROM curated_lsops.mailpiece_discharged
),

mp_discharged AS (
    SELECT
        Posttag,
        locationKey,
        CASE
            WHEN locationKey IN ('CAD','HRK','PRA','UTZ','VET','WAL')
```

```
                AND sourceId NOT LIKE 'w%'
                AND sourceId NOT LIKE 'W%'
                AND TRIM(sourceId) <> '' THEN 1
            WHEN locationKey IN ('BCH','RUE','OMU') THEN 1
            WHEN locationKey IN ('HAE','FRA','DAI','URD') THEN 1
            ELSE 0
        END AS relevant_discharged
    FROM mp_discharged_raw
),

discharged_agg AS (
    SELECT
        Posttag,
        locationKey,
        SUM(relevant_discharged) AS num_discharged
    FROM mp_discharged
    GROUP BY 1, 2
),

mp_submitted_raw AS (
    SELECT
        CAST(
            DATE_ADD(
                'hour', -6,
                submitted.timestamp AT TIME ZONE 'Europe/Zurich'
            ) AS DATE
        ) AS Posttag,
        submitted.sortingunit.locationKey AS locationKey,
        submitted.sortingProcess AS sortingProcess
    FROM curated_lsops.mailpiece_submitted
),

mp_submitted AS (
    SELECT
        Posttag,
        locationKey,
        CASE
            WHEN locationKey IN ('HRK','VET','CAD','WAL','UTZ','PRA')
                AND sortingProcess = 'AUTOMATIC' THEN 1
            WHEN locationKey IN ('BCH','RUE','OMU') THEN 1
            WHEN locationKey IN ('HAE','FRA','DAI','URD') THEN 1
            ELSE 0
        END AS relevant_submitted
    FROM mp_submitted_raw
),

submitted_agg AS (
    SELECT
        Posttag,
        locationKey,
        SUM(relevant_submitted) AS num_submitted
    FROM mp_submitted
    GROUP BY 1, 2
)

SELECT
    submitted.Posttag,
```

```
    submitted.locationKey,
    key_to_plz.plz,
    discharged.num_discharged,
    submitted.num_submitted
FROM submitted_agg submitted
INNER JOIN key_to_plz
    ON submitted.locationKey = key_to_plz.locationKey
INNER JOIN discharged_agg discharged
    ON submitted.Posttag = discharged.Posttag
   AND submitted.locationKey = discharged.locationKey
WHERE submitted.locationKey IN (
    'HRK','VET','HAE','CAD','WAL','UTZ',
    'PRA','FRA','DAI','URD','BCH','RUE','OMU'
);
```

A.3. New Star Shema DQM view query

The for star shema DQM checks. Similar structure to the previous full query.

```
CREATE OR REPLACE VIEW "sendungsverarbeitung_dqm_starschema_v" AS
WITH
  key_to_plz AS (
   SELECT
     'VET' locationKey
   , '196000' plz

UNION ALL    SELECT
     'HRK' locationKey
   , '462150' plz

UNION ALL    SELECT
     'CAD' locationKey
   , '659100' plz

UNION ALL    SELECT
     'WAL' locationKey
   , '830000' plz

UNION ALL    SELECT
     'HAE' locationKey
   , '462000' plz

UNION ALL    SELECT
     'UTZ' locationKey
   , '720000' plz

UNION ALL    SELECT
     'PRA' locationKey
   , '413400' plz

UNION ALL    SELECT
     'FRA' locationKey
   , '852000' plz
```

```
UNION ALL    SELECT
     'DAI' locationKey
   , '131000' plz

UNION ALL    SELECT
     'URD' locationKey
   , '892000' plz

UNION ALL    SELECT
     'BCH' locationKey
   , '503000' plz

UNION ALL    SELECT
     'RUE' locationKey
   , '815000' plz

UNION ALL    SELECT
     'OMU' locationKey
   , '307100' plz

)
, plz_to_key AS (
   SELECT
     '196000' plz
   , 'VET' locationKey

UNION ALL    SELECT
     '462150' plz
   , 'HRK' locationKey

UNION ALL    SELECT
     '659100' plz
   , 'CAD' locationKey

UNION ALL    SELECT
     '830470' plz
   , 'WAL' locationKey

UNION ALL    SELECT
     '830000' plz
   , 'WAL' locationKey

UNION ALL    SELECT
     '462000' plz
   , 'HAE' locationKey

UNION ALL    SELECT
     '720000' plz
   , 'UTZ' locationKey

UNION ALL    SELECT
     '413400' plz
   , 'PRA' locationKey

UNION ALL    SELECT
     '852000' plz
   , 'FRA' locationKey
```

```
UNION ALL    SELECT
     '131000' plz
   , 'DAI' locationKey

UNION ALL    SELECT
     '892000' plz
   , 'URD' locationKey

UNION ALL    SELECT
     '503000' plz
   , 'BCH' locationKey

UNION ALL    SELECT
     '815000' plz
   , 'RUE' locationKey

UNION ALL    SELECT
     '307100' plz
   , 'OMU' locationKey

)
, mp_discharged_raw AS (
   SELECT
     CAST(DATE_ADD('hour', -6, discharged.timestamp AT TIME ZONE 'Europe/Zurich') AS DATE) ↩
    ↪ Posttag
   , discharged.sortingunit.locationKey locationKey
   , discharged.sourceId sourceId
   FROM
     curated_lsops.mailpiece_discharged_v2
   WHERE (1 = 1)
)
, mp_discharged AS (
   SELECT
     Posttag Posttag
   , locationKey locationKey
   , (CASE WHEN ((locationKey IN ('CAD', 'HRK', 'PRA', 'UTZ', 'VET', 'WAL')) AND (NOT (↩
    ↪ sourceId LIKE 'w%')) AND (NOT (sourceId LIKE 'W%')) AND (NOT (trim(BOTH FROM sourceId) ↩
    ↪ = ''))) THEN 1 WHEN (locationKey IN ('BCH', 'RUE', 'OMU')) THEN 1 WHEN (locationKey IN ↩
    ↪ ('HAE', 'FRA', 'DAI', 'URD')) THEN 1 ELSE 0 END) relevant_discharged
   FROM
     mp_discharged_raw
)
, discharged_agg AS (
   SELECT
     Posttag Posttag
   , locationKey locationKey
   , SUM(relevant_discharged) num_discharged
   FROM
     mp_discharged
   GROUP BY 1, 2
)
, mp_submitted_raw AS (
   SELECT
     CAST(DATE_ADD('hour', -6, submitted.timestamp AT TIME ZONE 'Europe/Zurich') AS DATE) ↩
    ↪ Posttag
   , submitted.sortingunit.locationKey locationKey
```

```
    , submitted.sortingProcess sortingProcess
    FROM
      curated_lsops.mailpiece_submitted_v2
)
, mp_submitted AS (
    SELECT
      Posttag Posttag
    , locationKey locationKey
    , (CASE WHEN ((locationKey = 'HRK') AND (sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((↩
     ↪ locationKey = 'VET') AND (sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = '↩
     ↪ CAD') AND (sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'WAL') AND (↩
     ↪ sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'UTZ') AND (sortingProcess =↩
     ↪  'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'PRA') AND (sortingProcess = 'AUTOMATIC')) ↩
     ↪ THEN 1 WHEN (locationKey IN ('BCH', 'RUE', 'OMU')) THEN 1 WHEN (locationKey IN ('HAE', ↩
     ↪ 'FRA', 'DAI', 'URD')) THEN 1 ELSE 0 END) relevant_submitted
    FROM
      mp_submitted_raw
)
, submitted_agg AS (
    SELECT
      Posttag Posttag
    , locationKey locationKey
    , SUM(relevant_submitted) num_submitted
    FROM
      mp_submitted
    GROUP BY 1, 2
)
, mp_coded_raw AS (
    SELECT
      coded.sortingCycleId sortingCycleId
    , CAST(DATE_ADD('hour', -6, coded.timestamp AT TIME ZONE 'Europe/Zurich') AS DATE) Posttag
    , coded.sortingunit.locationKey locationKey
    , coded.sortingProcess sortingProcess
    FROM
      curated_lsops.mailpiece_coded_v2
)
, mp_coded AS (
    SELECT DISTINCT
      sortingCycleId sortingCycleId
    , Posttag Posttag
    , locationKey locationKey
    , (CASE WHEN ((locationKey = 'HRK') AND (sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((↩
     ↪ locationKey = 'VET') AND (sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = '↩
     ↪ CAD') AND (sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'WAL') AND (↩
     ↪ sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'UTZ') AND (sortingProcess =↩
     ↪  'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'PRA') AND (sortingProcess = 'AUTOMATIC')) ↩
     ↪ THEN 1 WHEN (locationKey IN ('BCH', 'RUE', 'OMU')) THEN 1 WHEN (locationKey IN ('HAE', ↩
     ↪ 'FRA', 'DAI', 'URD')) THEN 1 ELSE 0 END) relevant_coded
    FROM
      mp_coded_raw
)
, coded_agg AS (
    SELECT
      Posttag Posttag
    , locationKey locationKey
    , SUM(relevant_coded) num_coded
    FROM
```

```
    mp_coded
  GROUP BY 1, 2
)
, mp_destinationAssigned_raw AS (
  SELECT
    destinationAssigned.sortingCycleId sortingCycleId
  , CAST(DATE_ADD('hour', -6, destinationAssigned.timestamp AT TIME ZONE 'Europe/Zurich') AS
   ↪ DATE) Posttag
  , destinationAssigned.sortingunit.locationKey locationKey
  , destinationAssigned.sortingParameters.sortingProcess sortingProcess
  FROM
    curated_lsops.mailpiece_destinationassigned_v2
)
, mp_destinationAssigned AS (
  SELECT DISTINCT
    sortingCycleId sortingCycleId
  , Posttag Posttag
  , locationKey locationKey
  , (CASE WHEN ((locationKey = 'HRK') AND (sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((
   ↪ locationKey = 'VET') AND (sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = '
   ↪ CAD') AND (sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'WAL') AND (
   ↪ sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'UTZ') AND (sortingProcess =
   ↪  'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'PRA') AND (sortingProcess = 'AUTOMATIC')) 
   ↪ THEN 1 WHEN (locationKey IN ('BCH', 'RUE', 'OMU')) THEN 1 WHEN (locationKey IN ('HAE', 
   ↪ 'FRA', 'DAI', 'URD')) THEN 1 ELSE 0 END) relevant_destinationAssigned
  FROM
    mp_destinationAssigned_raw
)
, destinationassigned_agg AS (
  SELECT
    Posttag Posttag
  , locationKey locationKey
  , SUM(relevant_destinationAssigned) num_destinationassigned
  FROM
    mp_destinationAssigned
  GROUP BY 1, 2
)
, sendungsverarbeitung_raw AS (
  SELECT
    post_day posttag
  , plz_to_key.locationKey locationKey
  , dim_sor.sortierprozess_typ sortierprozess_typ
  FROM
    (((reporting_lsops.fact_sendungsverarbeitung_v sndv
  LEFT JOIN reporting_lsops.dim_sortierung_v dim_sor ON (sndv.sortierung_hash_key = dim_sor.
   ↪ sortierung_hash_key))
  LEFT JOIN reporting_lsops.dim_zentrum_v dim_zent ON (sndv.zentrum_hash_key = dim_zent.
   ↪ zentrum_hash_key))
  LEFT JOIN plz_to_key ON (plz_to_key.plz = dim_zent.zentrum_plz))
)
, sendungsverarbeitung AS (
  SELECT
    Posttag Posttag
  , locationKey locationKey
  , (CASE WHEN ((locationKey = 'HRK') AND (sortierprozess_typ = 'AUTOMATIC')) THEN 1 WHEN ((
   ↪ locationKey = 'VET') AND (sortierprozess_typ = 'AUTOMATIC')) THEN 1 WHEN ((locationKey 
   ↪ = 'CAD') AND (sortierprozess_typ = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'WAL') AND
```

```
    ↪  (sortierprozess_typ = 'AUTOMATIC')) THEN 1 WHEN ((lo-cationKey = 'UTZ') AND (↩
    ↪ sortierprozess_typ = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'PRA') AND (sor-↩
    ↪ tierprozess_typ = 'AUTOMATIC')) THEN 1 WHEN (locationKey IN ('BCH', 'RUE', 'OMU')) THEN↩
    ↪  1 WHEN (locationKey IN ('HAE', 'FRA', 'DAI', 'URD')) THEN 1 ELSE 0 END) relevant_sva
   FROM
     sendungsverarbeitung_raw
)
, sva_agg AS (
   SELECT
     Posttag Posttag
   , locationKey locationKey
   , SUM(relevant_sva) num_sva
   FROM
     sendungsverarbeitung
   GROUP BY 1, 2
)
, device_stats AS (
   SELECT
     Posttag Posttag
   , sorting_unit_location_key locationKey
   , sum(content_value) amount
   FROM
     "reporting_lsops"."anlagestatistik_curated_v"
   WHERE ((1 = 1) AND ((1 = 0) OR ((model = 'BEUMER') AND (content_type = 'INDUCTION') AND (↩
    ↪ con-tent_value_index = 1))))
   GROUP BY 1, 2
)
SELECT
  submitted.Posttag Posttag
, submitted.locationKey locationKey
, key_to_plz.plz plz
, discharged.num_discharged num_discharged
, submitted.num_submitted num_submitted
, coded.num_coded num_coded
, destinationassigned.num_destinationassigned num_destinationassigned
, sva.num_sva num_sva
, dvs.amount num_stats
FROM
  ((((((submitted_agg submitted
INNER JOIN key_to_plz ON ((1 = 1) AND (submitted.locationKey = key_to_plz.locationKey)))
INNER JOIN discharged_agg discharged ON ((1 = 1) AND (submitted.Posttag = discharged.Posttag) ↩
    ↪ AND (submit-ted.locationKey = discharged.locationKey)))
INNER JOIN coded_agg coded ON ((1 = 1) AND (submitted.Posttag = coded.Posttag) AND (submitted.↩
    ↪ locationKey = coded.locationKey)))
INNER JOIN destinationassigned_agg destinationassigned ON ((1 = 1) AND (submitted.Posttag = ↩
    ↪ destina-tionassigned.Posttag) AND (submitted.locationKey = destinationassigned.↩
    ↪ locationKey)))
INNER JOIN sva_agg sva ON ((1 = 1) AND (submitted.Posttag = sva.Posttag) AND (submitted.↩
    ↪ locationKey = sva.locationKey)))
LEFT JOIN device_stats dvs ON ((1 = 1) AND (submitted.Posttag = dvs.Posttag) AND (submitted.↩
    ↪ locationKey = dvs.locationKey)))
WHERE ((1 = 1) AND (submitted.locationKey IN ('HRK', 'VET', 'HAE', 'CAD', 'WAL', 'UTZ', 'PRA',↩
    ↪  'FRA', 'DAI', 'URD', 'BCH', 'RUE', 'OMU')))
```

A.4. Testing Monitoring Query

To enable testing of the monitoring data counts, the DQM query had to be adapted. The following shows the final version of this query.

```
CREATE OR REPLACE VIEW "sendungsverarbeitung_dqm_starschema_no_aggregation_small_v" AS
WITH
  key_to_plz AS (
   SELECT
     'VET' locationKey
   , '196000' plz

UNION ALL    SELECT
     'HRK' locationKey
   , '462150' plz

UNION ALL    SELECT
     'CAD' locationKey
   , '659100' plz

UNION ALL    SELECT
     'WAL' locationKey
   , '830000' plz

UNION ALL    SELECT
     'HAE' locationKey
   , '462000' plz

UNION ALL    SELECT
     'UTZ' locationKey
   , '720000' plz

UNION ALL    SELECT
     'PRA' locationKey
   , '413400' plz

UNION ALL    SELECT
     'FRA' locationKey
   , '852000' plz

UNION ALL    SELECT
     'DAI' locationKey
   , '131000' plz

UNION ALL    SELECT
     'URD' locationKey
   , '892000' plz

UNION ALL    SELECT
     'BCH' locationKey
   , '503000' plz

UNION ALL    SELECT
     'RUE' locationKey
```

```
   , '815000' plz

UNION ALL    SELECT
     'OMU' locationKey
   , '307100' plz

)
, plz_to_key AS (
   SELECT
     '196000' plz
   , 'VET' locationKey

UNION ALL    SELECT
     '462150' plz
   , 'HRK' locationKey

UNION ALL    SELECT
     '659100' plz
   , 'CAD' locationKey

UNION ALL    SELECT
     '830470' plz
   , 'WAL' locationKey

UNION ALL    SELECT
     '830000' plz
   , 'WAL' locationKey

UNION ALL    SELECT
     '462000' plz
   , 'HAE' locationKey

UNION ALL    SELECT
     '720000' plz
   , 'UTZ' locationKey

UNION ALL    SELECT
     '413400' plz
   , 'PRA' locationKey

UNION ALL    SELECT
     '852000' plz
   , 'FRA' locationKey

UNION ALL    SELECT
     '131000' plz
   , 'DAI' locationKey

UNION ALL    SELECT
     '892000' plz
   , 'URD' locationKey

UNION ALL    SELECT
     '503000' plz
   , 'BCH' locationKey

UNION ALL    SELECT
```

```
     '815000' plz
   , 'RUE' locationKey

UNION ALL    SELECT
     '307100' plz
   , 'OMU' locationKey

)

, mp_submitted_raw AS (
   SELECT
     CAST(DATE_ADD('hour', -6, submitted.timestamp AT TIME ZONE 'Europe/Zurich') AS DATE) ↩
    ↪ Posttag
   , submitted.sortingunit.locationKey locationKey
   , submitted.sortingProcess sortingProcess
   FROM
     curated_lsops.mailpiece_submitted_v2
)
, mp_submitted AS (
   SELECT
     Posttag Posttag
   , locationKey locationKey
   , (CASE WHEN ((locationKey = 'HRK') AND (sortingProcess = 'AUTOMATIC')) THEN 1
   WHEN ((locationKey = 'VET') AND (sortingProcess = 'AUTOMATIC')) THEN 1
   WHEN ((locationKey = 'CAD') AND (sortingProcess = 'AUTOMATIC')) THEN 1
   WHEN ((locationKey = 'WAL') AND (sortingProcess = 'AUTOMATIC')) THEN 1
   WHEN ((locationKey = 'UTZ') AND (sortingProcess = 'AUTOMATIC')) THEN 1
   WHEN ((locationKey = 'PRA') AND (sortingProcess = 'AUTOMATIC')) THEN 1
   WHEN (locationKey IN ('BCH', 'RUE', 'OMU')) THEN 1
   WHEN (locationKey IN ('HAE', 'FRA', 'DAI', 'URD')) THEN 1
   ELSE 0 END) relevant_submitted
   FROM
     mp_submitted_raw
)

SELECT
  submitted.Posttag Posttag
, submitted.locationKey locationKey
, key_to_plz.plz plz
, submitted.relevant_submitted relevant_submitted

FROM
  (mp_submitted submitted
INNER JOIN key_to_plz ON ((1 = 1) AND (submitted.locationKey = key_to_plz.locationKey)))
WHERE ((1 = 1) AND (submitted.locationKey IN ('HRK', 'VET', 'HAE', 'CAD', 'WAL', 'UTZ', 'PRA',↩
    ↪  'FRA', 'DAI', 'URD', 'BCH', 'RUE', 'OMU')))
```

B. Miro Usage

As a reference, three screenshots of the Miro board at different stages of this project are provided. They show the increase in information as well as the restructuring of information blocks. Three observations can be made. First, the zoom level required to display the entire board on one screen had to be increased multiple times due to the growing number of meetings, Kanban cards, and other elements. Second, the implementation and document writing section in the bottom right grew considerably compared to the first screenshot. Third, new information was added continuously, which can be seen mainly in the bottom right. Finally, after these reference screenshots, some sections are highlighted to show readable parts of the work carried out on the board.

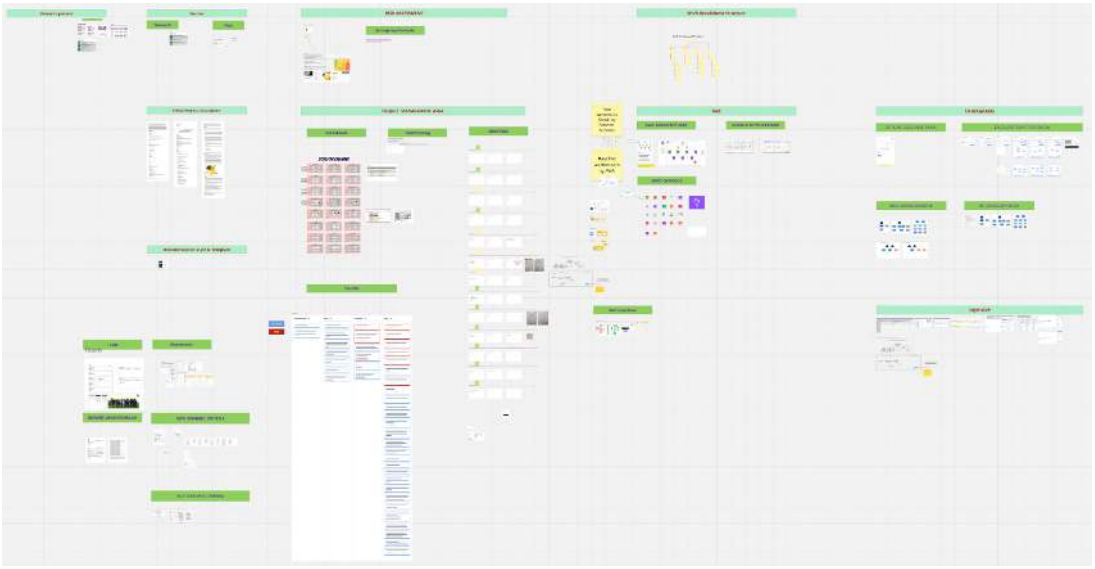


Figure B.1.: Miro board documentation, view of 10.11.2025

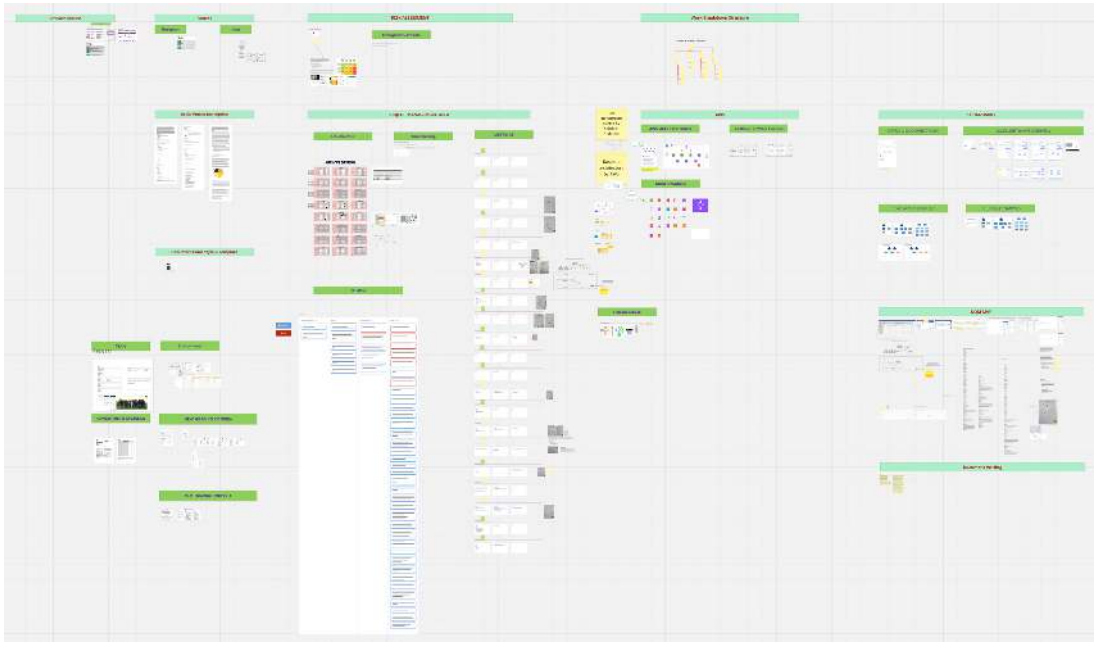


Figure B.2.: Miro board documentation, view of 25.11.2025

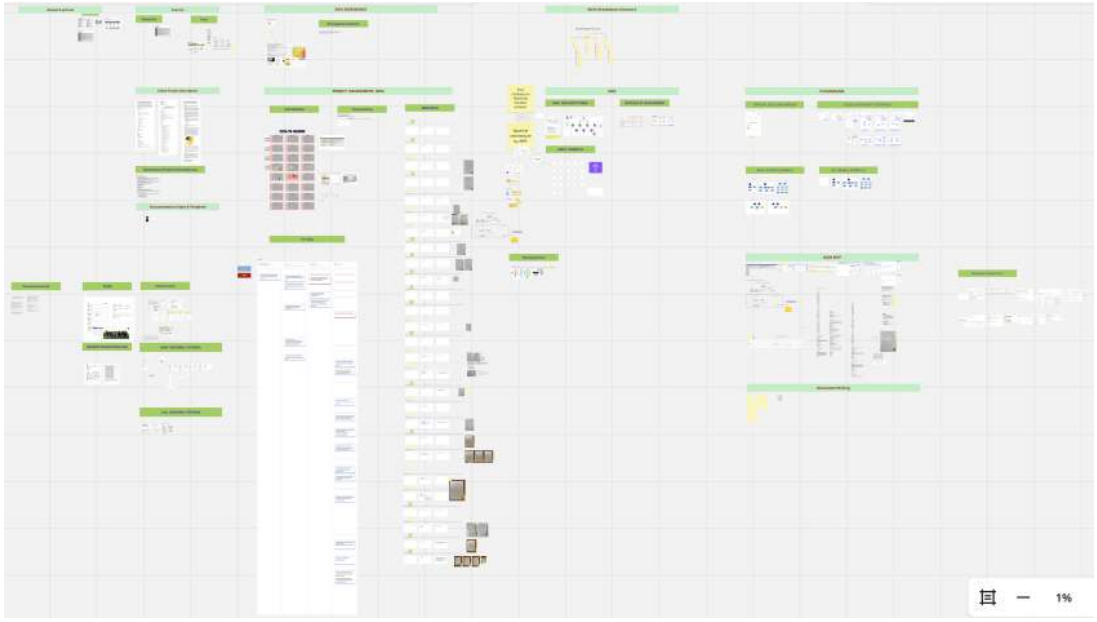


Figure B.3.: Miro board documentation, view of 23.12.2025

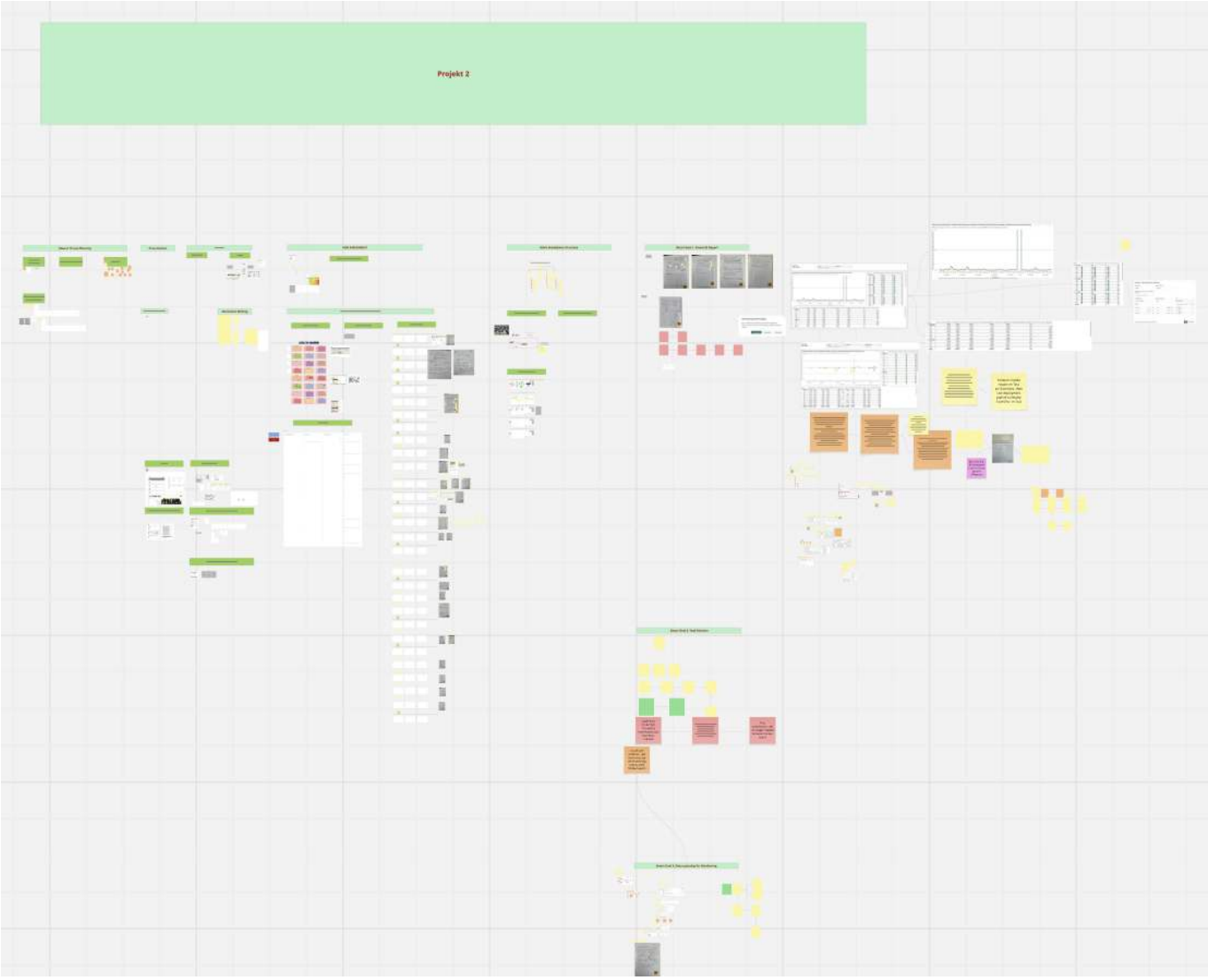


Figure B.4.: Complete Miro board documenting the evolution of the project. The board contains the complete collection of notes, diagrams, screenshots, milestones, implementation artefacts and planning information accumulated during the project. Screenshot taken on 24 July 2026.

B.1. Miro Details of SMART Goals

The following screenshots provide a high-level overview of the work carried out for the individual SMART Goals. These screenshots are intended only as a reference and are not meant to be readable in detail, but rather to provide a brief overview of the methods used.

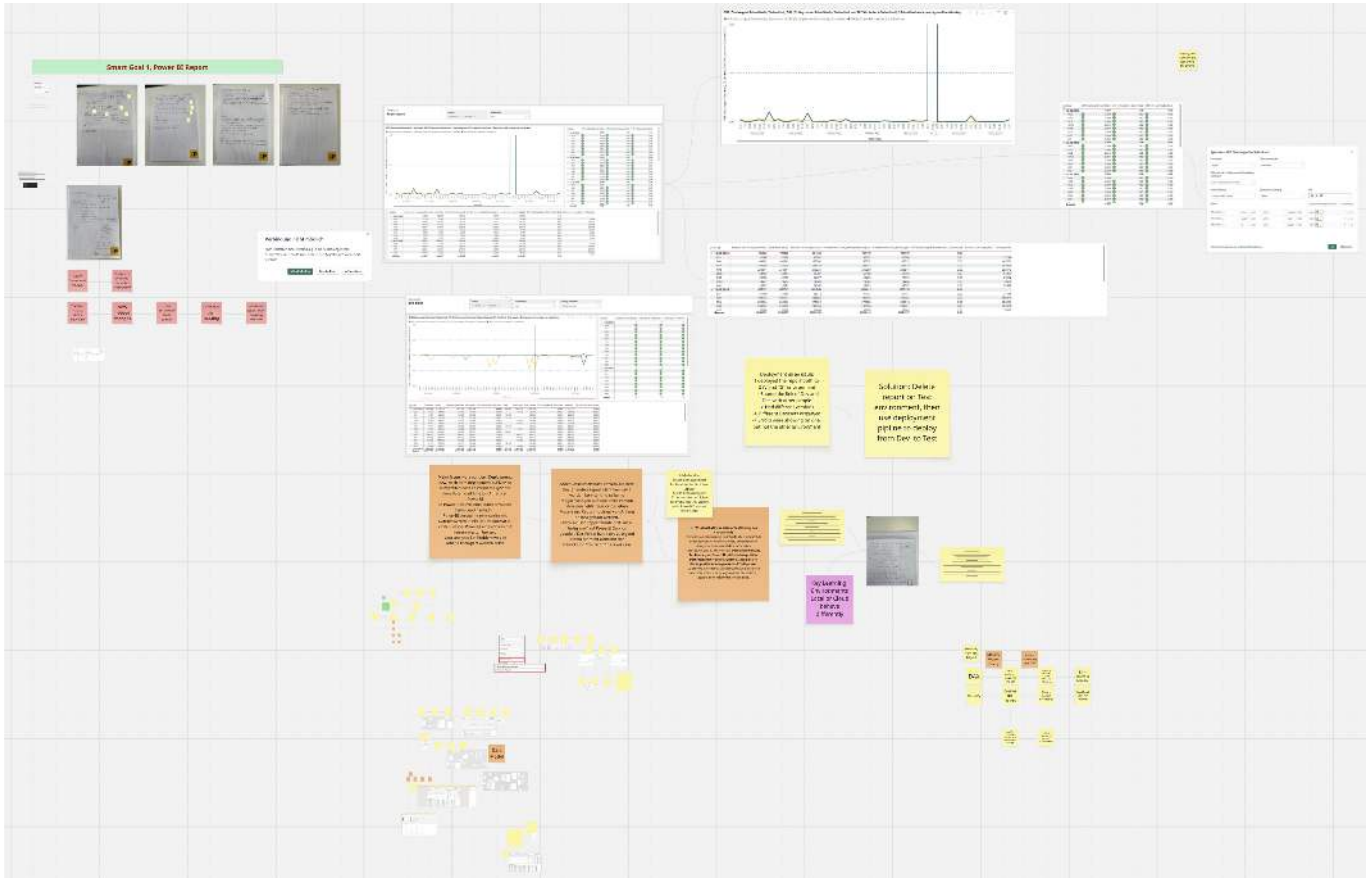


Figure B.5.: Extract of the Miro board related to Smart Goal 1. The board illustrates the planning and documentation of the reporting solution, including design concepts, implementation notes and supporting artefacts collected throughout the project.

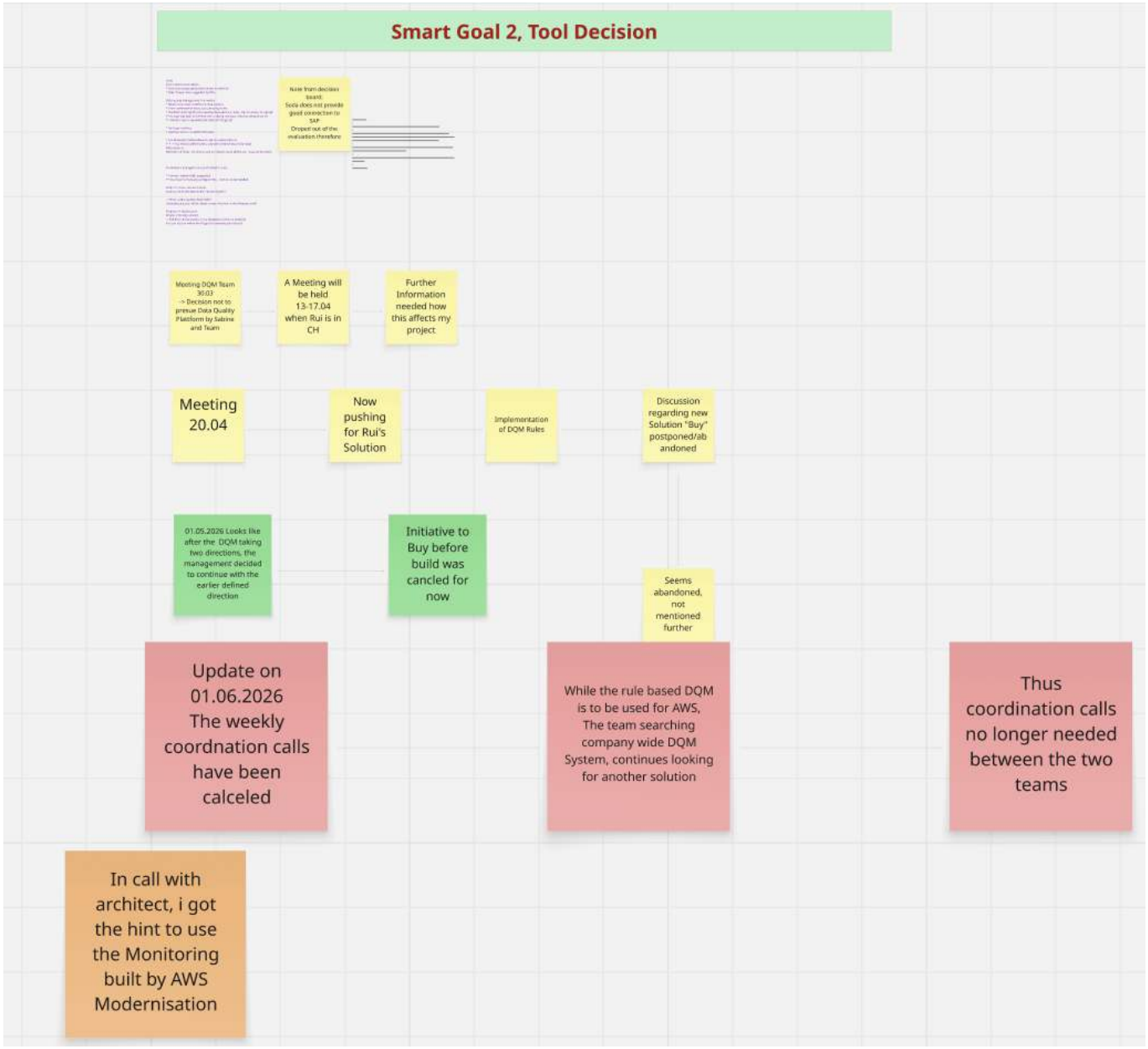


Figure B.6.: Extract of the Miro board related to Smart Goal 2. The material summarises the evaluation activities, comparisons, vendor discussions and supporting documentation used during the assessment of data quality management solutions.

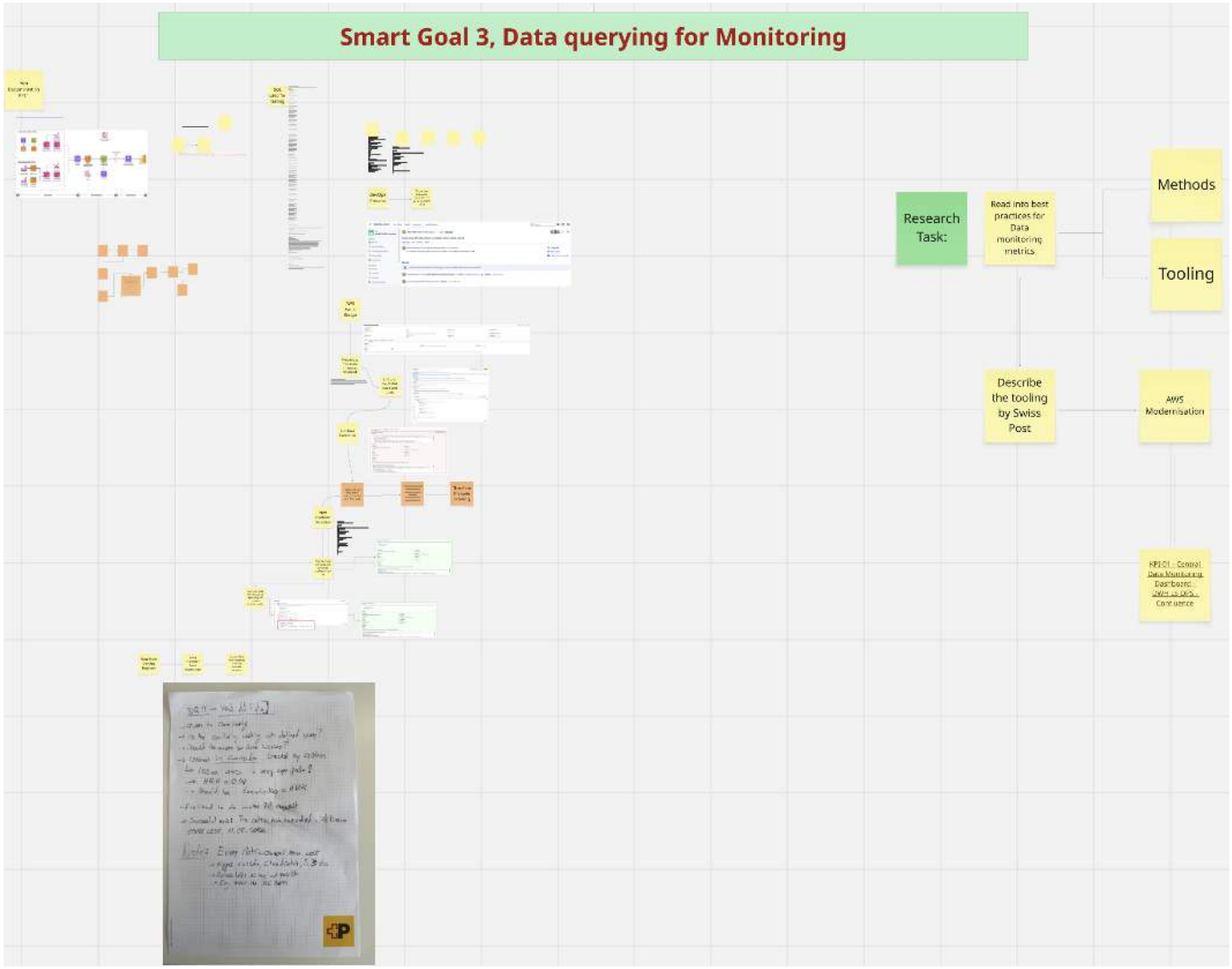


Figure B.7.: Extract of the Miro board related to Smart Goal 3. The board documents the collection of implementation notes, design ideas, intermediate results and decisions supporting the validation of the selected proof-of-concept.

C. Risk Monitoring

The risk Monitoring files are linked in the following tables.

July 2026

A final monitoring performed on 24.07.2026, concluded the writing phase on project 2.

Number	Risk	Occured	Detail	Why	Measures taken	Measure date
1	Unclear question	No			Measures taken	
2	Unclear goals	No		Team Members, PO in team		
3	Insufficient expertise	Yes	Latex usage to change document structure	Some parts of the used BFH template had to be changed due to amendments with tables and additional content	Used ChatGPT for review and changes	24.07.2026
4	Underestimated time/-effort	Yes	Efforts for writing were not committable before holiday	Took too much time in relation to needed cool down times	Moved writing partially to holiday	24.07.2026
5	Other study projects	Yes	No time invested	Focus on this project		24.07.2026
6	Work projects	No	In December much more work on the project was possible, no incidents	Full focus on new topics		

Number	Risk	Occured	Detail	Why	Measures taken	Measure date
7	Procrastination	No		Now time to finish	Working hard during beach holiday	24.07.2026
8	Unavailability of main project staff (e.g., illness)	No				
9	Unavailability of advisor (e.g., illness, accident)	No				
10	Software Version / Library / Dependency	No				
11	Hardware is damaged	No				
12	Data related	No				
13	Not reproducible work	No				
14	Plagiarism or wrong citation	No				
15	Copyright	No				
16	Bad communication with stakeholder or advisor	No				
17	Insufficient exchange with external partners	No				
18	Unforeseen familiar crisis	No				
19	General force majeure	No				
20	Unknown process	Yes	Finally after a lot of back and forth the invoice was handled through sending it via post mail	BFH could not sign the new Terms of Swiss Post	Next invoice to be sent by mail directly	01.07.2026

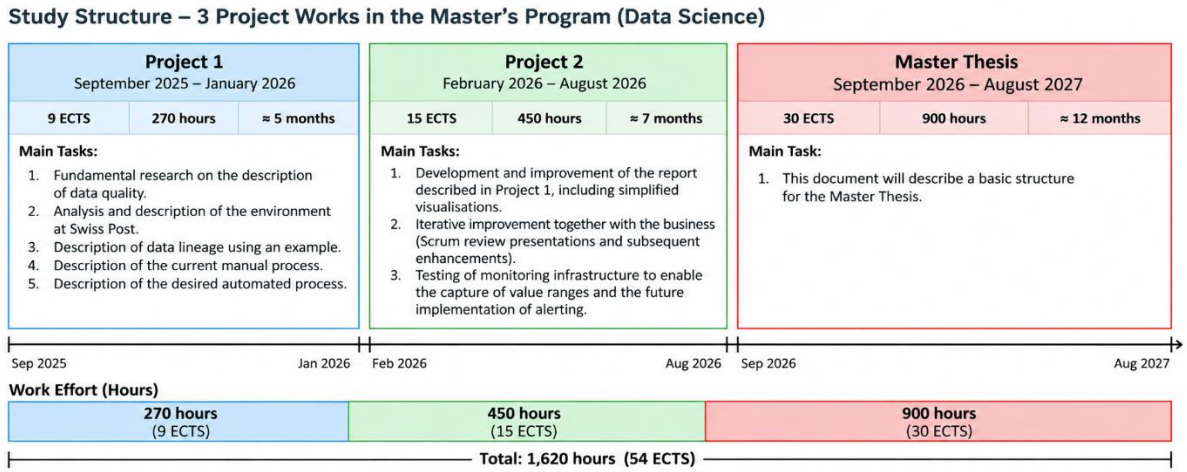
Master Thesis Outline Roland Roccaro

This document describes the outline of the Master Thesis of Roland Roccaro. This work is conducted with and for stakeholders at Swiss Post DWH LS OPS (Data warehouse Logistic Services Operations). The stakeholders define the standard reports to be used for all analytical reporting of all mail and parcel related workflows such as sorting, delivery etc.

It shows first the previous work, and then goes trough the research question, basic project management that will be applied. Following up with the current timeline throughout the Master Thesis using milestones and the outline of the chapters, which are partly based on the project 1 and 2.

Study structure in regard to project structure.

The study programme consists of three different project works. These projects do not necessarily have to build on each other.



- Project 1. September 2025 to January 2026 9 ECTS / 270 hours
  - Fundamental research on the description and concepts of data quality.
  - Analysis and description of the relevant environment at Swiss Post.
  - Description of the data lineage using an example.
  - Description of the current manual process.
  - Description of the desired automated process.
- Project 2. February 2026 to August 2026 15 ECTS / 450 hours
  - Development and improvement of the report described in Project 1, including the development of several simplified visualisations.
  - Iterative improvement in collaboration with business stakeholders, including Scrum Review presentations and subsequent improvements.
  - Testing of monitoring infrastructure to enable the collection of value ranges and support future alerting.
- Master Thesis. September 2026 to August 2027 30 ECTS / 900 hours

- This document describes the initial structure of the Master Thesis.

Research Question

Following research question is the core of this master thesis:

- To what extent can data quality in analytical reporting at DWH LS OPS be automatically validated at the reporting layer?

Hypotheses

Based on the research question the following hypotheses are defined.

- Relevant data quality errors in analytical reporting can be systematically identified, classified, and documented by the responsible stakeholders.

- An automated data quality validation approach can reliably detect the documented error classes at the analytical reporting layer.

Research Method

The research is carried out in phases.

- The project starts with an initial research phase, during which a baseline and direction are established.
- The results are documented as a baseline.
- At relevant phases or milestones during the project, additional research is performed to validate results.
- Furthermore, after completing the first iteration, further research is conducted to confirm topics for a second iteration.

Project Management Method

The following section describes the overall project management methods to be used during the master thesis. This does not mention all project management methods such as risk management.

- Overall Structure Based on Milestones
  - The milestones are used to structure the different stages of the research project, from the scientific foundation through implementation to the subsequent evaluation.
  - The milestones are defined at the beginning of the thesis.
  - They divide the work into blocks that provide an overall framework without predetermining the specific topics or approaches.

    - They also provide a basis for subsequently analysing and documenting the effort required for the individual work blocks.
- Weekly Meetings: Jour Fixe between Student and advisor
    - Ken as advisor and Roland as Student meet weekly for 1 hour
    - Topics of the meeting are:
        - Check-in, what is the projects topic as a refresh.
        - Check-in what deliverables are done.
        - Which deliverable am I working on?
        - Status of the deliverable
        - Next steps
        - Other
    - Upon reaching milestones or other specific dates, the topics can vary.
    - The expert has been invited to attend the meetings on an optional basis.
- Agile Project Management using Scrum.
    - Two-week sprints. Every uneven calendar week number sprints start and end.
    - The sprints are synchronised with those of the Swiss Post DWH LS OPS teams.
    - Sprint planning is conducted together with the relevant Swiss Post stakeholders.
    - The planning is reviewed and confirmed during the biweekly Jour Fixe meetings.
    - Scrum Reviews take place every four weeks within DWH LS OPS. These meetings can also be used to present and discuss the progress of the Master Thesis.
    - Retrospectives take place every two weeks within the DWH LS OPS teams. Within the academic team, retrospectives are planned upon completion of each milestone.

Phases / Milestones

The following phases define blocks which can be closed after completing each of them. The conclusion and start of each phase are used to review and adapt the current state.

The milestones are events during the course of the phases. Each milestone will be documented if it has been reached in time. Furthermore, any adaptations required to the milestone plans have to be documented.

- Phase 1, Research
    - Refine the problem statement.
    - Define and narrow the scope.
    - Define the use case.
        - **Milestone 1, use case defined.**
    - Conduct the literature review.
    - Define the basic outlines for P2 to P3 and specify possible alternatives.
    - **Closure conditions:**
        - Research based on problem statement initially closed.
        - Basis for P2 and P3 defined.
        - Methodology for thesis defined.
        - Review of baseline document parts from project 1 and 2 performed.
- Phase 2, Define the Baseline with Stakeholders

    - Adapt M2 based on the findings from P1.
    - Document the current stakeholder knowledge regarding error detection.
        - **Milestone 2, Stakeholder meetings held.**
    - Attempt to systematically identify and document relevant errors.
    - Define how the results will be evaluated against future iterations.
        - **Milestone 3, Error documentation by stakeholders started.**
    - **Closure Conditions**
        - Stakeholder knowledge documented
        - Stakeholders have means to document errors.
        - Defined test window during which manual and automatic errors are compared.
        - Error documentation setup so that it is comparable to future Phase 3 implementation.

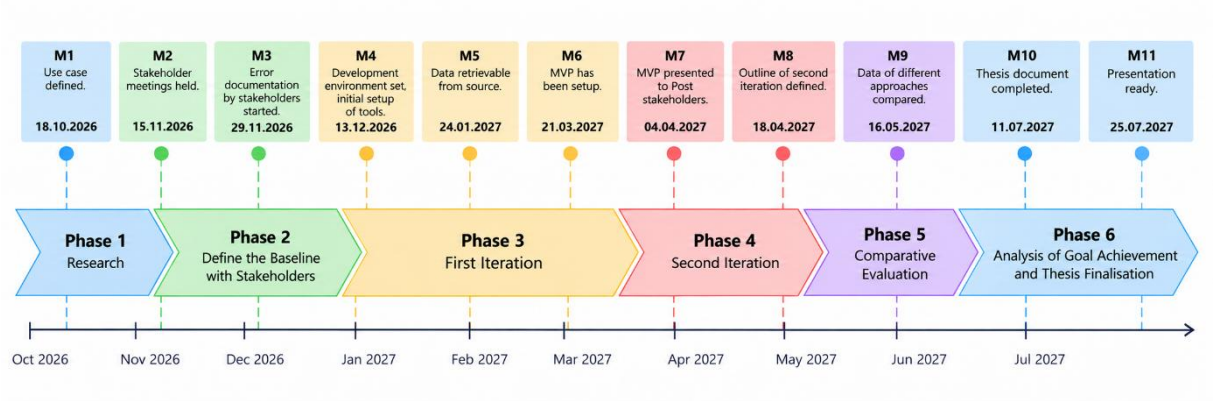
- Phase 3, First Iteration
    - Review the results of P2 and adapt P3 accordingly.
    - Define a minimal outline of the P3 objective while keeping the specific approach open to findings from the preceding research.
        - **Milestone 4, Development environment set, initial setup of tools**
    - Develop the data retrieval from AWS source.
        - **Milestone 5, Data retrievable from source**
    - Develop the first/basic implementation.
        - **Milestone 6, MVP has been set up.**
    - Summarise the results.
        - **Milestone 7, MVP presented to Post stakeholders.**
    - **Closure conditions**
        - The development environment is setup.
        - Permissions required to work have been setup
        - The MVP is setup and running.
        - Data for test window retrieved.
        - The Post stakeholders received presentation of current state.
        - Defined test slot and registered test results
- Phase 4, Second Iteration
    - Review the achievement of the objectives from P3 and, if necessary, pursue an alternative approach.
        - **Milestone 8, Outline of second iteration defined.**
    - Define a minimal outline of the phase objective while keeping the specific approach open to findings from the preceding research.
    - Develop an adapted implementation.
    - Summarise the results.
    - **Closure conditions**
        - Second iteration outline defined.
        - Data for test window retrieved.
- Phase 5, Comparative Evaluation
    - Compare the results of all investigated approaches.
        - **Milestone 9, Data of different approaches compared.**

    - Identify factors that may limit a direct or fair comparison, including specific differences between the approaches.
- Phase 6, Analysis of Goal Achievement and Thesis Finalisation
    - Describe which objectives were achieved and which were not.
    - Finalise the thesis document.
        - **Milestone 10, Thesis document completed.**
    - Finalise the required deliverables:
        - Video
        - Poster
        - Presentation and Defence
    - **Milestone 11, Presentation ready**
    - Thesis document delivered
    - Presentation held

Milestones Timeline

The milestone timelines are setup as follows before project start.

- Milestone 1, use case defined.
    - 18.10.2026
- Milestone 2, Stakeholder meetings held.
    - 15.11.2026
- Milestone 3, Error documentation by stakeholders started.
    - 29.11.2026
- Milestone 4, Development environment set, initial setup of tools
    - 13.12.2026
- Milestone 5, Data retrievable from source
    - 24.01.2027
- Milestone 6, MVP has been set up.
    - 21.03.2027
- Milestone 7, MVP presented to Post stakeholders.
    - 04.04.2027
- Milestone 8, Outline of second iteration defined.
    - 18.04.2027
- Milestone 9, Data of different approaches compared.
    - 16.05.2027
- Milestone 10, Thesis document completed.
    - 11.07.2027
- Milestone 11, Presentation ready
    - 25.07.2027



Chapter Structure

The chapter structure is a basic outline taken from projects 1 and 2 including additional topics for the master thesis such as the description of the milestones. Some existin content is to be [changed] or [renewed] which is shown in [brackets].

Contents

Abstract

1. Introduction
   1.1. Background
   1.2. Motivation
   1.3. Problem Statement
   1.4. Scope and Boundaries
   1.5. Expected Outcome
   1.6. Structure of the Document

2. Project Management
   2.1. Team Constellation
   2.2. Project Management Methodologies
      2.2.1. Agile and Iterative Approach [CHANGED]
      2.2.2. Project Plan and Work Packages [NEW]
      2.2.3. Milestones [CHANGED]

2.3. Tools to Support Project Management

2.4. Time Tracking and Effort Planning [CHANGED]

2.5. Risk Management

3. Topic of this Research

3.1. Previous Research and Problem Statement [CHANGED]

3.1.1. Project 1 – Conceptual and Analytical Foundation [NEW]

3.1.2. Project 2 – DQM Implementation and Validation [NEW]

3.1.3. Problem Generalisation for the Master Thesis [NEW]

3.2. Research Gap [CHANGED]

3.3. Research Objective [NEW]

3.4. Research Question [CHANGED]

3.5. Research Hypotheses [NEW]

3.6. Research Methodology [NEW]

3.6.1. Research Approach

3.6.2. Comparison of Detection Approaches

3.6.3. Experimental Design

3.6.4. Evaluation Criteria and Metrics

3.7. Success Definition [CHANGED]

3.8. S.M.A.R.T. Goals [CHANGED]

3.9. Expected Contributions [NEW]

3.9.1. Practical Contribution

3.9.2. Scientific Contribution

3.10. Expected Benefits for the Reporting Solution [CHANGED]

4. Solution Landscape

4.1. Data Quality Management at Swiss Post

4.2. Existing Rule-Based DQM Initiative

4.3. Heterogeneous Sorting Centre Environments

4.4. Data Lineage

4.5. Existing Reporting Pipeline

4.6. Existing Data Quality Reporting and Visualisation

4.7. Existing DQM Platform and Infrastructure [CHANGED]

4.8. Baseline for the Master Thesis [NEW]

5. Implementation

5.1. Technical and Experimental Setup [CHANGED]

5.2. First Basic Detection Approach (Eventually Rule based) [CHANGED]

5.2.1. Design

5.2.2. Implementation

5.2.3. Technical Validation

5.3. Improved Detection methods, (Statistical? Different tools? My very own implementation) [NEW]

5.3.1. Design

5.3.2. Implementation

5.3.3. Technical Validation

5.4. Advanced Detection Approach, (ML/Own / Extended first second approach / New Ideas... alternative implementation when 1 2 failed) [NEW]

5.4.1. Selection of Approach

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5.4.2. Design

5.4.3. Implementation

5.4.4. Technical Validation

6. Results

6.1. Introduction

6.2. Evaluation Criteria and Experimental Conditions [CHANGED]

6.3. Rule-Based Approach Results [CHANGED]

6.4. Statistical Approach Results [NEW]

6.5. Advanced / Machine-Learning Approach Results [NEW]

6.6. Comparative Evaluation [NEW]

6.7. Practical / Stakeholder Evaluation [NEW]

6.8. Summary

7. Discussion [CHANGED]

7.1. Interpretation of Results [NEW]

7.2. Evaluation of Research Hypotheses [NEW]

7.3. Answer to the Research Question [NEW]

7.4. Comparison with Existing Research [NEW]

7.5. Practical Implications [NEW]

7.6. Generalisability of the Results [NEW]

7.7. Threats to Validity and Limitations [NEW]

8. Conclusion and Outlook [CHANGED]

8.1. Summary of Findings

8.2. Scientific Contribution

8.3. Practical Contribution

8.4. Future Work

9. Statement of Authorship

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Appendices

A. Queries and Technical Artefacts

B. Project and Experiment Documentation

C. Risk Monitoring

D. Supplementary Experimental Results

E. Stakeholder Evaluation Material